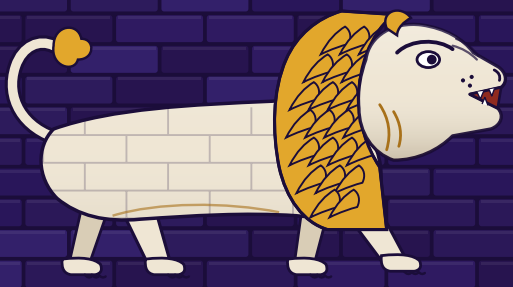
PHYSICS & ENERGY TECHNOLOGY · LEVEL 3

L3

THE COMPLETE STUDY BOOK

Eight subjects of ideas, facts and problems



82 sections · Edition of November 2026 · 242 pages

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How to read this handout

This handout follows the method of Jaan Kalda's physics handouts. The text is short; the work is in the numbered items. Read each section once, then return to its items until you can use them without looking.

Idea 1. An *idea* is a way of thinking that unlocks a whole family of problems. Ideas are framed in violet.

Fact 1. A *fact* is a law or relationship you are expected to know, with the meaning of every symbol and the conditions under which it holds.

Method 1. A *method* is a fixed sequence of steps for a standard task.

Pr 1. Problems are numbered through the handout. A problem followed by a *Solution* is worked for you; cover the solution and try first. For the remaining problems, attempt the problem, then use the *Hints* section, and only then the *Answers* section, both at the back of the subject.

Physical quantities are set in italic, units in roman, with a thin space between number and unit. The index at the end lists every topic and symbol with the page on which it is introduced.

Mechanics and Heat

Motion, vectors, forces, work, energy, temperature and heat. New merged lectures are the main mechanics source; the older heat lecture supplies thermal topics.

1 Choose units and describe motion

Aims. Distinguish distance from signed displacement; Calculate average speed and average velocity from a route.
Before this section: Add and subtract lengths; divide by elapsed time; convert minutes to seconds.

Position tells you where an object is. Displacement is final position minus initial position; distance counts the entire path. Average velocity uses displacement, while average speed uses distance. A negative velocity gives direction, not a negative speed.

Idea 1. Imagine laying a measuring tape along the whole route, then stretching a separate arrow directly from start to finish. The tape represents distance; the arrow represents displacement. Turning back adds tape length while shortening the arrow, so speed and velocity can describe the same journey differently.

Fact 1.

$$\bar{v} = \Delta x/\Delta t$$
$$\text{average speed} = \text{distance}/\Delta t$$

where Δx is the final position minus initial position (m); Δt is the elapsed time (s); $\bar{v}$ is the average velocity (m/s).

Why it holds. Distance adds each part of the route because every part was travelled. Displacement depends only on endpoints, so reversing direction changes its sign contribution.

Method 1.

1. Choose east or right as positive and sketch the route.
2. Add every path length for distance; subtract start position from finish for displacement.
3. Divide distance and displacement separately by the same total time.
4. Attach a direction to velocity and a unit to every result.

Recognising it. Use this distinction whenever a route includes a turn, a return or a request for both speed and velocity.

Pr 1. A student walks 30 m east and 10 m west in 20 s.

Solution. Take east as positive: the two legs are +30 m and −10 m. Distance counts both legs: 30 + 10 = 40 m; displacement is 30 − 10 = +20 m. Over 20 s, average speed is 40/20

$= 2$ m/s and average velocity is $+20/20 = +1$ m/s, east.

Common error. Do not replace displacement with distance in a velocity question.

In the examination. Write the two numerators before dividing: distance for speed, displacement for velocity. This earns method marks even if arithmetic slips.

Pr 2. Can average speed be nonzero while average velocity is zero?

Pr 3. A cart travels 12 m right then 4 m left in 4 s. Find distance, displacement, average speed and average velocity.

Pr 4. Two walkers start and finish together in the same elapsed time but take routes of different lengths. Which averages must match?

(a) Their average velocities

(b) Their average speeds

(c) Both averages

2 Constant acceleration and graphs

Aims. Select a constant-acceleration equation from the known quantities; Interpret slope and area on a velocity-time graph.
Before this section: Signed velocity; rectangle and triangle areas; rearranging a simple equation.

Acceleration measures the change of velocity per second. On a position-time graph, slope is velocity. On a velocity-time graph, slope is acceleration and signed area is displacement. The constant-acceleration equations require acceleration to remain constant.

Idea 2. Acceleration tells how much the velocity changes during each second. A velocity-time graph is like a running record of how quickly position accumulates: each thin time strip contributes velocity times time. The slope describes velocity change, whereas the accumulated signed area describes displacement.

Fact 2.

$$v = u + at$$
$$s = ut + \tfrac{1}{2}at^2$$
$$v^2 = u^2 + 2as$$

where u is the initial velocity (m/s); v is the final velocity (m/s); a is the constant acceleration (m/s^2); s is the displacement (m); t is the elapsed time (s).
Conditions. The equations in this lesson assume constant acceleration throughout the stated interval.

Why it holds. Constant acceleration makes velocity change evenly, so the velocity-time graph is a straight line. With positive forward velocity and negative acceleration, the car initially slows because the change opposes its motion.

Method 2.

1. List the known quantities with signs and identify the unknown.
2. Check that the acceleration is constant before using the equations.
3. Choose an equation without an extra unknown, then substitute with units.
4. Use graph slope or area as an independent interpretation of the result.

Recognising it. Look for a constant acceleration, a uniform slowing statement or a straight velocity-time segment.

Pr 5. A car starts from rest and accelerates at 3 m/s^2 for 4 s.

Solution. Starts from rest means $u = 0$; the example gives $a = 3\ m/s^2$ and $t = 4$ s. The velocity gain is $at = 12$ m/s, so $v = 12$ m/s. The displacement is $\frac{1}{2} \times 3 \times 4^2 = 24$ m. On the velocity-time graph the triangle has base 4 s and height 12 m/s; its area is also 24 m.

Common error. A straight horizontal velocity-time graph means zero acceleration, not zero velocity.

In the examination. For a graph, state which axes you used before calculating slope or area; their units identify acceleration or displacement.

Pr 6. A car has velocity +8 m/s and acceleration $-2\ m/s^2$. Is it initially speeding up?

Pr 7. A car slows from 20 to 8 m/s in 4 s with constant acceleration. Find acceleration and displacement.

Pr 8. A car moves forward while its velocity-time line slopes downward but stays above zero. What happens?

(a) It is stationary because acceleration is negative

(b) It moves forward and slows down

(c) It moves backward and speeds up

3 Resolve vectors before adding

Aims. Add perpendicular vectors using signed components; Give a resultant direction in the correct quadrant.
Before this section: Right-triangle trigonometry; Pythagoras; compass directions.

A vector has size and direction. Two vectors must be added through their signed components, not by simply adding magnitudes. The resultant points from the initial tail to the final head in a head-to-tail drawing.

Idea 3. Components are the separate instructions for moving horizontally and vertically. A component pair describes one arrow even when one instruction is negative. Magnitude measures the arrow length, while signs tell which quadrant it occupies; a calculator angle cannot replace that directional information.

Fact 3.

$$A_x = A\cos\theta$$
$$A_y = A\sin\theta$$
$$R = \sqrt{R_x^2 + R_y^2}$$

where A_x is the horizontal vector component (same as vector); A_y is the vertical vector component (same as vector); R is the resultant magnitude (same as vector); θ is the angle from the stated axis (degrees or radians).

Why it holds. Perpendicular components form a right triangle, so their squares combine to give magnitude. Squaring removes signs, so the quadrant must be recovered from the original component signs.

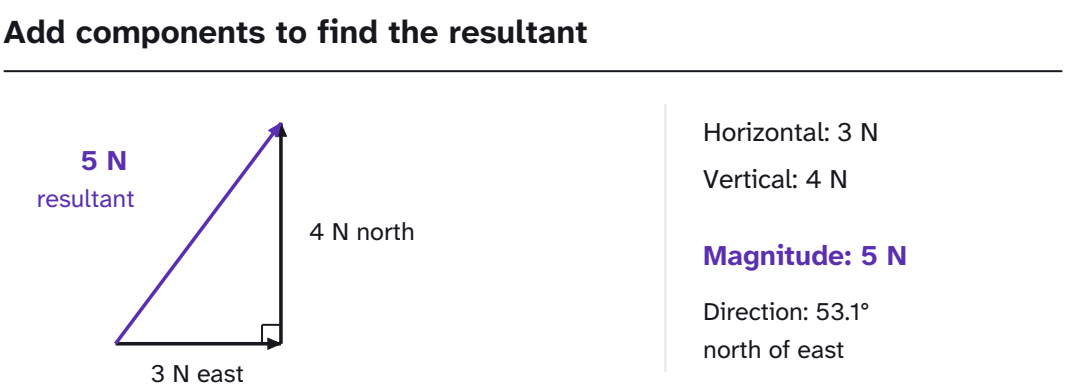


Figure 1. Figure 01. A 3 N east force and a 4 N north force give a 5 N resultant at 53.1° north of east. The force arrows use one common scale; the balancing force points in the opposite direction.

Method 3.

1. Draw axes and identify the reference axis for each angle.
2. Resolve each vector and assign signs to its components.
3. Add horizontal components and vertical components separately.
4. Calculate magnitude and use component signs to choose the direction.

Recognising it. Use components when vectors point in different directions, especially when an angle alone could be ambiguous.

Pr 9. A 3 N force east and a 4 N force north produce a 5 N resultant, 53.1 degrees north of east.

Solution. East and north are perpendicular, so the example has Rx = 3 N and Ry = 4 N. The magnitude is the square root of $3^2 + 4^2$, giving 5 N. The angle satisfies $\tan\theta = 4/3$, so $\theta = 53.1°$ north of east. A balancing force reverses both components: 3 N west and 4 N south, with the same 5 N magnitude.

Common error. The inverse tangent alone can return the wrong quadrant.

In the examination. Report both an angle and its reference direction; an angle alone can leave the quadrant ambiguous.

Pr 10. Does a vector with negative horizontal and positive vertical components point northeast?

Pr 11. Find the magnitude and direction of a vector with components -3 m and $+4$ m.

Pr 12. Two vectors have components $(3,4)$ and $(-3,4)$. What do they share?

(a) Equal direction but different magnitude

(b) Both magnitude and direction

(c) Equal magnitude but different direction

4 Projectiles: one time, two directions

Aims. Separate a projectile into horizontal and vertical motions; Use the shared flight time to calculate horizontal range.

Before this section: Constant-acceleration motion; squaring and square roots; horizontal and vertical directions.

Ignoring air resistance, horizontal acceleration is zero and vertical acceleration is downward. Horizontal and vertical motion share the same time. At the top, only the vertical velocity becomes zero; horizontal velocity continues.

Idea 4. Imagine filming a dropped ball and a horizontally launched ball from the side. With equal starting height and no air drag, their vertical motions match. The launched ball also moves sideways, but that sideways motion does not delay gravity; both directions share one clock.

Fact 4.

$$v_x = u\cos\theta$$
$$v_y = u\sin\theta - gt$$
$$y = u\sin\theta\, t - \tfrac{1}{2}gt^2$$

where u_x is the initial horizontal velocity (m/s); u_y is the initial vertical velocity (m/s); g is the gravitational acceleration magnitude (m/s^2); t is the flight time (s).
Conditions. Ignore air resistance and take gravity as uniform; launch and landing heights need not match.

Why it holds. Gravity changes vertical velocity, while zero horizontal force leaves horizontal velocity constant. Landing time is determined by the vertical fall; that same time determines the sideways range.

Method 4.

1. Draw the launch point and select a positive vertical direction.
2. Write horizontal motion with zero acceleration and vertical motion with gravity.
3. Use the height information to find the positive physical flight time.
4. Use this same time in the horizontal displacement equation.

Recognising it. A horizontal launch from a table or cliff signals a vertical free-fall calculation followed by horizontal motion.

Pr 13. A ball leaves a horizontal table at 3 m/s and takes 0.4 s to land.

Solution. A horizontal launch has initial vertical velocity zero; its horizontal velocity remains 3 m/s. The stated time is 0.4 s, so horizontal range is $3 \times 0.4 = 1.2$ m. The vertical drop is $\frac{1}{2} \times 9.8 \times 0.4^2 = 0.784$ m. The unequal range and drop are consistent because the two directions obey different motion equations.

Common error. Do not set the total velocity to zero at the highest point.

In the examination. If you choose upward positive, the downward displacement and acceleration must both carry negative signs.

Pr 14. At the highest point of an angled projectile, which velocity component is zero?

Pr 15. A ball is projected horizontally at 5 m/s from a height of 4.9 m. Ignore air resistance and use $g = 9.8$ m/s^2. Find flight time and horizontal range.

Pr 16. Two balls leave the same height at the same time, one dropped and one launched horizontally. Ignoring air drag, which lands first?

(a) They land together

(b) The horizontally launched ball lands later

(c) The dropped ball lands later

5 Newton laws and friction

Aims. Construct a free-body diagram for one selected object; Use net force and friction to find acceleration.
Before this section: Signed addition; mass and weight; perpendicular directions.

A free-body diagram shows only forces acting on the selected object. Net force causes acceleration. Action-reaction forces act on different objects and therefore do not cancel on one free-body diagram. Static friction adjusts up to its maximum; kinetic friction applies during sliding.

Idea 5. A free-body diagram is an inventory of pushes and pulls on one object. The object accelerates according to the unbalanced part of that inventory. Friction opposes the sliding direction, and the normal force adjusts to the perpendicular force balance; neither is an additional forward driving force.

Fact 5.

$$\sum F = ma$$
$$W = mg$$
$$f_k = \mu_k N$$
$$f_s \leq \mu_s N$$

where m is the object mass (kg); N is the normal contact force (N); μ_k is the coefficient of kinetic friction (dimensionless); a is the acceleration ($\mathrm{m/s^2}$).
Conditions. The example treats the block as sliding and the given 4 N as kinetic friction.

Why it holds. The horizontal pull and kinetic friction oppose, so only their difference produces horizontal acceleration. With a horizontal pull and no vertical acceleration, the floor normal balances weight in this particular setup.

Method 5.

1. Isolate one object and draw only the forces acting on it.
2. Choose axes and use the perpendicular force balance to obtain the normal force if needed.
3. Choose static or kinetic friction from the stated motion.
4. Add signed forces along the motion and divide the net force by mass.

Recognising it. Use a force diagram before calculating whenever a problem includes a pull, friction, contact or an incline.

Pr 17. A 2 kg block on a horizontal surface is pulled with 10 N.

Solution. Choose the 10 N pull as positive; the 4 N sliding friction acts in the opposite direction. The horizontal net force is $10 - 4 = 6$ N. For the 2 kg block, $a = 6/2 = 3\ \mathrm{m/s^2}$ in the direction of the pull. Weight and normal force balance vertically under the horizontal-surface assumptions, so they do not enter the horizontal sum.

Common error. The normal force is not always mg: angled pulls and inclines change it.

In the examination. Show the signed force equation before substitution; do not write the applied force alone as ma.

Pr 18. A sliding block has a 7 N pull and 7 N friction. Must it be at rest?

Pr 19. A 5 kg block slides to the right on a horizontal floor with coefficient of kinetic friction 0.2. It is pulled horizontally to the right with 20 N. Use $g = 10\ \mathrm{m/s^2}$. Find normal force, friction and acceleration.

Pr 20. A block already slides right. Its pull becomes smaller than the leftward kinetic friction. What happens initially?

(a) It accelerates right because it is moving right

(b) It keeps moving right but slows

(c) It immediately moves left

6 Circular motion and connected objects

Aims. Identify the real forces that supply inward acceleration; Set up separate force equations for connected masses.
Before this section: Newton second law; force diagrams; uniform circular motion directions.

Changing direction is acceleration even at constant speed. Centripetal means inward; it names the required net force, not an extra physical force. In an ideal taut string over a massless frictionless pulley, connected objects share acceleration magnitude.

Idea 6. Changing direction is a change in velocity even when speed stays fixed. For circular motion, the real forces must supply that inward change. For connected masses, internal string forces can be eliminated by considering the whole system, while separate diagrams are needed to recover the tension.

Fact 6.

$$a_r = v^2/r$$
$$\sum F_{\text{inward}} = mv^2/r$$

where v is the speed along the circle (m/s); r is the circle radius (m); a_r is the inward acceleration (m/s^2); T is the string tension (N).
Conditions. Connected-object formulas assume a taut massless string and an ideal frictionless pulley when stated.

Why it holds. Tension acts on both connected masses, but it cancels from the whole-system driving-force calculation. The heavier side supplies the weight imbalance; both masses contribute inertia and therefore belong in the total mass.

Method 6.

1. For circular motion mark the centre and select inward as a positive radial direction.
2. Sum the actual radial force components and equate them to mv^2/r.
3. For connected objects draw a diagram for each mass and use compatible acceleration directions.
4. Use the ideal string constraint to share acceleration magnitude, then solve for acceleration and tension.

Recognising it. For circular paths look for the inward net force; for a pulley pair use one system equation and one mass equation.

Pr 21. A 2 kg object moves at 3 m/s in a circle of radius 1.5 m.

Solution. In the circular example, v = 3 m/s and r = 1.5 m. Squaring speed first gives ar = 9/1.5 = 6 m/s^2, directed toward the centre. Multiplying by the 2 kg mass gives 12 N inward net force. The problem would need a physical setup to decide whether tension, gravity or another force provides those 12 N.

Common error. Do not add a separate centripetal force to tension or gravity.

In the examination. Do not put tension and an extra centripetal force on the same diagram; centripetal describes the inward net force.

Pr 22. If the speed remains fixed and radius doubles, what happens to inward acceleration?

Pr 23. An ideal Atwood machine has masses 2 kg and 3 kg. Use $g = 10\ \mathrm{m/s^2}$. Find acceleration and tension.

Pr 24. For an ideal Atwood machine, why is the total mass used in finding acceleration?

(a) Only the heavier mass moves

(b) Tension adds extra mass to the system

(c) Both connected masses must accelerate

7 Work and kinetic energy

Aims. Determine the sign of work from force and displacement directions; Calculate net work from a change in kinetic energy.
Before this section: Cosine of common angles; squaring speed; positive and negative energy transfers.

Work transfers energy when a force has a component along displacement. A perpendicular force does no work. Net work equals the change in kinetic energy. For a varying force along a line, work is the signed area under the force-position graph.

Idea 7. Work measures how a force transfers energy along a displacement. Carrying a force at an angle is like splitting it into a useful along-path component and a sideways component. Net work concerns all forces together, so it is directly linked to the change in speed through kinetic energy.

Fact 7.

$$W = Fd\cos\theta$$
$$K = \tfrac{1}{2}mv^2$$
$$W_{\text{net}} = K_f - K_i$$

where W is the work transferred by a force (J); F is the force magnitude (N); d is the displacement magnitude (m); K is the kinetic energy (J).

Why it holds. Only the component parallel to displacement contributes to work because the perpendicular component has no along-path displacement. Kinetic energy depends on speed squared, so equal speed increases at different starting speeds require different net work.

Method 7.

1. Identify the displacement and the angle between it and the force.
2. Use only the force component along displacement to calculate work.
3. Add work from all forces for net work, or use the kinetic-energy change.
4. Check that positive net work corresponds to an increase in speed.

Recognising it. Use the work-energy theorem when speeds and mass are given but the detailed force or time history is unnecessary.

Pr 25. A 10 N force pulls through 3 m at 60 degrees to displacement.

Solution. The example states an angle of 60° between the 10 N force and 3 m displacement. Only $10 \cos 60° = 5$ N acts along the displacement. Work by this force is $5 \times 3 = 15$ J. The remaining perpendicular component contributes zero work; 15 J is not automatically net work if other forces also do work.

Common error. Use the force-displacement angle, not an unrelated angle in the drawing.

In the examination. When using the energy theorem, square each speed before subtracting; $(\mathrm{vf} - \mathrm{vi})^2$ is not $\mathrm{vf}^2 - \mathrm{vi}^2$.

Pr 26. Can a nonzero force do zero work during a displacement?

Pr 27. A 2 kg object speeds up from 3 to 5 m/s. Find the net work done.

Pr 28. The same object goes from speed 1 to 2 m/s, then from 2 to 3 m/s. Which stage requires more net work?

(a) The second stage

(b) The first stage

(c) Both require equal net work

8 Energy conservation and springs

Aims. Track kinetic, gravitational and spring energy between two states; Identify when a loss term is required in an energy balance.
Before this section: Work and kinetic energy; choosing a height reference; spring compression or extension.

Energy can move between kinetic, gravitational and elastic stores. Mechanical energy stays constant only when the model excludes losses or other transfers. Choose a height reference; only height changes affect the result.

Idea 8. Energy bookkeeping compares two snapshots rather than following every instant of motion. A compressed spring initially stores energy; when it returns to its natural length, that store can become kinetic energy. The exchange is exact only after accounting for any friction, air drag or other transfer.

Fact 8.

$$K_i + U_i + W_{\text{other}} = K_f + U_f$$
$$U_g = mgh$$
$$U_s = \tfrac{1}{2}kx^2$$

where U_g is the gravitational potential energy (J); U_s is the elastic potential energy (J); k is the spring stiffness (N/m); x is the spring deformation from natural length (m).
Conditions. The falling-mass example neglects air resistance and other energy transfers; spring-energy formulas require the proportional linear range.

Why it holds. At natural length the ideal spring deformation is zero, so its elastic energy term vanishes. On a level frictionless surface gravitational energy does not change, allowing spring energy to become block kinetic energy.

Method 8.

1. Name initial and final states and choose a zero-height level.
2. Write the energy balance before removing terms that are zero.
3. Include friction or other transfers if the model contains them.
4. Solve for the unknown and check the physical sign and units.

Recognising it. Choose an energy balance when initial and final positions or spring deformations are given and time is not needed.

Pr 29. A mass drops 5 m from rest without losses.

Solution. Set the final height to zero; the initial height is then 5 m. The object starts from rest, so its initial energy is gravitational: mg × 5. With no losses, mg × 5 equals one half mv^2; dividing by mass shows why mass cancels. Using g = 10 m/s^2 gives $v^2 = 100$ and speed v = 10 m/s.

Common error. Spring extension is measured from natural length, not total length.

In the examination. For spring energy convert centimetres to metres before squaring the deformation.

Pr 30. Would the same loss-free 5 m fall give a heavier object a larger final speed?

Pr 31. A spring of constant 200 N/m is compressed by 0.10 m and releases a 0.50 kg block on a frictionless horizontal surface. Find its speed when the spring returns to natural length.

Pr 32. The same compressed spring releases two blocks separately on a frictionless surface. Which block leaves faster?

(a) Both have the same speed

(b) The lighter block

(c) The heavier block

9 Temperature, calorimetry and phase changes

Aims. Separate temperature change from phase change in a heating problem; Apply an energy balance with consistent mass and temperature units.

Before this section: Energy in joules; mass conversion; subtraction of temperatures.

Temperature describes thermal state; heat is energy transferred because of a temperature difference. During a phase change of a pure substance at fixed pressure, energy can transfer without a temperature change. In an insulated mixing model, heat lost equals heat gained.

Idea 9. Specific heat tells how much energy a material needs for a temperature rise. Latent heat describes a different job: changing phase. A heating problem becomes manageable when treated as a sequence of stages, each assigned the relation that matches what is physically changing.

Fact 9.

$$Q = mc\Delta T$$
$$Q = mL$$
$$T_K = T_C + 273.15$$

where Q is the energy transferred as heat (J); c is the specific heat capacity (J/(kg K)); L is the specific latent heat (J/kg); ΔT is the temperature change (K or °C difference).

Why it holds. A liquid that warms without a phase change needs energy proportional to its mass and temperature rise. The Celsius and kelvin scales have equally sized intervals, so a temperature difference has the same numerical value in both.

Method 9.

1. Identify the initial and final phase and temperature.
2. Split the process into warming, phase-change and further-warming stages where needed.
3. Use mcΔT for each warming stage and mL for each phase change.
4. Add the energy requirements or balance heat gained and lost in an insulated mixture.

Recognising it. Use specific heat when temperature changes within one phase; add latent heat only when the process crosses a phase transition.

Pr 33. Heating 0.20 kg of water by 10 degrees C needs Q = 0.20 × 4200 × 10 = 8400 J.

Solution. The water remains liquid while warming, so the relevant relation is Q = mcΔT. A rise of 10°C is a temperature difference of 10 K. For 0.20 kg and c = 4200 J/(kg K), Q = 0.20 × 4200 × 10 = 8400 J. If the process also melts ice, that extra stage needs mL; it is not included in the 8400 J.

Common error. A temperature difference of 10 degrees C equals 10 K; an absolute temperature does not.

In the examination. On a heating curve label sloping segments mcΔT and horizontal phase-change segments mL.

Pr 34. Does liquid water warming from 20°C to 30°C have ΔT = 283.15 K?

Pr 35. How much energy heats 0.50 kg of water from 20 to 30 degrees C? Use c = 4200 J/(kg K).

Pr 36. Two samples of the same liquid have the same temperature rise. One has twice the mass. How do their required heats compare?

(a) Both need the same energy

(b) The larger sample needs four times the energy

(c) The larger sample needs twice the energy

10 Heat transfer and thermal expansion

Aims. Classify heat transfer as conduction, convection or radiation; Predict rate changes and calculate small linear expansion.

Before this section: Direct and inverse proportionality; temperature differences; metres and millimetres.

Conduction transfers energy through matter without bulk transport; convection uses fluid motion; radiation uses electromagnetic waves and can cross vacuum. Most solids expand on warming. A thermal insulator reduces transfer but does not create cold.

Idea 10. Thermal expansion is a small fractional change spread over an original length. A longer rod contains more length over which the same fractional change acts. Heat-transfer equations describe energy flow instead; first decide whether the question asks for a size change or a rate of energy transfer.

Fact 10.

$$\Delta L = \alpha L_0 \Delta T$$
$$P_{\text{cond}} = kA\Delta T/L$$
$$P_{\text{rad}} = \epsilon\sigma A(T^4 - T_s^4)$$

where P is the heat-transfer rate (W); k is the thermal conductivity (W/(m K)); A is the transfer surface area (m^2); L is the wall thickness (m); α is the linear expansion coefficient (K^{-1}).

Conditions. The wall comparison uses steady one-dimensional conduction with unchanged material and temperature difference.

Why it holds. For fixed material, expansion grows with both original length and temperature rise. The coefficient gives fractional expansion per degree, so its unit is inverse

kelvin rather than metres.

Method 10.

1. Identify the physical mechanism and select the matching relation.
2. For a comparison, hold every unstated quantity fixed and change only the named factor.
3. For expansion use the original length and the temperature rise.
4. Use kelvin for absolute temperatures in radiation and convert the final length if requested.

Recognising it. Words such as rod length, expansion coefficient and temperature rise indicate a linear-expansion calculation.

Pr 37. Doubling the thickness of an otherwise unchanged wall halves its conductive heat-transfer rate.

Solution. For conduction, the same wall material and temperature difference give P proportional to A/L. Doubling thickness replaces L by 2L, so P becomes P/2. Doubling area instead gives 2P. If both area and thickness double together, the factors cancel and the rate is unchanged.

Common error. Use kelvin in fourth-power radiation equations; Celsius produces incorrect results.

In the examination. For a ratio question explicitly state what stays constant; otherwise the proportionality is incomplete.

Pr 38. A wall is made three times thicker with all other quantities unchanged. What is its new conduction rate?

Pr 39. A rod is 2 m long and has expansion coefficient 0.000012 per K. It warms by 50 K. Find the increase in length in metres and millimetres.

Pr 40. Two equal-length rods undergo the same temperature rise. The first has twice the expansion coefficient. What follows?

(a) The first length increase is twice as large

(b) Their length increases must match

(c) The first final length is twice as large

Hints

Use one hint at a time, then return to the problem.

Pr 3 (p. 3) (1) Sketch the two legs and choose right as positive. Which quantity adds lengths and which adds signed changes? (2) Use 12 m and 4 m positively for distance; use a negative contribution for the leftward leg in displacement. (3) Divide each separate total by 4 s. Keep the direction on velocity and check that speed is nonnegative.

Pr 7 (p. 4) (1) List initial velocity, final velocity and elapsed time. Decide the sign of the velocity change before calculating. (2) Find acceleration from final minus initial velocity divided by 4 s. (3) For displacement use the average of 20 and 8 m/s multiplied by 4 s; this is the area of the trapezoid under the graph.

Pr 11 (p. 6) (1) Place the horizontal component to the left and the vertical component upward. Identify the quadrant first. (2) Use the right triangle for length and find the acute reference angle from the ratio 4/3. (3) Express that angle north of west, or convert it to a counterclockwise angle from positive x. State both magnitude and direction.

Pr 15 (p. 7) (1) Separate the height information from the sideways speed. A horizontal launch has zero initial vertical velocity. (2) Use the fall-distance relation with height 4.9 m and gravitational acceleration 9.8 $\mathrm{m/s^2}$ to find the positive time. (3) Multiply that flight time by 5 m/s for range. Do not put the horizontal speed into the fall equation.

Pr 19 (p. 8) (1) Draw weight, normal force, the rightward pull and leftward sliding friction. Separate vertical from horizontal equations. (2) Use vertical balance to obtain normal force, then multiply it by the kinetic coefficient 0.2. (3) Subtract friction from the 20 N pull and divide the net force by 5 kg. State the acceleration direction.

Pr 23 (p. 10) (1) Decide which mass descends. Draw one force diagram for each mass without adding any extra driving force. (2) Use the difference of the two weights as the system driving force and the sum of masses as system inertia. (3) After finding acceleration, apply the upward force equation to the rising 2 kg mass to obtain tension; it is not simply either weight.

Pr 27 (p. 11) (1) This question gives mass and two speeds rather than a force and distance. Which energy theorem connects them directly? (2) Calculate final and initial kinetic energies separately, retaining each squared speed. (3) Subtract the initial energy from the final energy. A speed increase should give positive net work in joules.

Pr 31 (p. 12) (1) Identify the initial energy store and the final energy store. Is there friction or a change in height? (2) Set the initial spring energy equal to the final block kinetic energy, using the compression in metres. (3) Insert spring constant 200 N/m, compression 0.10 m and mass 0.50 kg. Solve for the positive speed after isolating its square.

Pr 35 (p. 14) (1) Check whether the water changes phase. Identify the initial and final temperatures rather than using either as the temperature rise. (2) Subtract 20°C from 30°C and use the given liquid-water specific heat. (3) Multiply mass 0.50 kg, specific heat 4200 J/(kg K) and the temperature difference. Report energy in joules.

Pr 39 (p. 15) (1) Identify original length and temperature rise; the question asks for the increase rather than the final length. (2) Multiply coefficient by original length 2 m and temperature rise 50 K. (3) The product is in metres. Multiply by 1000 to express the same increase in millimetres; do not add the original length unless asked.

Answers

Pr 2 (p. 3) Yes. A return trip has zero displacement but a positive total distance.

Pr 3 (p. 3) Distance = 16 m; displacement = +8 m; speed = 16/4 = 4 m/s; velocity = 8/4 = +2 m/s. Award one mark for each quantity.

Pr 4 (p. 3) (a) Correct. Matching endpoints give equal displacement and matching times give equal average velocity.

Pr 6 (p. 4) No. Acceleration opposes its current velocity, so its speed initially decreases.

Pr 7 (p. 4) a = (8−20)/4 = −3 m/s^2. s = [(20+8)/2] × 4 = 56 m. Two marks for each calculation, including the sign and unit.

Pr 8 (p. 4) (b) Correct. Positive velocity means forward motion; a decreasing positive velocity means lower speed.

Pr 10 (p. 6) No. It points northwest, in quadrant II.

Pr 11 (p. 6) Magnitude = 5 m. The vector is in quadrant II: 126.9 degrees counterclockwise from positive x, or 53.1 degrees north of west. Two marks for magnitude and two for a correct direction.

Pr 12 (p. 6) (c) Correct. Squared components give equal lengths, while horizontal signs place them in different quadrants.

Pr 14 (p. 7) Only the vertical component; horizontal velocity remains constant in this model.

Pr 15 (p. 7) 4.9 = 0.5 × 9.8 times t^2, so t = 1 s. Range = 5 × 1 = 5 m. Two marks per result with the equation shown.

Pr 16 (p. 7) (a) Correct. They share initial vertical velocity and vertical acceleration, so their fall times match.

Pr 18 (p. 8) No. Zero net force gives zero acceleration; it may continue moving at constant velocity.

Pr 19 (p. 8) N = 50 N, friction = 0.2 × 50 = 10 N, net force = 10 N, a = 10/5 = 2 m/s^2. One mark for each step.

Pr 20 (p. 8) (b) Correct. Velocity initially remains rightward, but the leftward net force reduces that velocity.

Pr 22 (p. 10) It halves because v^2/r is inversely proportional to radius at fixed speed.

Pr 23 (p. 10) Net driving force on the two-mass system is (3−2) × 10 = 10 N. Total mass = 5 kg, so a = 2 m/s^2. For the rising 2 kg mass, T−20 = 2 × 2, hence T = 24 N. Two marks per result.

Pr 24 (p. 10) (c) Correct. Both masses share an acceleration magnitude, so both contribute to the system inertia.

Pr 26 (p. 11) Yes. A force perpendicular to the displacement does zero work.

Pr 27 (p. 11) Net work = 0.5 × 2 × (25−9) = 16 J. Give marks for the energy theorem, both squared speeds, substitution and unit.

Pr 28 (p. 11) (a) Correct. The squared-speed increase is larger during the second stage.

Pr 30 (p. 12) No. Mass cancels from mgh = one half mv^2.

Pr 31 (p. 12) Spring energy = 0.5 × 200 × 0.10 squared = 1 J. Set 1 = 0.5 × 0.50 × v squared to get v = 2 m/s. Two marks for each stage.

Pr 32 (p. 12) (b) Correct. The same spring energy gives a larger speed when shared by a smaller mass.

Pr 34 (p. 14) No. $\Delta T = 10$ K. The Celsius-to-kelvin offset cancels when subtracting two temperatures.

Pr 35 (p. 14) Temperature rise = 10 K. Q = mc times temperature rise = 0.50 × 4200 × 10 = 21000 J. Four marks: model, temperature difference, substitution, result/unit.

Pr 36 (p. 14) (c) Correct. With unchanged specific heat and temperature rise, heat is proportional to mass.

Pr 38 (p. 15) One third of the original rate.

Pr 39 (p. 15) Increase = 0.000012 × 2 × 50 = 0.0012 m = 1.2 mm. Award marks for formula, substitution, metres and conversion.

Pr 40 (p. 15) (a) Correct. Length increase is proportional to the coefficient with the other factors fixed.

Mechanics and Heat, Test A

25 marks. Suggested time 25 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. A runner completes one lap and returns to the start. Average velocity is:

   (a) Always positive
   (b) Impossible to define
   (c) Zero
   (d) Equal to average speed

2. The area under a velocity-time graph gives:

   (a) Displacement
   (b) Force
   (c) Speed at the final instant
   (d) Acceleration

3. Two perpendicular forces of 6 N and 8 N have resultant magnitude:

   (a) 10 N
   (b) 48 N
   (c) 2 N
   (d) 14 N

4. At the top of a projectile path, acceleration is:

   (a) Zero
   (b) Horizontal
   (c) Upward
   (d) Downward with magnitude g

5. An object moving at constant velocity has:

   (a) A forward net force
   (b) Increasing acceleration
   (c) Zero net force
   (d) No forces acting at all

Part II. Show your reasoning.

1. A cart travels 12 m right then 4 m left in 4 s. Find distance, displacement, average speed and average velocity.

2. A car slows from 20 to 8 m/s in 4 s with constant acceleration. Find acceleration and displacement.

3. Find the magnitude and direction of a vector with components −3 m and +4 m.

4. A ball is projected horizontally at 5 m/s from a height of 4.9 m. Ignore air resistance and use $g = 9.8\ \text{m/s}^2$. Find flight time and horizontal range.

5. A 5 kg block slides to the right on a horizontal floor with coefficient of kinetic friction 0.2. It is pulled horizontally to the right with 20 N. Use $g = 10\ \text{m/s}^2$. Find normal force, friction and acceleration.

Mechanics and Heat, Test B

25 marks. Suggested time 25 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. Doubling speed at fixed radius changes centripetal acceleration by a factor of:

   (a) Two
   (b) Four
   (c) Eight
   (d) One half

2. A force perpendicular to displacement does work equal to:

   (a) Fd
   (b) Zero
   (c) -Fd
   (d) Half of Fd

3. If the extension of a Hooke-law spring doubles while k stays constant, its stored energy becomes:

   (a) Four times as large
   (b) Twice as large
   (c) Unchanged
   (d) Half as large

4. During melting at fixed pressure, added heat mainly:

   (a) Raises temperature continuously
   (b) Changes phase
   (c) Decreases mass
   (d) Disappears

5. Which heat-transfer mechanism can operate through a vacuum?

   (a) Fluid mixing
   (b) Convection
   (c) Conduction through air
   (d) Radiation

Part II. Show your reasoning.

1. An ideal Atwood machine has masses 2 kg and 3 kg. Use $g = 10\ \mathrm{m/s^2}$. Find acceleration and tension.

2. A 2 kg object speeds up from 3 to 5 m/s. Find the net work done.

3. A spring of constant 200 N/m is compressed by 0.10 m and releases a 0.50 kg block on a frictionless horizontal surface. Find its speed when the spring returns to natural length.

4. How much energy heats 0.50 kg of water from 20 to 30 degrees C? Use c = 4200 J/(kg K).

5. A rod is 2 m long and has expansion coefficient 0.000012 per K. It warms by 50 K. Find the increase in length in metres and millimetres.

Mechanics and Heat, Test A: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (c) The net displacement is zero even though the distance travelled is not.

Written part. Distance = 16 m; displacement = +8 m; speed = 16/4 = 4 m/s; velocity = 8/4 = +2 m/s. Award one mark for each quantity. (Section: p. 2.)

Question 2. (a) Area has units (m/s) × s = m and includes the sign of velocity.

Written part. a = (8−20)/4 = −3 m/s^2. s = [(20+8)/2] × 4 = 56 m. Two marks for each calculation, including the sign and unit. (Section: p. 3.)

Question 3. (a) The Pythagorean calculation gives the square root of 36+64 = 10 N.

Written part. Magnitude = 5 m. The vector is in quadrant II: 126.9 degrees counterclockwise from positive x, or 53.1 degrees north of west. Two marks for magnitude and two for a correct direction. (Section: p. 4.)

Question 4. (d) Gravity continues to act at the top.

Written part. 4.9 = 0.5 × 9.8 times t^2, so t = 1 s. Range = 5 × 1 = 5 m. Two marks per result with the equation shown. (Section: p. 6.)

Question 5. (c) Constant velocity means zero acceleration, so forces balance.

Written part. N = 50 N, friction = 0.2 × 50 = 10 N, net force = 10 N, a = 10/5 = 2 m/s^2. One mark for each step. (Section: p. 7.)

Mechanics and Heat, Test B: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (b) The speed is squared.

Written part. Net driving force on the two-mass system is $(3-2) \times 10 = 10$ N. Total mass = 5 kg, so a = 2 m/s^2. For the rising 2 kg mass, T$-$20 = 2 × 2, hence T = 24 N. Two marks per result. (Section: p. 9.)

Question 2. (b) The cosine of 90 degrees is zero.

Written part. Net work = 0.5 × 2 × (25$-$9) = 16 J. Give marks for the energy theorem, both squared speeds, substitution and unit. (Section: p. 10.)

Question 3. (a) Spring energy is proportional to extension squared.

Written part. Spring energy = 0.5 × 200 × 0.10 squared = 1 J. Set 1 = 0.5 × 0.50 × v squared to get v = 2 m/s. Two marks for each stage. (Section: p. 11.)

Question 4. (b) Energy changes the phase while temperature remains at the melting point.

Written part. Temperature rise = 10 K. Q = mc times temperature rise = 0.50 × 4200 × 10 = 21000 J. Four marks: model, temperature difference, substitution, result/unit. (Section: p. 13.)

Question 5. (d) Electromagnetic radiation needs no material medium.

Written part. Increase = 0.000012 × 2 × 50 = 0.0012 m = 1.2 mm. Award marks for formula, substitution, metres and conversion. (Section: p. 14.)

Facts at a glance

1. Choose units and describe motion p. 2

$$\bar{v} = \Delta x / \Delta t$$
$$\text{average speed} = \text{distance}/\Delta t$$

2. Constant acceleration and graphs p. 3

$$v = u + at$$
$$s = ut + \tfrac{1}{2}at^2$$
$$v^2 = u^2 + 2as$$

3. Resolve vectors before adding p. 4

$$A_x = A\cos\theta$$
$$A_y = A\sin\theta$$
$$R = \sqrt{R_x^2 + R_y^2}$$

4. Projectiles: one time, two directions p. 6

$$v_x = u\cos\theta$$
$$v_y = u\sin\theta - gt$$
$$y = u\sin\theta\, t - \tfrac{1}{2}gt^2$$

5. Newton laws and friction p. 7

$$\sum F = ma$$
$$W = mg$$
$$f_k = \mu_k N$$
$$f_s \le \mu_s N$$

6. Circular motion and connected objects p. 9

$$a_r = v^2/r$$
$$\sum F_{\text{inward}} = mv^2/r$$

7. Work and kinetic energy p. 10

$$W = Fd\cos\theta$$
$$K = \tfrac{1}{2}mv^2$$
$$W_{\text{net}} = K_f - K_i$$

8. Energy conservation and springs p. 11

$$K_i + U_i + W_{\text{other}} = K_f + U_f$$
$$U_g = mgh$$
$$U_s = \tfrac{1}{2}kx^2$$

9. Temperature, calorimetry and phase changes p. 13

$$Q = mc\Delta T$$
$$Q = mL$$
$$T_K = T_C + 273.15$$

10. Heat transfer and thermal expansion p. 14

$$\Delta L = \alpha L_0 \Delta T$$
$$P_{\text{cond}} = kA\Delta T/L$$
$$P_{\text{rad}} = \epsilon\sigma A(T^4 - T_s^4)$$

Notation

Symbols are listed by section. The same letter may name different quantities in different sections; read the section's Fact.

1. Choose units and describe motion

Δx	final position minus initial position (m)
Δt	elapsed time (s)
$\bar{v}$	average velocity (m/s)

2. Constant acceleration and graphs

u	initial velocity (m/s)
v	final velocity (m/s)
a	constant acceleration ($\mathrm{m/s^2}$)
s	displacement (m)
t	elapsed time (s)

3. Resolve vectors before adding

A_x	horizontal vector component (same as vector)
A_y	vertical vector component (same as vector)
R	resultant magnitude (same as vector)
θ	angle from the stated axis (degrees or radians)

4. Projectiles: one time, two directions

u_x	initial horizontal velocity (m/s)
u_y	initial vertical velocity (m/s)
g	gravitational acceleration magnitude ($\mathrm{m/s^2}$)
t	flight time (s)

5. Newton laws and friction

m	object mass (kg)
N	normal contact force (N)
μ_k	coefficient of kinetic friction (dimensionless)
a	acceleration ($\mathrm{m/s^2}$)

6. Circular motion and connected objects

v	speed along the circle (m/s)
r	circle radius (m)
a_r	inward acceleration ($\mathrm{m/s^2}$)
T	string tension (N)

7. Work and kinetic energy

W	work transferred by a force (J)
F	force magnitude (N)
d	displacement magnitude (m)
K	kinetic energy (J)

8. Energy conservation and springs

U_g	gravitational potential energy (J)
U_s	elastic potential energy (J)
k	spring stiffness (N/m)
x	spring deformation from natural length (m)

9. Temperature, calorimetry and phase changes

Q	energy transferred as heat (J)
c	specific heat capacity (J/(kg K))
L	specific latent heat (J/kg)
ΔT	temperature change (K or °C difference)

10. Heat transfer and thermal expansion

P	heat-transfer rate (W)
k	thermal conductivity (W/(m K))
A	transfer surface area (m^2)
L	wall thickness (m)
α	linear expansion coefficient (K^{-1})

Sources

Motion, vectors, forces, work, energy, temperature and heat. New merged lectures are the main mechanics source; the older heat lecture supplies thermal topics.

The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. Check lecturer-specific derivations and conventions. The practice tests are original study material, not predictions of an examination.

Electricity and Magnetism

Concept-first coverage of the course plan. Chapters 23-31 have lecture support; inductance, AC and EM waves are course-plan extensions without separate lecture files in this folder. Advanced field integrals are recognition topics, not worked derivations.

1 Charges, conductors and Coulomb force

Aims. Predict attraction or repulsion from charge signs; Use simple Coulomb-force ratios without unnecessary arithmetic.
Before this section: Positive and negative charge; squares; micro means one millionth.

Charge is conserved and occurs in integer multiples of the elementary charge. Ordinary charging transfers electrons. Like charges repel and unlike charges attract. Conductors let charge move readily; insulators restrict it. A neutral object can still be attracted by polarization.

Idea 1. Coulomb-force ratios let you compare two situations without knowing every charge value. Imagine moving the same two charges farther apart: the effect spreads according to the square of separation. Charge signs tell attraction or repulsion, while charge sizes and separation set the magnitude.

Fact 1.

$$F = k|q_1 q_2|/r^2$$
$$k = 8.99 \times 10^9 \ \mathrm{N\,m^2/C^2}$$

where F is the electrostatic force magnitude (N); q_1, q_2 is the source charges (C); r is the separation of charges (m).
Conditions. Coulomb calculations here use point charges, or a stated model equivalent to point charges.

Why it holds. Doubling separation squares the distance factor, making the force one quarter as large. Changing only one charge magnitude scales the force directly without changing the separation factor.

Method 1.

1. Mark the signs and positions of both charges.
2. Predict attraction or repulsion before calculating magnitude.
3. For a ratio, compare the charge product and the square of the separation.
4. For a numerical force, convert charges and lengths to SI units first.

Recognising it. Use a ratio when a force is already known and only a charge or separation changes.

Pr 1. Two charges at distance r exert force F.

Solution. Keep both charges fixed and replace the separation r by 2r. The denominator becomes $(2r)^2 = 4r^2$, so the force becomes F/4. If instead one charge doubles at the original separation, its product with the other charge doubles, so force becomes 2F. Changing separation or charge magnitude alone does not change attraction into repulsion.

Common error. A negative charge does not mean a negative force magnitude.

In the examination. A magnitude answer is positive; state attraction or repulsion separately to communicate direction.

Pr 2. Tripling separation with fixed charges changes the force by what factor?

Pr 3. Two charges exert 12 N at separation r. What is the force at separation 2r, and at r if one charge is tripled?

Pr 4. Two unlike charges are moved farther apart without changing their signs. What changes?

(a) The attraction becomes weaker

(b) Attraction becomes repulsion

(c) The force keeps the same magnitude

2 Electric field and superposition

Aims. Combine electric fields by their directions at an observation point; Explain why a negative test charge feels force opposite the field.

Before this section: Vector addition; positive and negative charges; multiplication.

The electric field gives force per unit positive test charge. Field arrows point away from positive sources and toward negative sources. The field exists whether or not a test charge is present. Add fields as vectors, so equal magnitudes can cancel.

Idea 2. A field is a direction-and-strength instruction already present at a location. Its arrow says how a positive test charge would be pushed. An actual negative charge reads the same instruction with the direction reversed, so finding a field and finding a charge force are separate steps.

Fact 2.

$$E = F/q_{\text{test}}$$
$$E = k|Q|/r^2$$
$$\vec{F} = q\vec{E}$$

where E is the electric field magnitude (N/C); q is the charge experiencing the force (C); F is the force magnitude (N).

Why it holds. Multiplying by a negative charge reverses direction but does not make the force magnitude negative. The field is defined by the source arrangement and does not require the test charge to be present.

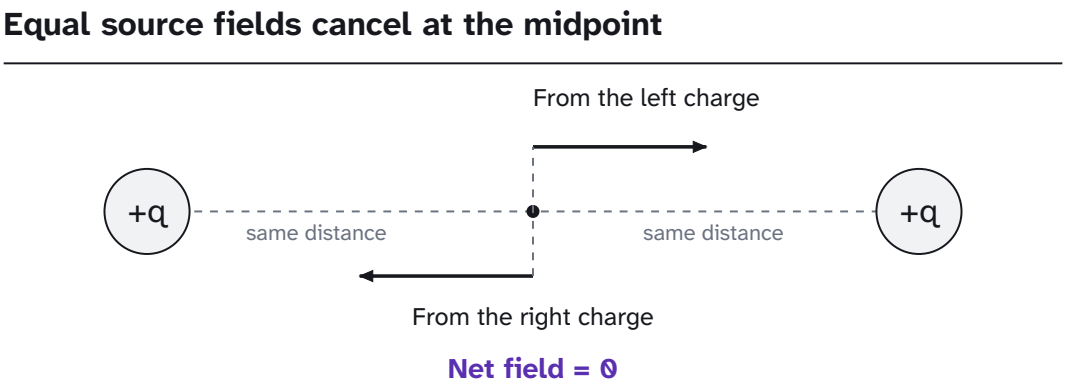


Figure 2. Figure 02. Equal positive charges give zero net field at the midpoint, so the force on a test charge there is zero. The two field arrows are offset vertically only for clarity; both represent fields at the midpoint.

Method 2.

1. Choose the observation point and draw the field arrow from each source there.
2. Point away from a positive source and toward a negative source.
3. Add signed collinear fields or use vector components when needed.
4. Only after finding the net field, use the test-charge sign to obtain force direction.

Recognising it. Separate field direction from force direction whenever the question supplies a negative charge.

Pr 5. At the midpoint between equal positive charges, the two electric fields oppose and cancel.

Solution. At the midpoint between equal positive charges, both source distances are the same, so field magnitudes match. The left positive charge points the field right; the right positive charge points it left. The net field is zero. Replace the right source by an equal negative charge: its field now points toward it, also to the right. The two fields therefore reinforce, pointing from the positive source toward the negative source.

Common error. Zero net field at a point does not mean each source field is zero.

In the examination. Draw field arrows before multiplying by a test charge; otherwise a negative test charge can lead to two direction reversals.

Pr 6. If the net field is zero at a point, is a charge placed there forced in either direction by that field?

Pr 7. A field of 200 N/C points east. Find the force on a charge of −0.003 C.

Pr 8. Two small test charges, one positive and one negative with equal magnitudes, are placed separately at the same point in a nonzero electric field. Their forces have:

(a) Different magnitudes solely because their signs differ

(b) Equal magnitudes and opposite directions

(c) Equal magnitudes and the same direction

3 Flux and Gauss law

Aims. Use the surface normal to determine electric flux; Distinguish enclosed charge from outside charge in Gauss law.
Before this section: Surface area; normal means perpendicular; cosine at 0° and 90°.

Electric flux measures the field passing through an oriented surface. The angle is measured from the surface normal, not the surface itself. Net flux through a closed surface depends on enclosed charge. An outside charge can affect the field at the surface but contributes zero net closed-surface flux.

Idea 3. Flux counts the field crossing a surface rather than the field lying along it. Imagine rain falling through a hoop: turning the hoop changes how much rain crosses its opening even when the rain itself stays the same. A surface normal records that orientation.

Fact 3.

$$\Phi = EA\cos\theta$$

$$\Phi_{\text{closed}} = Q_{\text{enclosed}}/\epsilon_0$$

where Φ is the electric flux ($\mathrm{N\,m^2/C}$); E is the electric field magnitude (N/C); A is the surface area ($\mathrm{m^2}$); Q_{enclosed} is the net charge inside a closed surface (C).

Why it holds. A field parallel to the normal crosses the full effective area. A field tangent to the surface has no crossing component, even if its magnitude is large.

Method 3.

1. Decide whether the question concerns one flat surface or a complete closed surface.
2. For a flat surface, draw the normal and identify its angle to the field.
3. For a closed surface, add only the enclosed charges with signs.
4. Interpret positive closed flux as net outward flux and negative as net inward flux.

Recognising it. Use flat-surface flux when area, a uniform field and orientation are given; use enclosed charge for net closed-surface flux.

Pr 9. A cube enclosing +q has net outward flux $\frac{q}{\varepsilon_0}$.

Solution. The cube is a closed surface enclosing net charge +q. Gauss law fixes its total outward flux as q divided by $\epsilon 0$. Moving q off-centre leaves enclosed charge unchanged, so total flux is unchanged. The nearby faces can now have stronger local fields; Gauss law does not say each face receives an equal share after the move.

Common error. Zero net closed-surface flux does not imply zero field everywhere.

In the examination. The angle in EA cos θ is to the normal; a field parallel to the surface gives zero flux.

Pr 10. Can an external charge produce a nonzero field on a box while contributing zero net flux through it?

Pr 11. A uniform field of 50 N/C passes through a flat area of 0.20 square metres. The chosen surface normal points in the same direction as the field. Find flux. What if the normal is perpendicular to the field?

Pr 12. A flat surface rotates from normal parallel to the field to field tangent to the surface. What happens to flux magnitude?

(a) It stays constant because field strength is unchanged

(b) It increases because the surface turns

(c) It falls from its maximum to zero

4 Symmetry and conductors

Aims. State the electrostatic properties of conducting material; Use the inverse-square field rule outside a spherical conductor.

Before this section: Electric field; conductor versus insulator; inverse-square ratios.

Gauss law is always valid, but it easily gives field magnitude only when symmetry makes the surface calculation simple. In electrostatic equilibrium the field inside conducting material is zero and excess charge lies on its surface. A conductor is an equipotential. For the outside-sphere inverse-square rule, assume an isolated conducting sphere with no external fields.

Idea 4. On an isolated conducting sphere, charge can rearrange until the conductor is in electrostatic equilibrium. Outside this symmetric arrangement, its field follows the same distance pattern as one central point charge. The relevant distance therefore starts at the centre, not at the surface.

Fact 4.

$$E_{\text{outside sphere}} = k|Q|/r^2$$
$$(r > R)$$

where Q is the net sphere charge (C); R is the sphere radius (m); r is the distance from sphere centre (m); E is the electric field magnitude (N/C).
Conditions. The sphere is isolated with spherical symmetry and is in electrostatic equilibrium.

Why it holds. Spherical symmetry makes all outside points at the same centre distance equivalent. The inverse-square outside field weakens by a factor of four when centre distance doubles.

Method 4.

1. Check that the conductor is in electrostatic equilibrium.
2. Distinguish a point in the conducting material from a point outside the sphere.
3. For an outside point measure distance from the centre, not from the surface.
4. Use the point-charge field or a simple ratio, then describe the radial direction.

Recognising it. Use the outside-sphere comparison only when the sphere is isolated as stated and both points lie outside it.

Pr 13. A charged conducting sphere behaves outside like a point charge at its centre.

Solution. Outside a spherically symmetric charged conducting sphere, the field has the same form as a charge at its centre. Compare two outside distances r and 2r. The square of the distance grows by a factor of four, so the field magnitude becomes E/4. This comparison uses centre distances; doubling the gap from the surface would generally give a different ratio.

Common error. Do not extend the zero-field-in-metal statement to every cavity containing a charge.

In the examination. Write r > R beside the outside-sphere formula to show that its domain has been checked.

Pr 14. A field is 80 N/C at 0.20 m from the centre. What is it at 0.40 m if both points are outside?

Pr 15. An isolated charged conducting sphere, with no external fields, has field magnitude 80 N/C at r = 0.20 m. Find the field at r = 0.40 m. Both distances are from the centre and outside the sphere.

Pr 16. Why can doubling the gap above a sphere surface differ from doubling the distance used in its field formula?

(a) The formula measures distance from the centre

(b) The formula always uses distance from the nearest charge on the surface

(c) Outside fields never depend on distance

5 Potential is not field

Aims. Add scalar potentials with their charge signs; Relate potential difference to energy change and field work.

Before this section: Scalar versus vector; signed energy changes; potential energy.

Electric potential is energy per unit charge, a scalar. Electric field is a vector. Add potentials with charge signs, but add fields with directions. Work by the electric field equals the decrease in electric potential energy.

Idea 5. Potential is a location-based energy-per-charge value, rather like height in a gravitational comparison. A change in potential tells how the potential energy of a particular charge changes. The electric field work has the opposite sign because energy given to kinetic motion comes from a decrease in potential energy.

Fact 5.

$$V = kQ/r \text{ (zero at infinity)}$$
$$\Delta U = q\Delta V$$
$$W_{\text{field}} = -\Delta U$$

where V is the electric potential (V = J/C); ΔV is the final minus initial potential (V); ΔU is the change in potential energy (J); q is the moving charge (C).

Why it holds. A positive charge moving to higher potential gains electric potential energy. Positive potential-energy gain means the electric field does negative work over that move.

Method 5.

1. Separate a field question, requiring direction, from a potential question, requiring signed scalar addition.
2. Choose the potential reference supplied by the problem.
3. Calculate energy change from charge times final-minus-initial potential.
4. Reverse the sign of that energy change to obtain work by the electric field.

Recognising it. When a question gives a charge and a potential difference, use the energy change directly rather than inventing a field or travel distance.

Pr 17. Equal positive charges can give zero field at their midpoint but positive potential there.

Solution. At the midpoint of two equal positive charges, equal field arrows point in opposite directions and cancel. Each positive charge contributes a positive potential at that point when zero potential is chosen at infinity. The two potentials add rather than cancel because potential has no directional arrow. Thus zero net field and positive potential can occur at the same point without contradiction.

Common error. A charge moving to lower potential does not always lose potential energy; a negative charge does the opposite.

In the examination. Write ΔV = Vfinal - Vinitial before calculating; swapping the endpoints reverses both energy and work signs.

Pr 18. A +2 C charge moves through ΔV = +3 V. What are ΔU and work by the field?

Pr 19. A +2 C charge moves through a potential difference of +3 V. Find its potential-energy change and the work done by the field.

Pr 20. A negative charge moves through a positive potential difference. What is the sign of its potential-energy change?

(a) Always zero

(b) Negative

(c) Positive

6 Capacitors and dielectrics

Aims. Distinguish series and parallel capacitor connections; Calculate stored charge and identify the effect of a dielectric.

Before this section: Potential difference; charge; reciprocals and addition.

A capacitor stores separated charge and electric-field energy. Capacitance describes charge stored per volt. Parallel capacitors share voltage and their capacitances add. Series capacitors carry equal charge magnitude and have a smaller equivalent capacitance.

Idea 6. Capacitance describes how much separated charge corresponds to each volt across two plates. A capacitor with a larger capacitance can store more charge at the same voltage. It is the connection to a battery or isolation of the capacitor that decides which quantity stays fixed during a change.

Fact 6.

$$Q = CV$$
$$U = \tfrac{1}{2}CV^2$$
$$C_{\parallel} = C_1 + C_2$$
$$1/C_s = 1/C_1 + 1/C_2$$

where C is the capacitance (F); Q is the stored charge magnitude on either plate (C); V is the potential difference between plates (V); U is the stored electric energy (J).

Why it holds. At fixed voltage, increasing capacitance increases stored charge. A microfarad multiplied by a volt gives a microcoulomb, so compatible prefixes can simplify a calculation.

Method 6.

1. Identify shared nodes: parallel capacitors share voltage, while ideal series capacitors carry equal charge magnitudes.
2. Add capacitances for parallel or add their reciprocals for series.
3. Use Q = CV with compatible prefixes such as μF times V giving μC.
4. For dielectric questions first check whether voltage or charge is held fixed.

Recognising it. Use the charge-capacitance-voltage relation when any two of those three quantities are supplied.

Pr 21. Two 6 μF capacitors give 12 μF in parallel and 3 μF in series.

Solution. For two 6 μF capacitors in parallel, capacitances add: $6 + 6 = 12$ μF. For series, $1/C_s = 1/6 + 1/6 = 1/3$ in reciprocal μF, so $C_s = 3$ μF. The series result is smaller than either individual capacitor, providing a quick plausibility check. Inserting a suitable dielectric increases capacitance; the resulting charge or voltage change depends on the connection to a battery.

Common error. Do not use resistor series/parallel rules for capacitors.

In the examination. For series, invert the reciprocal sum at the end; stopping at 1/C is a common lost result mark.

Pr 22. What charge magnitude is stored by 4 μF at 12 V?

Pr 23. A 4 μF capacitor is connected to 12 V. Find the charge in microcoulombs.

Pr 24. A capacitor stays connected to an ideal battery while its capacitance increases. What happens?

(a) Charge stays fixed and voltage increases

(b) Both voltage and charge must stay fixed

(c) Voltage stays fixed and stored charge increases

Fact 7. What changes a capacitor's capacitance?. These simple comparisons cover the geometry and dielectric ideas in the lecture without a derivation.

$$C = \frac{\varepsilon A}{d}$$

$$Q = CV$$

1. Use the first formula for parallel plates with a small gap compared with plate size, so edge effects can be ignored.
2. Here A is overlapping plate area, d is separation, and ε describes the material filling the gap.
3. Doubling area doubles capacitance. Doubling separation halves it.
4. A dielectric filling the gap increases capacitance. Battery attached: voltage stays fixed. Disconnected and isolated: charge stays fixed.

Pr 25. An isolated charged capacitor has capacitance 2 μF and voltage 12 V. A dielectric triples its capacitance. Find the new charge and voltage.

7 Current, resistance and electrical power

Aims. Find current, voltage and power for an ohmic resistor; Predict how wire length and area change resistance.

Before this section: Charge flow; multiplication and division; circuit voltage.

Current is the rate of charge flow. Conventional current follows positive-charge flow, opposite electron drift in a metal. Resistance depends on material and geometry. Ohm law describes an ohmic element under fixed physical conditions.

Idea 7. Voltage is the energy transferred per unit charge, while current counts charge passing each second. Multiplying them gives energy per second, which is power. Resistance connects the current through a resistor with the voltage across that same resistor; those locations matter in a circuit.

Fact 8.

$$I = \Delta Q/\Delta t$$
$$V = IR$$
$$R = \rho L/A$$
$$P = VI$$

where I is the conventional current (A); R is the resistance (Ω); V is the potential difference (V); P is the electrical power (W); ρ is the resistivity (Ω m).
Conditions. Ohm law is used for an ohmic resistor under unchanged physical conditions.

Why it holds. For a given resistor current, the voltage drop follows its resistance. Power follows both the amount of charge passing each second and the energy transferred by each unit charge.

Method 7.

1. Label voltage across the resistor and current through it.
2. Use V = IR to find the missing electrical quantity.
3. Calculate power using VI or an equivalent resistor relation.
4. For wire comparisons hold material and temperature fixed before changing length or area.

Recognising it. When current and resistance are supplied, find the resistor voltage first, then use voltage times current for power.

Pr 26. A 6 Ω resistor across 12 V carries 2 A and dissipates 24 W.

Solution. The 6 Ω resistor has 12 V across it, so I = 12/6 = 2 A. Its electrical power is VI = 12 × 2 = 24 W. Power of 24 W means 24 joules of energy are transferred each second. A longer wire with unchanged material, area and temperature has larger resistance because R is proportional to length.

Common error. More resistance lowers power at fixed voltage but raises power at fixed current.

In the examination. Do not confuse power in watts with energy in joules; multiply power by time when energy is requested.

Pr 27. What happens to resistance if a wire length doubles but material, area and temperature stay fixed?

Pr 28. A 3 Ω resistor carries 2 A. Find voltage and power.

Pr 29. The current through the same fixed resistor doubles. What happens to its power?

(a) It becomes four times as large

(b) It doubles

(c) It stays fixed because resistance stays fixed

8 DC networks, Kirchhoff rules and RC behaviour

Aims. Reduce simple resistor networks and recover branch currents; Describe the initial and final behaviour of an RC charging circuit.

Before this section: Ohm law; series and parallel connections; charge and energy conservation.

Series resistors share current; parallel resistors share voltage. Kirchhoff junction rules express charge conservation, and loop rules express energy conservation. An initially uncharged capacitor charges gradually through a resistor and eventually blocks steady DC in the ideal model.

Idea 8. In a series path, charge has no branch in which to split, so the same current passes through each resistor. The source voltage is shared as energy drops along the path. A larger resistance takes a larger voltage share because the current is common.

Fact 9.

$$R_s = R_1 + R_2$$
$$1/R_p = 1/R_1 + 1/R_2$$
$$\tau = RC$$

where R_{eq} is the equivalent resistance (Ω); I is the current (A); V is the voltage (V); τ is the RC time constant (s).

Why it holds. Series resistances add because their voltage drops add at the same current. Checking that the individual drops sum to the source voltage tests the energy bookkeeping.

Method 8.

1. Identify series elements by shared current and parallel elements by shared voltage.
2. Reduce the network and calculate total source current.
3. Return to the original circuit to calculate branch currents or voltage drops.
4. Check junction current sums and loop voltage sums; for RC questions distinguish transient from long-time behaviour.

Recognising it. Use series reduction when every charge passing one element must pass the next without a junction.

Pr 30. Two 6 Ω resistors in parallel give 3 Ω.

Solution. Both 6 Ω resistors are connected across the same two nodes, so they are parallel. Their equal-resistor equivalent is $6/2 = 3\ \Omega$. The 12 V source therefore supplies $12/3 = 4$ A total. Each branch still has 12 V and carries $12/6 = 2$ A; the branch currents sum to the 4 A source current.

Common error. Adjacent components in a drawing are not necessarily in series.

In the examination. After finding an equivalent resistance, redraw or relabel the original circuit before assigning individual voltage drops.

Pr 31. After a long time, does an ideal capacitor in a DC series RC circuit carry a steady current?

Pr 32. A 4 Ω and 8 Ω resistor are in series across 24 V. Find total current and each voltage drop.

Pr 33. A 4-ohm resistor and an 8-ohm resistor carry the same series current. Which has the larger voltage drop?

(a) The drops must be equal because current is equal

(b) The 8-ohm resistor

(c) The 4-ohm resistor

Fact 10. One time constant: the 63% and 37% rules. Recognize how quickly a capacitor charges or discharges without deriving an exponential.

$$\tau = RC$$

$$1 - e^{-1} \approx 0.632$$

$$e^{-1} \approx 0.368$$

1. Use an ideal series resistor-capacitor circuit with constant R and C. The time constant τ is measured in seconds.
2. Charging from zero: after one time constant, capacitor voltage and charge reach about 63% of their final values.
3. Discharging: after one time constant, about 37% of the starting voltage and charge remain.
4. A larger resistance or capacitance makes the change slower.

Pr 34. A capacitor charges from zero toward 10 V. Its time constant is 2 s. Estimate its voltage after 2 s.

Fact 11. Kirchhoff signs and a real battery. Choosing consistent signs is the main skill. A negative current answer tells you the actual direction.

$$\text{current in} = \text{current out}$$

$$\sum \Delta V = 0$$

$$V_{\text{terminal}} = \mathcal{E} - Ir \quad \text{(battery delivering current)}$$

1. At a junction, charge does not keep building up: add entering and leaving currents separately.
2. Choose a loop direction. Across a resistor, travelling with the assumed current gives $-IR$; travelling against it gives $+IR$.
3. Across a battery, going from its negative to positive terminal gives $+\mathcal{E}$; the reverse gives $-\mathcal{E}$.
4. For a battery supplying current, internal resistance r causes the terminal voltage to be less than its emf $\mathcal{E}$.

Pr 35. A battery has emf 12 V and internal resistance 1 Ω. It supplies a 5 Ω resistor. Find current and terminal voltage.

9 Magnetic force: direction before calculation

Aims. Determine when magnetic force vanishes or is largest; Explain circular motion without a change in kinetic energy.
Before this section: Perpendicular directions; circular acceleration; charge sign.

A magnetic field exerts force on a moving charge when its motion has a perpendicular component. The force is perpendicular to velocity and therefore does no work on a point charge. It can change direction without changing speed.

Idea 9. Magnetic force acts sideways relative to moving charge. For a straight current-carrying wire, the current direction plays the corresponding role. Geometry matters: a wire across the field has maximum magnetic force, whereas a wire along the field has no perpendicular component and no magnetic force.

Fact 12.

$$F = |q|vB\sin\theta$$
$$F_{\text{wire}} = BIL\sin\theta$$

where q is the particle charge (C); v is the particle speed (m/s); B is the magnetic field magnitude (T); F is the magnetic force magnitude (N); θ is the angle between velocity and field (degrees or radians).

Why it holds. A perpendicular arrangement uses the full field-current-length product because the sine factor is one. Reversing current reverses force direction but does not change magnitude when other factors stay fixed.

Method 9.

1. Decide whether the object is a moving charge or a current-carrying wire.
2. Identify the angle to the magnetic field and the perpendicular motion or wire length.
3. Use the right-hand rule for positive charge or conventional current; reverse for negative charge.
4. Calculate magnitude and check that the force is perpendicular to motion.

Recognising it. Use the wire-force relation when current, straight wire length and magnetic field are given; do not substitute particle speed.

Pr 36. A positive charge moving perpendicular to a uniform magnetic field follows a circular path if no other force acts.

Solution. For the positive charge in the example, velocity is perpendicular to the uniform field, so the magnetic force magnitude is qvB. At each point the force is perpendicular to velocity and turns the particle inward. Because the force does no work, speed and kinetic energy remain constant. With no other forces, the continual turning gives a circular path in the plane perpendicular to the field.

Common error. Do not apply the right-hand direction unchanged to an electron.

In the examination. A direction answer needs the charge sign; a right-hand-rule result for a positive charge must be reversed for an electron.

Pr 37. Does a charge moving exactly along a magnetic field feel magnetic force?

Pr 38. A straight 0.50 m wire carries 2 A perpendicular to a 0.30 T field. Find force magnitude.

Pr 39. A straight wire initially lies perpendicular to a uniform field. It rotates until parallel while current stays fixed. Its magnetic force magnitude:

(a) Stays unchanged

(b) Doubles

(c) Falls to zero

10 Sources of magnetic fields

Aims. Relate a current to the direction of the magnetic field it produces; Use simple distance and current ratios for a long wire.
Before this section: Current; inverse proportionality; circular field-line diagrams.

Currents produce magnetic fields. Around a long straight wire the field forms circles. Inside an ideal long solenoid the field is approximately uniform. For two parallel wires, currents in the same direction attract and opposite currents repel.

Idea 10. A straight current-carrying wire has magnetic field lines that circle the wire. Moving farther away weakens the field, but its distance pattern differs from the field of a point charge. Recognizing the source geometry prevents using the familiar but incorrect inverse-square rule here.

Fact 13.

$$B_{\text{wire}} = \mu_0 I/(2\pi r)$$
$$B_{\text{solenoid}} = \mu_0 n I$$

where B is the magnetic field magnitude (T); I is the wire current (A); r is the perpendicular distance from wire (m); n is the solenoid turns per unit length (m^{-1}).
Conditions. The wire is sufficiently long and the observation point sufficiently far from its ends for the long-wire approximation.

Why it holds. For an ideal long wire, magnetic field magnitude is inversely proportional to perpendicular distance. At unchanged current, a threefold increase in distance therefore gives a threefold decrease rather than a ninefold decrease.

Method 10.

1. Identify whether the source is a straight wire, a solenoid or a pair of wires.

2. For a straight wire, use the right-hand grip rule to draw field circles.
3. Apply the matching field relation; for ratios hold unstated factors fixed.
4. For paired wires determine attraction or repulsion from relative current directions.

Recognising it. The phrase long straight wire signals the current-over-distance rule, not the point-charge inverse-square rule.

Pr 40. Doubling distance from a long wire halves its magnetic field; doubling current doubles the field.

Solution. For the long wire, B is proportional to I/r. Moving from r to 2r at the same current gives half the original field. Doubling current at fixed r instead doubles the field. If current and distance both double, the two factors cancel and field magnitude stays the same.

Common error. The wire field varies as inverse distance, not inverse distance squared.

In the examination. Use $1/r$ for the long-wire field; the $1/r^2$ rule belongs to a different source geometry.

Pr 41. A wire gives 30 μT at 2 cm. What field does it give at 6 cm with the same current?

Pr 42. The field is 30 microtesla at 2 cm from a long straight wire. Find the field at 6 cm at the same current.

Pr 43. Current and perpendicular distance from a long wire both double. What happens to magnetic field magnitude?

(a) It stays unchanged

(b) It doubles

(c) It halves

11 Induction and Lenz law

Aims. Identify which change produces an induced emf; Use Lenz law to choose the induced current direction.

Before this section: Magnetic field direction; loop area; clockwise versus counterclockwise viewed from a stated side.

An induced emf requires a change in magnetic flux, which can arise from changing field, area or orientation. Lenz law sets the direction: the induced effect opposes the change of flux. A static magnet near a stationary loop does not produce a sustained induced current.

Idea 11. Induction responds to a changing magnetic flux through a loop, rather than merely to the presence of a magnet. A faster change gives a larger average emf. A coil repeats the same flux-change effect through many turns, so their induced emfs add when the turns are linked alike.

Fact 14.

$$|\mathcal{E}| = N|\Delta\Phi_B|/\Delta t$$
$$\Phi_B = BA\cos\theta$$

where Φ_B is the magnetic flux through one turn (Wb); N is the number of coil turns (dimensionless); $\mathcal{E}$ is the induced emf (V); Δt is the time interval of flux change (s).

Why it holds. More turns multiply the emf when each turn experiences the same flux change. Spreading the same flux change over a longer time reduces the average emf.

Method 11.

1. State the original flux direction and whether its magnitude increases or decreases.
2. Choose an induced field that opposes that change.
3. Use the grip rule to convert the required field to current direction, naming the viewing side.
4. For magnitude, multiply the flux change per turn by the number of turns and divide by elapsed time.

Recognising it. Look for a change in field, loop area or loop orientation over a stated time; identify whether flux is per turn.

Pr 44. If outward flux increases, the induced field points inward.

Solution. In the example, outward flux through the loop is increasing. The induced effect must oppose the increase, so its own field points inward. Viewed from the side toward which the original flux points outward, the inward field requires clockwise current. If the outward flux were decreasing instead, the induced field and current direction would both reverse.

Common error. Opposes the change is not the same as always opposes the original field.

In the examination. Say oppose the change in flux, not oppose the flux: a decreasing field produces a reinforcing induced field.

Pr 45. Does a stationary loop near an unchanging stationary magnet have a sustained induced emf?

Pr 46. A 20-turn coil has flux per turn changing by 0.03 Wb in 0.10 s. Find average emf magnitude.

Pr 47. The same flux change per turn occurs in half the time. With turn count unchanged, average emf magnitude:

(a) Stays unchanged

(b) Doubles

(c) Halves

12 Inductance, AC and electromagnetic waves

Aims. Convert sinusoidal peak voltage to RMS and calculate resistive heating; Relate electromagnetic frequency to vacuum wavelength.
Before this section: Square roots; frequency and wavelength; electrical power.

Course-plan extension: an inductor resists changes of current and stores magnetic energy. AC reverses direction periodically. RMS values describe equivalent heating for resistive loads. In vacuum an EM wave has mutually perpendicular electric field, magnetic field and travel direction.

Idea 12. An alternating voltage has a changing instantaneous value, so its peak is not its steady heating equivalent. RMS voltage compresses that varying waveform into the DC voltage that would produce the same average heating in a resistor. The familiar peak conversion depends on the waveform being sinusoidal.

Fact 15.

$$U_L = \tfrac{1}{2}LI^2$$
$$V_{\text{rms}} = V_{\text{peak}}/\sqrt{2}\ \text{(sinusoidal)}$$
$$c = f\lambda\ \text{(vacuum)}$$

where V_{peak} is the maximum sinusoidal voltage magnitude (V); V_{rms} is the root-mean-square voltage (V); L is the inductance (H); f is the wave frequency (Hz); λ is the wavelength (m). *Conditions.* The peak-to-RMS conversion assumes a sinusoid; the wavelength calculation assumes propagation in vacuum.

Why it holds. Resistive heating depends on voltage squared, so simply averaging positive and negative voltage would miss the heating. A sinusoid has RMS voltage below its peak because it spends only part of each cycle near the peak.

Method 12.

1. Identify whether the question concerns an inductor, sinusoidal AC or an electromagnetic wave.
2. For sinusoidal AC divide peak voltage by the square root of two.
3. Use RMS values for average power in a pure resistor.
4. For vacuum waves calculate wavelength as c divided by frequency and convert metres to the requested unit.

Recognising it. Use RMS values for average resistive power; use the peak-to-RMS conversion only for a specified sinusoid.

Pr 48. A sinusoidal supply with peak voltage 141.4 V has RMS voltage 100 V.

Solution. For the stated sinusoidal supply, 141.4 divided by $\sqrt{2}$ is approximately 100 V RMS. This RMS value matches the heating effect of 100 V DC in the same pure resistor.

For the separate light example, wavelength is 3.00×10^8 divided by $6 \times 10^{14} = 5 \times 10^{-7}$ m. Since one nanometre is 10^{-9} m, the wavelength is 500 nm.

Common error. An ideal transformer does not continuously transform steady DC after switching transients.

In the examination. Keep this extension conceptual: use RMS for resistor power and $c = f\lambda$ for waves; do not mix peak and RMS values.

Pr 49. A 20 V peak sinusoid is applied to 10 Ω. What is the average power?

Pr 50. A sinusoidal voltage has peak value 20 V. Find its RMS value, then the average power in a 10 Ω resistor.

Pr 51. A symmetric sinusoidal voltage has zero average voltage over a cycle. Does a resistor receive zero average power?

(a) Yes; positive and negative heating cancel

(b) Only the positive half-cycle heats it

(c) No; both voltage polarities produce heating

Hints

Use one hint at a time, then return to the problem.

Pr 3 (p. 31) (1) Treat the two requested situations separately, each starting from the original 12 N force. (2) For doubled separation, square the separation-change factor before applying it to the force. (3) For one charge tripled at the original separation, apply only the charge-change factor. State both results with newtons.

Pr 7 (p. 33) (1) The field direction is east. Decide the force direction using the sign of the supplied charge before calculating. (2) Multiply the field magnitude 200 N/C by the charge magnitude 0.003 C. (3) State the positive force magnitude in newtons and its compass direction. The minus sign belongs to the directional reasoning, not a magnitude.

Pr 11 (p. 34) (1) Draw the chosen normal as an arrow. In the first case it points with the field, so identify their angle. (2) Multiply field 50 N/C by area 0.20 m^2 and the cosine of that normal angle. (3) For the second case change only the cosine factor. Keep flux units distinct from field units.

Pr 15 (p. 35) (1) Check that both stated distances start at the centre and exceed the sphere radius as specified. (2) Compare 0.40 m with 0.20 m and square that distance ratio. (3) Divide the original 80 N/C by the squared distance factor. No unknown total charge is needed.

Pr 19 (p. 37) (1) Treat the stated +3 V as final potential minus initial potential. Identify the moving charge sign. (2) Multiply charge +2 C by the potential difference to obtain the signed potential-energy change. (3) Reverse that energy-change sign for work done by the field. Report each quantity in joules and label them separately.

Pr 23 (p. 38) (1) The capacitor is connected across 12 V. Identify whether charge follows by multiplying or dividing capacitance and voltage. (2) Use capacitance 4 microfarads and the supplied voltage. (3) A result from microfarads times volts is in microcoulombs. If using farads instead, convert the final coulombs back consistently.

Pr 25 (p. 39) (1) Initial charge: $Q = CV = 2 \times 12 = 24\,\mu$C. (2) Isolation keeps the charge at 24 μC. (3) New capacitance: 6 μF; new voltage: $V = 24/6 = 4$ V.

Pr 28 (p. 40) (1) Mark the given 2 A as current through the 3-ohm resistor. Which relation gives its voltage drop? (2) Multiply current by resistance to obtain voltage. (3) Use that voltage with the same current to obtain power. Check that the units are volts for the first result and watts for the second.

Pr 32 (p. 41) (1) Confirm there is one series path. Add the two resistances before finding source current. (2) Divide the 24 V source voltage by the total resistance. (3) Use that same current with each resistance separately. Verify that the two voltage drops add to 24 V.

Pr 34 (p. 42) (1) The elapsed time is one time constant. (2) Multiply final voltage by 0.632: $10 \times 0.632 = 6.32$ V.

Pr 35 (p. 42) (1) Use one loop: $12 - I(1) - I(5) = 0$. (2) Total resistance is 6 Ω, so $I = 12/6 = 2$ A. (3) Terminal voltage is $12 - (2)(1) = 10$ V. Check the external resistor: $IR = (2)(5) = 10$ V.

Pr 38 (p. 44) (1) This is a current-carrying wire, not a single moving particle. Choose the relation involving field, current and length. (2) The wire is perpendicular to the field, so

the angle factor is one. (3) Multiply 0.30 T, 2 A and 0.50 m. The question asks for magnitude, so no charge sign is needed.

Pr 42 (p. 45) (1) Keep the current fixed as stated. Compare the new perpendicular distance with the original distance. (2) Use an inverse-distance ratio for a long wire; do not square the ratio. (3) Apply that factor to 30 microtesla. Centimetres can remain in the ratio because both distances use the same unit.

Pr 46 (p. 46) (1) The problem gives flux change per turn. Decide where the 20 turns enter the total effect. (2) Multiply turn count by flux change per turn, then divide by the 0.10 s interval. (3) Use the magnitude of the change because magnitude is requested. A current direction would require separate Lenz-law information.

Pr 50 (p. 48) (1) Separate the request into RMS voltage and average resistor power. The supplied 20 V is a peak value. (2) Divide the peak by the square root of two to obtain the sinusoidal RMS value. (3) Square that RMS voltage and divide by 10 ohms. Keep a few extra digits until the final rounding.

Answers

Pr 2 (p. 31) It becomes one ninth as large.

Pr 3 (p. 31) At 2r: 12/4 = 3 N. Tripling one charge at r: 36 N. Two marks for each ratio and result.

Pr 4 (p. 31) (a) Correct. Opposite signs still attract, but larger separation reduces magnitude.

Pr 6 (p. 32) No. The electric force qE is zero there in the stated source-field model.

Pr 7 (p. 33) Magnitude = 0.003 × 200 = 0.60 N. Direction is west because the charge is negative. Two marks for magnitude and two for direction with reasoning.

Pr 8 (p. 33) (b) Correct. The field is the same and charge magnitudes match; opposite signs reverse force direction.

Pr 10 (p. 34) Yes. Field enters and leaves; the total closed-surface flux from that external charge cancels.

Pr 11 (p. 34) Parallel normal: flux = 50 × 0.20 = 10 N m squared/C. Perpendicular normal: cosine 90 degrees = 0, so flux = 0. Two marks per case.

Pr 12 (p. 34) (c) Correct. The crossing component decreases to zero as the normal becomes perpendicular to the field.

Pr 14 (p. 35) 20 N/C: the centre distance doubles and the field is divided by four.

Pr 15 (p. 35) Inverse-square scaling gives 80 × (0.20/0.40) squared = 20 N/C. Credit the outside-sphere condition and inverse-square reasoning.

Pr 16 (p. 35) (a) Correct. Centre distance equals sphere radius plus the surface gap, so the two doubling operations differ.

Pr 18 (p. 37) ΔU = +6 J and work by the field = −6 J.

Pr 19 (p. 37) Energy change = 2 × 3 = +6 J. Work by field = −6 J. Two marks each; the signs are essential.

Pr 20 (p. 37) (b) Correct. Charge times potential difference is negative for a negative charge and positive difference.

Pr 22 (p. 38) 48 μC, from Q = CV.

Pr 23 (p. 38) Q = CV = 4 × 12 = 48 microcoulombs. In SI this is 0.000048 C. Credit formula, substitution, prefix and result.

Pr 24 (p. 38) (c) Correct. The battery fixes voltage and supplies the extra charge associated with larger capacitance.

Pr 25 (p. 39) Charge stays 24 μC; voltage falls to 4 V.

Pr 27 (p. 40) Resistance doubles.

Pr 28 (p. 40) V = IR = 6 V. P = VI = 12 W (or I^2R). Two marks per result.

Pr 29 (p. 40) (a) Correct. Both current and its corresponding voltage double, so their product quadruples.

Pr 31 (p. 41) No. Once charged, it blocks steady DC in the ideal model.

Pr 32 (p. 41) Total resistance = 12 Ω; current = 2 A; drops are 8 V and 16 V. One mark each; the drops must sum to 24 V.

Pr 33 (p. 41) (b) Correct. At common current, voltage drop is proportional to resistance.

Pr 34 (p. 42) About 6.3 V. It is not fully charged after one time constant.

Pr 35 (p. 42) Current 2 A; terminal voltage 10 V.

Pr 37 (p. 44) No. The angle is 0° or 180°, so its sine is zero.

Pr 38 (p. 44) F = BIL = 0.30 × 2 × 0.50 = 0.30 N. Four marks for formula, perpendicular condition, substitution and unit.

Pr 39 (p. 44) (c) Correct. At parallel alignment the perpendicular component and sine factor vanish.

Pr 41 (p. 45) 10 μT because distance triples.

Pr 42 (p. 45) B is inversely proportional to r, so B = 30 × 2/6 = 10 microtesla. Credit model, ratio, arithmetic and unit.

Pr 43 (p. 45) (a) Correct. Doubling current and distance multiplies the numerator and denominator by the same factor.

Pr 45 (p. 46) No. Its magnetic flux is not changing.

Pr 46 (p. 46) Magnitude = 20 × 0.03/0.10 = 6 V. The direction needs the direction of the flux change; it cannot be inferred from these magnitudes alone.

Pr 47 (p. 46) (b) Correct. Emf depends on the rate of flux change, so halving time doubles that rate.

Pr 49 (p. 48) 20 W: $V_{rms}^2/R = (20^2/2)/10$.

Pr 50 (p. 48) RMS = 20/square root of 2 = 14.14 V. Average power = RMS squared/10 = 20 W. Two marks per stage.

Pr 51 (p. 48) (c) Correct. The heating depends on voltage squared, which is nonnegative for either polarity.

Electricity and Magnetism, Test A

30 marks. Suggested time 30 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. A neutral object loses electrons. It becomes:

   (a) Negative
   (b) Positive
   (c) Uncharged forever
   (d) Magnetic only

2. A negative test charge in a field directed right feels force:

   (a) Left
   (b) Zero
   (c) Right
   (d) Up

3. A charge outside a closed empty box gives net flux through the box of:

   (a) $\frac{q}{\varepsilon_0}$
   (b) Always positive
   (c) Always negative
   (d) Zero

4. Inside the material of a conductor in electrostatic equilibrium, the electric field is:

   (a) Equal to its surface value
   (b) Infinite
   (c) Zero
   (d) Always outward

5. For equal and opposite point charges, take potential as zero at infinity. At their midpoint the potential is:

   (a) Zero
   (b) Undefined
   (c) Positive
   (d) Negative

6. Adding a dielectric to an isolated charged capacitor increases C. Its voltage:

   (a) Decreases
   (b) Increases
   (c) Must stay fixed
   (d) Becomes infinite

Part II. Show your reasoning.

1. Two charges exert 12 N at separation r. What is the force at separation 2r, and at r if one charge is tripled?

2. A field of 200 N/C points east. Find the force on a charge of -0.003 C.

3. A uniform field of 50 N/C passes through a flat area of 0.20 square metres. The chosen surface normal points in the same direction as the field. Find flux. What if the normal is perpendicular to the field?

4. An isolated charged conducting sphere, with no external fields, has field magnitude 80 N/C at $r = 0.20$ m. Find the field at $r = 0.40$ m. Both distances are from the centre and outside the sphere.

5. A +2 C charge moves through a potential difference of +3 V. Find its potential-energy change and the work done by the field.

6. A 4 μF capacitor is connected to 12 V. Find the charge in microcoulombs.

Electricity and Magnetism, Test B

30 marks. Suggested time 30 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. At fixed voltage, doubling resistance makes power:

   (a) Unchanged
   (b) Double
   (c) Half
   (d) Four times

2. After a long time in an ideal DC charging circuit, capacitor current is:

   (a) Maximum
   (b) Negative forever
   (c) Zero
   (d) Equal to the battery voltage divided by the resistor resistance

3. A charge moving parallel to a magnetic field experiences magnetic force:

   (a) qvB
   (b) Twice qvB
   (c) Opposite its velocity
   (d) Zero

4. Two parallel wires carrying currents in the same direction:

   (a) Repel
   (b) Attract
   (c) Have no interaction
   (d) Must carry zero field

5. Which situation produces an induced emf?

   (a) No change of area, field or angle
   (b) Changing magnetic flux
   (c) A stationary unchanging field only
   (d) Constant flux

6. In vacuum, electric and magnetic fields of a plane EM wave are:

   (a) Parallel to each other
   (b) Longitudinal only
   (c) Zero
   (d) Perpendicular to each other and to travel

Part II. Show your reasoning.

1. A 3 Ω resistor carries 2 A. Find voltage and power.

2. A 4 Ω and 8 Ω resistor are in series across 24 V. Find total current and each voltage drop.

3. A straight 0.50 m wire carries 2 A perpendicular to a 0.30 T field. Find force magnitude.

4. The field is 30 microtesla at 2 cm from a long straight wire. Find the field at 6 cm at the same current.

5. A 20-turn coil has flux per turn changing by 0.03 Wb in 0.10 s. Find average emf magnitude.

6. A sinusoidal voltage has peak value 20 V. Find its RMS value, then the average power in a 10 Ω resistor.

Electricity and Magnetism, Test A: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (b) Removing negative charge leaves a positive net charge.

Written part. At 2r: 12/4 = 3 N. Tripling one charge at r: 36 N. Two marks for each ratio and result. (Section: p. 30.)

Question 2. (a) The negative charge reverses the field direction in $F = qE$.

Written part. Magnitude = 0.003 × 200 = 0.60 N. Direction is west because the charge is negative. Two marks for magnitude and two for direction with reasoning. (Section: p. 31.)

Question 3. (d) No charge is enclosed, so total entering and leaving flux cancels.

Written part. Parallel normal: flux = 50 × 0.20 = 10 N m squared/C. Perpendicular normal: cosine 90 degrees = 0, so flux = 0. Two marks per case. (Section: p. 33.)

Question 4. (c) Free charges redistribute until the internal field is cancelled.

Written part. Inverse-square scaling gives 80 × (0.20/0.40) squared = 20 N/C. Credit the outside-sphere condition and inverse-square reasoning. (Section: p. 34.)

Question 5. (a) The signed scalar potentials cancel when distances and charge magnitudes match.

Written part. Energy change = 2 × 3 = +6 J. Work by field = −6 J. Two marks each; the signs are essential. (Section: p. 36.)

Question 6. (a) Charge remains fixed and $V = \frac{Q}{C}$.

Written part. Q = CV = 4 × 12 = 48 microcoulombs. In SI this is 0.000048 C. Credit formula, substitution, prefix and result. (Section: p. 37.)

Electricity and Magnetism, Test B: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (c) $P = \frac{V^2}{R}$, so doubling the denominator halves it.

Written part. V = IR = 6 V. P = VI = 12 W (or I^2R). Two marks per result. (Section: p. 39.)

Question 2. (c) After a long time, the capacitor voltage equals the battery voltage. The resistor voltage and circuit current are zero. During charging, charge moves in the wires onto and off the plates; it does not cross the ideal dielectric.

Written part. Total resistance = 12 Ω; current = 2 A; drops are 8 V and 16 V. One mark each; the drops must sum to 24 V. (Section: p. 40.)

Question 3. (d) The sine of the angle is zero.

Written part. F = BIL = 0.30 × 2 × 0.50 = 0.30 N. Four marks for formula, perpendicular condition, substitution and unit. (Section: p. 43.)

Question 4. (b) Each current experiences the magnetic field of the other wire.

Written part. B is inversely proportional to r, so B = 30 × 2/6 = 10 microtesla. Credit model, ratio, arithmetic and unit. (Section: p. 44.)

Question 5. (b) Faraday law depends on the rate of flux change.

Written part. Magnitude = 20 × 0.03/0.10 = 6 V. The direction needs the direction of the flux change; it cannot be inferred from these magnitudes alone. (Section: p. 45.)

Question 6. (d) A plane electromagnetic wave in vacuum is transverse.

Written part. RMS = 20/square root of 2 = 14.14 V. Average power = RMS squared/10 = 20 W. Two marks per stage. (Section: p. 47.)

Facts at a glance

1. Charges, conductors and Coulomb force p. 30

$$F = k|q_1 q_2|/r^2$$
$$k = 8.99 \times 10^9 \ \mathrm{N\,m^2/C^2}$$

2. Electric field and superposition p. 31

$$E = F/q_{\text{test}}$$
$$E = k|Q|/r^2$$
$$\vec{F} = q\vec{E}$$

3. Flux and Gauss law p. 33

$$\Phi = EA\cos\theta$$
$$\Phi_{\text{closed}} = Q_{\text{enclosed}}/\epsilon_0$$

4. Symmetry and conductors p. 34

$$E_{\text{outside sphere}} = k|Q|/r^2$$
$$(r > R)$$

5. Potential is not field p. 36

$$V = kQ/r \ \text{(zero at infinity)}$$
$$\Delta U = q\Delta V$$
$$W_{\text{field}} = -\Delta U$$

6. Capacitors and dielectrics p. 37

$$Q = CV$$
$$U = \tfrac{1}{2}CV^2$$
$$C_{\parallel} = C_1 + C_2$$
$$1/C_s = 1/C_1 + 1/C_2$$

7. Current, resistance and electrical power p. 39

$$I = \Delta Q/\Delta t$$
$$V = IR$$
$$R = \rho L/A$$
$$P = VI$$

8. DC networks, Kirchhoff rules and RC behaviour p. 40

$$R_s = R_1 + R_2$$
$$1/R_p = 1/R_1 + 1/R_2$$
$$\tau = RC$$

9. Magnetic force: direction before calculation p. 43

$$F = |q|vB\sin\theta$$
$$F_{\text{wire}} = BIL\sin\theta$$

10. Sources of magnetic fields p. 44

$$B_{\text{wire}} = \mu_0 I/(2\pi r)$$
$$B_{\text{solenoid}} = \mu_0 nI$$

11. Induction and Lenz law p. 45

$$|\mathcal{E}| = N|\Delta\Phi_B|/\Delta t$$
$$\Phi_B = BA\cos\theta$$

12. Inductance, AC and electromagnetic waves p. 47

$$U_L = \tfrac{1}{2}LI^2$$
$$V_{\text{rms}} = V_{\text{peak}}/\sqrt{2}\ \text{(sinusoidal)}$$
$$c = f\lambda\ \text{(vacuum)}$$

Notation

Symbols are listed by section. The same letter may name different quantities in different sections; read the section's Fact.

1. Charges, conductors and Coulomb force

F	electrostatic force magnitude (N)
q_1, q_2	source charges (C)
r	separation of charges (m)

2. Electric field and superposition

E	electric field magnitude (N/C)
q	charge experiencing the force (C)
F	force magnitude (N)

3. Flux and Gauss law

Φ	electric flux (N m^2/C)
E	electric field magnitude (N/C)
A	surface area (m^2)
Q_{enclosed}	net charge inside a closed surface (C)

4. Symmetry and conductors

Q	net sphere charge (C)
R	sphere radius (m)
r	distance from sphere centre (m)
E	electric field magnitude (N/C)

5. Potential is not field

V	electric potential (V = J/C)
ΔV	final minus initial potential (V)
ΔU	change in potential energy (J)
q	moving charge (C)

6. Capacitors and dielectrics

C	capacitance (F)
Q	stored charge magnitude on either plate (C)
V	potential difference between plates (V)
U	stored electric energy (J)

7. Current, resistance and electrical power

I	conventional current (A)
R	resistance (Ω)
V	potential difference (V)
P	electrical power (W)
ρ	resistivity (Ω m)

8. DC networks, Kirchhoff rules and RC behaviour

R_{eq}	equivalent resistance (Ω)
I	current (A)
V	voltage (V)
τ	RC time constant (s)

9. Magnetic force: direction before calculation

q	particle charge (C)
v	particle speed (m/s)
B	magnetic field magnitude (T)
F	magnetic force magnitude (N)
θ	angle between velocity and field (degrees or radians)

10. Sources of magnetic fields

B	magnetic field magnitude (T)
I	wire current (A)
r	perpendicular distance from wire (m)
n	solenoid turns per unit length (m^{-1})

11. Induction and Lenz law

Φ_B	magnetic flux through one turn (Wb)
N	number of coil turns (dimensionless)
$\mathcal{E}$	induced emf (V)
Δt	time interval of flux change (s)

12. Inductance, AC and electromagnetic waves

V_{peak}	maximum sinusoidal voltage magnitude (V)
V_{rms}	root-mean-square voltage (V)
L	inductance (H)
f	wave frequency (Hz)
λ	wavelength (m)

Sources

Concept-first coverage of the course plan. Chapters 23-31 have lecture support; inductance, AC and EM waves are course-plan extensions without separate lecture files in this folder. Advanced field integrals are recognition topics, not worked derivations.

The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. Check lecturer-specific derivations and conventions. The practice tests are original study material, not predictions of an examination.

Optics and Waves

The introduction and four current sections: wave motion; electromagnetic theory, photons and light; geometrical optics; physical optics. Derivations are replaced by plain-language interpretation and short exam calculations.

1 Wave language and simple harmonic motion

Aims. Convert an oscillation count into frequency and period; Relate displacement, speed and acceleration at key SHM positions.
Before this section: One oscillation is one complete cycle; reciprocal quantities; displacement from equilibrium.

A wave carries a disturbance and energy. Mechanical waves need a medium; electromagnetic waves can travel through vacuum. Transverse displacement is perpendicular to travel; longitudinal displacement is parallel. Simple harmonic motion has a restoring acceleration proportional and opposite to displacement.

Idea 1. One complete oscillation returns the motion to the same position and direction. Period asks how long that return takes; frequency asks how many returns occur each second. They describe the same repeating motion from opposite viewpoints.

Fact 1.

$$T = 1/f$$
$$f = N/t$$
$$a = -\omega^2 x$$

where N is the number of complete cycles (dimensionless); t is the total elapsed time (s); f is the cycles per second (Hz); T is the time for one cycle (s); ω is the angular frequency (rad/s).

Why it holds. More cycles within the same time means larger frequency. A larger frequency leaves less time for each cycle, so period decreases.

Method 1.

1. Count complete cycles rather than individual crossings of equilibrium.
2. Divide cycle count by total time for frequency.
3. Find period as total time per cycle or the reciprocal of frequency.
4. For SHM compare the point with equilibrium and use restoring acceleration toward equilibrium.

Recognising it. A counted number of complete oscillations over a measured time signals a frequency and period calculation.

Pr 1. If 20 oscillations take 5 s, frequency is 4 Hz and period is 0.25 s.

Solution. Twenty complete oscillations in 5 s means $20/5 = 4$ cycles per second, so $f = 4$ Hz. One cycle takes $5/20 = 0.25$ s, giving $T = 0.25$ s. At an extreme the object must turn around, so its instantaneous speed is zero. Its displacement magnitude is largest there; restoring acceleration therefore has maximum magnitude and points toward equilibrium.

Common error. The direction of particle motion is not necessarily the direction of wave travel.

In the examination. Write $f = N/t$ and $T = t/N$ side by side to prevent interchanging hertz and seconds.

Pr 2. At SHM equilibrium is acceleration magnitude maximum?

Pr 3. A source makes 50 complete oscillations in 10 s. Find frequency and period.

Pr 4. A source doubles its period. How does its frequency change?

(a) It halves

(b) It doubles

(c) It stays unchanged

2 Read a travelling wave without deriving it

Aims. Read amplitude, wavelength and frequency from a travelling-wave expression; Use $v = f\lambda$ and distinguish pattern motion from particle motion.

Before this section: Sine-wave shape; π; coefficients of x and t; SI units.

Amplitude is maximum displacement, wavelength is crest-to-crest separation, and frequency counts cycles per second. Wave speed is the speed of the pattern, not the oscillating particle. In a sine wave, the coefficients of position and time reveal wavelength and period.

Idea 2. The source decides how often a wave repeats, while the medium sets its pattern speed under the stated conditions. Wavelength is the distance the pattern advances during one cycle. At fixed speed, more cycles each second mean each cycle occupies less distance.

Fact 2.

$$v = f\lambda$$
$$k = 2\pi/\lambda$$
$$\omega = 2\pi f$$
$$y = A\sin(kx - \omega t + \phi)$$

where A is the amplitude or maximum displacement (m); k is the wave number (rad/m); ω is the angular frequency (rad/s); λ is the wavelength (m); f is the frequency (Hz).
Conditions. The example takes x and y in metres and t in seconds.

Why it holds. Frequency times wavelength gives distance travelled each second. In the same nondispersive medium, a frequency increase changes wavelength while speed stays fixed.

Method 2.

1. Match the wave to y = A sin(kx - ωt + ϕ).
2. Read A directly, then divide 2π by k to obtain wavelength.
3. Divide ω by 2π to obtain frequency.
4. Multiply frequency by wavelength for pattern speed and check the units.

Recognising it. Use the fixed-speed relationship when frequency changes but the same nondispersive medium is retained.

Pr 5. For $y = 0.02\sin(4\pi x - 20\pi t)$, amplitude is 0.02 m, wavelength is 0.50 m, frequency is 10 Hz and speed is 5 m/s.

Solution. Compare y = 0.02 sin(4πx - 20πt) with the standard form: A = 0.02 m, k = 4π rad/m and $\omega = 20\pi$ rad/s. Wavelength is $2\pi/(4\pi) = 0.50$ m. Frequency is $20\pi/(2\pi) = 10$ Hz. Pattern speed is $10 \times 0.50 = 5$ m/s; this is not the maximum transverse speed of an individual particle.

Common error. The coefficient of time is angular frequency, not ordinary frequency.

In the examination. Identify coefficients before calculating; treating k as wavelength or ω as ordinary frequency loses the factor 2π.

Pr 6. Is the coefficient 20π in the example the frequency in hertz?

Pr 7. A wave has frequency 8 Hz and wavelength 0.75 m. Find speed. If frequency becomes 12 Hz in the same nondispersive medium, find wavelength.

Pr 8. A wave enters a medium where its speed is lower while frequency stays fixed. Its wavelength:

(a) Must remain fixed

(b) Becomes shorter

(c) Becomes longer

Fact 3. Phase means position within a cycle. A phase difference tells whether two oscillations are at the same or different stages of a cycle.

$$\Delta\phi = 2\pi\frac{\Delta x}{\lambda} \quad \text{(phase-difference magnitude at one time)}$$

$$\Delta\phi = 2\pi\frac{\Delta t}{T} \quad \text{(phase-change magnitude at one place)}$$

1. One full cycle is 2π radians. Half a cycle is π radians.
2. At a fixed time, points one wavelength apart are in phase. Points half a wavelength apart are opposite in phase.
3. At one place, a full period brings the oscillation back to the same phase.
4. These formulas give the size of the phase difference. Its sign depends on the order of comparison and the travelling-wave convention.

Pr 9. Two points on a sinusoidal wave are separated by $\lambda/4$. Find their phase-difference magnitude at the same time.

3 Wavefronts, spreading and group velocity

Aims. Predict intensity and amplitude changes for spherical spreading; Distinguish wavefront, ray, phase velocity and group velocity.
Before this section: Surface area grows with radius squared; square roots; amplitude versus intensity.

A ray indicates propagation direction and is normal to a wavefront. Plane wavefronts are flat, spherical wavefronts spread from a point source, and cylindrical wavefronts spread around a line source. Phase velocity tracks a crest; group velocity tracks the envelope of a narrow wave packet.

Idea 3. Picture fixed source power being shared over an expanding spherical shell. The larger shell carries the same total power when nothing is absorbed, but each square metre receives a smaller share. Geometric spreading can therefore reduce intensity without dissipating the total power.

Fact 4.

$$I \propto A^2$$

$$I_{\text{spherical}} \propto 1/r^2$$

where I is the energy-transfer rate per unit area ($\mathrm{W/m^2}$); A is the wave amplitude (depends on disturbance); r is the distance from point source (m).
Conditions. The spreading example assumes an isotropic point source, no absorption and unchanged medium properties.

Why it holds. Sphere area grows with radius squared. With total power fixed, intensity decreases by the same area factor.

Method 3.

1. Identify whether the source produces plane, spherical or cylindrical spreading.
2. For an isotropic point source use intensity proportional to $1/r^2$.
3. Convert an intensity ratio to an amplitude ratio by taking its square root.
4. In description questions, identify a crest for phase velocity and the packet envelope for group velocity.

Recognising it. An isotropic point source with negligible absorption permits inverse-square spreading; other geometries need their own model.

Pr 10. For an isotropic point source without absorption, doubling distance reduces intensity to one quarter and amplitude to one half.

Solution. The same source power crosses a sphere around the source at each distance. Doubling radius makes the sphere area four times as large. The available power per square metre is therefore one quarter as large. Because intensity is proportional to amplitude squared, the amplitude ratio is $\sqrt{(1/4)} = 1/2$.

Common error. The ideal plane-wave model does not have spherical inverse-square spreading.

In the examination. Do not apply the inverse-square intensity rule automatically to a plane wave or line source.

Pr 11. An intensity of 18 W/m^2 is measured at r. What is it at 3r?

Pr 12. At distance r an isotropic source has intensity 18 W per square metre. Find intensity at 3r, ignoring absorption.

Pr 13. Without absorption, which quantity stays the same on complete spheres farther from an ideal point source?

(a) Intensity per square metre

(b) Wave amplitude at each point

(c) Total power crossing each sphere

4 Superposition, standing waves and beats

Aims. Calculate beat frequency from two nearby frequencies; Connect fixed-string harmonics with allowed wavelengths.
Before this section: Adding displacements; wavelength; frequency; fixed endpoints.

Overlapping disturbances add algebraically. In-phase equal waves reinforce; opposite-phase equal waves cancel. A standing wave forms when matching waves travel in opposite directions. Nodes stay at zero displacement. Beats are slow changes in loudness from nearby frequencies.

Idea 4. A string fixed at both ends must have nodes at its endpoints. Only patterns fitting whole numbers of loops between them can persist as normal modes. One loop spans half a wavelength, so a string length is not automatically a complete wavelength.

Fact 5.

$$f_{\text{beat}} = |f_1 - f_2|$$
$$\lambda_n = 2L/n \text{ (string fixed at both ends)}$$

where f_{beat} is the number of loudness cycles per second (Hz); L is the vibrating string length (m); n is the harmonic number, 1, 2, 3, ... (dimensionless); λ_n is the wavelength of harmonic n (m).

Why it holds. Fixed endpoints restrict which wavelengths fit. At fixed wave speed, fitting more loops shortens wavelength and increases frequency.

Method 4.

1. For superposition add instantaneous signed displacements at the same point.
2. For beats take the positive difference of the two nearby frequencies.
3. For a string fixed at both ends draw nodes at both ends.
4. Fit n half-wavelengths into the length, then use wave speed to find each frequency.

Recognising it. For a string fixed at both ends, draw its node-and-loop pattern before selecting the harmonic wavelength.

Pr 14. Two tones at 256 and 260 Hz produce 4 beats per second.

Solution. The tones have frequencies 256 Hz and 260 Hz. Their frequency difference is 4 Hz, so there are four loudness beats each second. For the separate string example, the fundamental has one loop between two end nodes. That loop occupies half a wavelength, so $\lambda 1 = 2\text{L}$ and $\text{f1} = \text{v}/(2\text{L})$.

Common error. A node is not a point where two travelling waves are individually absent.

In the examination. A beat frequency is a difference, not the average of the two tone frequencies.

Pr 15. For a fixed string, how does the second-harmonic frequency compare with the fundamental?

Pr 16. A 1 m string fixed at both ends supports waves at 40 m/s. Find its fundamental frequency and second harmonic.

Pr 17. Adjacent nodes on an ideal standing wave are separated by:

(a) Half a wavelength

(b) One wavelength

(c) One quarter wavelength

5 Light, photons and the electromagnetic picture

Aims. Compare photon energy using frequency or vacuum wavelength; Distinguish beam brightness from energy per photon.

Before this section: Direct and inverse proportionality; wave frequency; energy transfer.

Light behaves as an electromagnetic wave in propagation and exchanges energy in photons. Photon energy increases with frequency. In the photoelectric effect, emission requires enough energy per photon; increasing brightness alone cannot overcome a frequency below threshold.

Idea 5. Photon energy belongs to an individual packet, whereas intensity describes energy arriving per area each second. Increasing frequency makes each packet more energetic. Making a fixed-colour beam brighter can instead increase the number of packets arriving, without changing the energy in each.

Fact 6.

$$E_\gamma = hf = hc/\lambda_0$$
$$\lambda_0 = \text{vacuum wavelength}$$
$$c = 3.00 \times 10^8 \text{ m/s}$$

where E_γ is the energy of one photon (J); h is the Planck constant (J s); f is the light frequency (Hz); λ is the vacuum wavelength (m).

Why it holds. Photon energy increases directly with frequency. Vacuum wavelength decreases as frequency increases because vacuum propagation speed is fixed.

Method 5.

1. Decide whether the question asks about one photon or the whole beam.
2. Use Eγ proportional to frequency, or inversely proportional to vacuum wavelength.
3. For brightness comparisons separate photon energy from the number arriving each second.

4. For photoelectric emission compare energy per photon with the threshold requirement.

Recognising it. Use frequency ratios to compare photon energies and vacuum wavelengths without inserting unnecessary constants.

Pr 18. Blue light has shorter wavelength than red light and therefore higher photon energy.

Solution. Blue light has a shorter vacuum wavelength than red light. Since $E\gamma = hc/\lambda$, the smaller wavelength corresponds to a larger energy per photon. Increasing red-beam brightness at the same colour increases the energy arriving per second, usually through more photons. Each red photon still has the same lower energy; brightness alone does not change its frequency.

Common error. Higher intensity does not mean higher energy per photon. In E = hc/lambda, use the vacuum wavelength. At an ordinary stationary boundary, frequency and photon energy stay the same even though wavelength in the medium changes.

In the examination. Use the phrase energy per photon when discussing threshold effects; total beam energy does not answer the same question.

Pr 19. If light A has twice light B frequency, how do photon energy and vacuum wavelength compare?

Pr 20. Light A has twice the frequency of light B. Compare photon energy and vacuum wavelength.

Pr 21. Light crosses a stationary ordinary air-glass boundary. The transmitted photon energy:

(a) Increases solely because the ray bends

(b) Stays the same because frequency stays the same

(c) Decreases because wavelength in glass is shorter

6 Reflection, refraction and total internal reflection

Aims. Track speed, frequency and wavelength across a refracting boundary; Recognize the two conditions for total internal reflection.

Before this section: Normal to a surface; sine; reciprocal refractive index.

Angles of incidence, reflection and refraction are measured from the normal. At a stationary interface frequency stays the same, while speed and wavelength may change. When light enters a higher refractive index at a nonzero angle, the ray bends toward the normal. Total internal reflection needs travel from higher to lower index and an incidence angle greater than the critical angle.

Idea 6. A ray entering a lower-index medium bends away from the normal. As incidence angle increases, the transmitted direction approaches the boundary. At the critical angle the limiting transmitted ray runs along that boundary; beyond it the ordinary transmitted ray cannot propagate away into the second medium.

Fact 7.

$$\theta_i = \theta_r$$
$$n = c/v$$
$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$
$$\sin\theta_c = n_2/n_1$$

where n is the refractive index (dimensionless); c is the vacuum light speed (m/s); v is the light speed in medium (m/s); θ_c is the critical incidence angle (degrees or radians).
Conditions. The boundary is stationary and the media are treated as uniform; air index is approximated as 1.

Why it holds. Only higher-to-lower index travel can reach the critical-angle condition. Total internal reflection requires incidence above, rather than merely equal to, the critical angle.

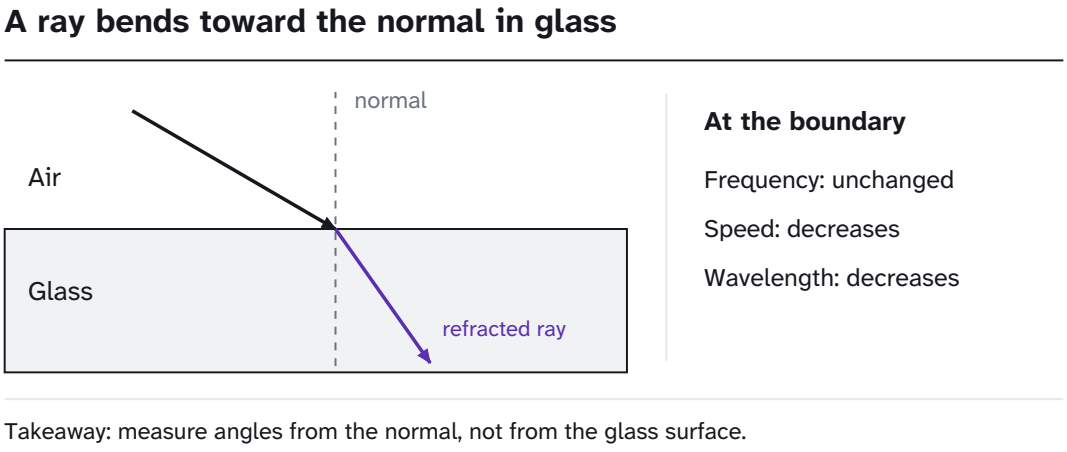


Figure 3. Figure 03. At oblique incidence from air into glass, the ray bends toward the normal. At a stationary boundary its frequency is unchanged, while speed and wavelength decrease.

Method 6.

1. Draw the surface normal and measure all ray angles from it.
2. Use v = c/n to compare speeds; frequency stays fixed at a stationary boundary.
3. Use wavelength = speed/frequency to compare wavelengths.
4. For total internal reflection check higher-to-lower index travel, then compare incidence with the critical angle.

Recognising it. The words critical angle signal a check of travel direction followed by the lower-to-higher index sine ratio.

Pr 22. From air into glass with index 1.5, light slows to 2×10^8 m/s.

Solution. Air is approximated by n = 1 and glass has n = 1.5. In glass, speed is $3.00 \times 10^8 / 1.5 = 2.00 \times 10^8$ m/s. The frequency is set by the source and remains the same across the boundary. Because speed drops while frequency stays fixed, wavelength in glass is two thirds of its air value.

Common error. The critical angle is an incidence angle in the higher-index medium.

In the examination. At the critical angle the refracted ray runs along the boundary; total internal reflection occurs above that angle.

Pr 23. Can light travelling from air into glass undergo total internal reflection at that interface?

Pr 24. Find the critical angle for glass of index 1.5 into air of index 1.0.

Pr 25. At exactly the critical angle, the refracted ray travels:

(a) Along the normal

(b) Back along the incident ray

(c) Along the boundary

7 Mirrors, lenses and image signs

Aims. Use the stated image-distance sign convention consistently; Infer image reality, orientation and relative size from di and m.

Before this section: Reciprocals; positive and negative numbers; a simple ray sketch.

A real image is formed where rays actually meet and can be projected on a screen. A virtual image is where rays appear to originate. Use one sign convention consistently: real object distance positive, real image distance positive; converging focal length positive and diverging focal length negative.

Idea 7. A lens equation locates where rays meet or appear to originate. Magnification then compares image height with object height, including orientation. A negative magnification records an inverted image; it does not mean the physical image has a negative length.

Fact 8.

$$1/f = 1/d_o + 1/d_i$$
$$m = -d_i/d_o$$

where f is the signed focal length (cm or m consistently); d_o is the object distance (same length unit as f); d_i is the signed image distance (same length unit as f); m is the transverse magnification (dimensionless).
Conditions. Use the lesson convention: real object and real image distances positive; converging focal length positive.

Why it holds. A converging lens with an object at twice focal length gives a real image at twice focal length on the other side. Magnitude of magnification gives relative size, while its sign gives orientation.

Method 7.

1. Write the sign convention and assign f and do before substitution.
2. Rearrange the lens equation to isolate 1/di and then invert.
3. Calculate m = -di/do.
4. Use di sign for real or virtual, m sign for orientation, and magnitude of m for size.

Recognising it. Use one consistent sign convention when focal length and object distance are given, then interpret the numerical result.

Pr 26. For a converging lens with focal length 10 cm and object distance 30 cm, image distance is 15 cm and magnification is -0.5: real, inverted, reduced.

Solution. A converging lens has f = +10 cm and the real object has do = +30 cm. Then 1/di = 1/10 − 1/30 = 1/15 per cm, so di = +15 cm. Magnification is $-15/30 = -0.5$. Positive image distance means real; negative magnification means inverted; magnitude 0.5 means half the object height.

Common error. Do not call every image behind a lens virtual; interpret actual ray convergence and your convention.

In the examination. After subtraction, invert 1/di to get di; include all three requested image descriptions.

Pr 27. What does m = +2 tell you about an image?

Pr 28. A converging lens has focal length 12 cm and an object at 24 cm. Find image distance, magnification and image type.

Pr 29. Why can a real image be projected onto a screen?

(a) Actual rays meet at its position

(b) Its magnification must be positive

(c) All images lie on the lens surface

8 Optical systems, prisms and instruments

Aims. Explain the different roles of objective and eyepiece; Calculate telescope angular magnification for normal adjustment.

Before this section: Focal length; image formation; ratios of lengths.

A prism refracts light at two surfaces; dispersion separates colours because refractive index depends on wavelength. In a multi-lens system, the first image becomes the object for the next lens. A microscope and a telescope both use an objective and eyepiece, but they serve different object distances.

Idea 8. A telescope objective forms an intermediate image of a distant scene, and an eyepiece enlarges the angle under which it is viewed. Angular magnification compares viewing angles, so it is different from the height ratio for one ordinary lens image.

Fact 9.

$$m_{\text{total}} = m_1 m_2$$
$$|M_{\text{telescope}}| = f_o/f_e \ \text{(normal adjustment)}$$

where f_o is the objective focal length (cm or m consistently); f_e is the eyepiece focal length (same length unit as fo); M is the angular magnification (dimensionless); m_1, m_2 is the transverse magnifications of successive stages (dimensionless).

Conditions. The telescope is in normal adjustment, giving the final image at infinity.

Why it holds. For normal adjustment, a shorter eyepiece focal length gives larger angular magnification with the same objective. A focal-length ratio is dimensionless when both lengths use the same unit.

Method 8.

1. Identify the instrument and whether it views a distant or nearby object.
2. For a normal-adjustment astronomical telescope divide objective focal length by eyepiece focal length for magnification magnitude.
3. State image orientation separately.
4. For a general multi-lens system locate the first image before using it as the next object.

Recognising it. Use the focal-length ratio for an astronomical telescope in normal adjustment, not for an arbitrary two-lens system.

Pr 30. An astronomical telescope with objective focal length 80 cm and eyepiece 4 cm has angular magnification magnitude 20.

Solution. The telescope objective focal length is 80 cm and the eyepiece focal length is 4 cm. Their ratio is $80/4 = 20$, so the angular magnification magnitude is 20. An astronomical telescope with this usual two-converging-lens arrangement gives an inverted image. The ratio needs no length conversion because both focal lengths use the same unit.

Common error. Linear image magnification and angular magnification are different quantities.

In the examination. Use the telescope focal-length ratio only for the stated telescope model, not as a universal magnification formula.

Pr 31. Keeping the objective fixed, does doubling the eyepiece focal length increase or decrease telescope magnification?

Pr 32. A telescope objective has focal length 100 cm and eyepiece 5 cm. Find angular magnification magnitude. What happens if the eyepiece is replaced by 10 cm?

Pr 33. Which eyepiece gives greater magnification with the same objective in normal adjustment?

(a) Both give equal magnification

(b) The shorter-focal-length eyepiece

(c) The longer-focal-length eyepiece

9 Interference and diffraction

Aims. Predict how slit geometry changes an interference or diffraction pattern; Distinguish double-slit bright fringes from single-slit dark fringes.

Before this section: Wavelength; coherent sources; nanometres and millimetres; small angles.

Interference occurs when overlapping waves combine. A stable visible pattern needs a fixed phase relationship (coherence). Diffraction is spreading associated with an opening or obstacle. Narrower slits spread light more. Young double-slit bright fringes are evenly spaced when the angles are small.

Idea 9. Coherent slits create bright and dark directions through path differences between overlapping waves. A screen turns those angles into distances. Moving the screen farther away enlarges the visible spacing even though the slit geometry and bright angular directions remain unchanged.

Fact 10.

$$\Delta y = \lambda L/d$$

$$d \sin\theta = m\lambda \text{ (double-slit bright)}$$

$$a \sin\theta = m\lambda \text{ (single-slit dark, } m \neq 0\text{)}$$

where Δy is the double-slit fringe spacing (m); L is the slit-to-screen distance (m); d is the separation of double slits (m); a is the single-slit width (m); λ is the wavelength (m).
Conditions. Even double-slit spacing uses the small-angle, distant-screen approximation and coherent illumination.

Why it holds. Longer wavelength increases the angular change needed between neighbouring bright fringes. A more distant screen maps the same angular separation to a larger

physical distance.

Method 9.

1. Identify whether the setup is double-slit interference or single-slit diffraction.
2. Label d as slit separation and a as slit width; they describe different distances.
3. For double-slit spacing use $\lambda L/d$ with all lengths in consistent units.
4. For a single slit remember that a smaller width moves the first minima outward and broadens the centre.

Recognising it. For two coherent narrow slits and a distant screen, use wavelength, slit separation and screen distance to find spacing.

Pr 34. Doubling screen distance doubles double-slit fringe spacing.

Solution. Double-slit spacing is proportional to screen distance when wavelength and slit separation stay fixed. Replacing L by 2L therefore doubles fringe spacing. In the separate single-slit comparison, the first dark direction obeys $\sin\theta = \lambda/a$. Halving a makes that angle larger, so the bright central maximum broadens.

Common error. The double-slit bright condition is not the single-slit bright condition.

In the examination. Label whether you are locating a bright double-slit fringe or a dark single-slit fringe before using the order number.

Pr 35. If double-slit separation doubles with screen distance and wavelength fixed, what happens to fringe spacing?

Pr 36. In a double-slit experiment, wavelength is 500 nm, screen distance 2 m and slit separation 0.50 mm. Find fringe spacing.

Pr 37. Only the double-slit screen distance doubles. The bright-fringe angular directions:

(a) Double along with screen spacing

(b) All become zero

(c) Stay unchanged while screen spacing grows

Fact 11. Recognize gratings, Michelson paths and thin films. All three appear in the physical-optics section. Start by identifying which arrangement the question describes.

$$\Delta\text{path} = 2\Delta x \quad \text{(one Michelson mirror moves)}$$

1. A diffraction grating has many equally spaced slits. It makes sharp bright directions that separate wavelengths.
2. A Michelson interferometer splits light into two arms. Each beam travels to a mirror and back, so mirror movement changes a round-trip path.
3. A thin film produces interference between reflections from its two surfaces. Reflection can add a phase reversal, so do not copy the ordinary double-slit bright rule.
4. For fringe counting in a simple Michelson setup, each one-wavelength change in path difference shifts the pattern by one fringe.

Pr 38. In a Michelson interferometer using 500 nm light in air, one mirror moves 250 nm along its arm. What path change and fringe shift result?

10 Polarization and Malus law

Aims. Apply the initial half-intensity rule to unpolarized light; Apply Malus law using the angle between polarization and analyzer axes.
Before this section: Cosine of common angles; intensity versus field amplitude; sequential fractions.

Polarization describes the electric-field direction of a transverse light wave. An ideal polarizer transmits half the intensity of unpolarized light. An analyzer then transmits a fraction set by the angle between its axis and the incident polarization. Linear, circular and elliptical polarization describe different field-tip motions.

Idea 10. The first ideal polarizer turns unpolarized light into linearly polarized light while transmitting half its intensity. The analyzer receives that output. It then passes a fraction determined by the angle to the already established polarization, so the first and second stages use different rules.

Fact 12.

$$I = I_0 \cos^2 \theta$$
$$I_{\text{after first}} = I_{\text{unpolarized}}/2$$

where I_0 is the intensity incident on the analyzer ($\mathrm{W/m^2}$ or stated units); I is the intensity transmitted by analyzer (same as I0); θ is the angle between incident polarization and analyzer axis (degrees or radians).
Conditions. The calculation assumes ideal polarizers and initially unpolarized illumination.

Why it holds. The half-intensity rule applies to initially unpolarized input. For linearly polarized input, the analyzer transmits a squared-cosine fraction.

Method 10.

1. Check whether the first incident beam is unpolarized or already linearly polarized.
2. For unpolarized light through an ideal first polarizer divide intensity by two.
3. Use that transmitted intensity as I0 for the analyzer.
4. Multiply by $\cos^2\theta$, where θ is between the two relevant axes.

Recognising it. For successive polarizers, identify input polarization and process each element in order.

Pr 39. Unpolarized intensity 100 units passes a polarizer then an analyzer at 60 degrees.

Solution. The initial 100 intensity units are unpolarized, so the first ideal polarizer transmits 50 units. The analyzer axis is at 60° to the first polarizer axis. Cos 60° is 0.5 and its square is 0.25. The analyzer therefore transmits $50 \times 0.25 = 12.5$ units, one eighth of the original intensity.

Common error. Do not halve already polarized intensity again before applying Malus law.

In the examination. Square the cosine, not the angle; and do not apply an extra half factor to an already polarized analyzer input.

Pr 40. Two ideal polarizers have perpendicular axes. What does the second transmit from the first?

Pr 41. Unpolarized intensity 80 W per square metre passes through a polarizer and an analyzer at 45 degrees. Find the final intensity.

Pr 42. A second ideal polarizer is aligned with the first. What fraction of the first output passes it?

(a) All of it

(b) Half of it

(c) None of it

11 Scattering, apparent depth and lens power

Aims. Calculate apparent depth near normal viewing and signed lens power; Explain why small-particle scattering favours shorter wavelengths.
Before this section: Refraction; inverse relationships; centimetres-to-metres conversion.

Small particles compared with wavelength produce Rayleigh scattering, which favours shorter wavelengths strongly. Droplets comparable to or larger than visible wavelengths can scatter a broad range of colours, helping explain white clouds. Refraction makes a submerged object appear shallower when viewed near the normal from air.

Idea 11. Lens power describes how strongly a lens converges or diverges rays. A shorter focal-length magnitude means stronger bending and greater power magnitude. The negative sign of a diverging lens communicates the direction of its optical effect, rather than the physical size of the lens.

Fact 13.

$$I_{\text{Rayleigh}} \propto 1/\lambda^4$$
$$d_{\text{apparent}} \approx d_{\text{real}}/n$$
$$P_{\text{lens}} = 1/f \ \ (\text{f in metres})$$

where d_{real} is the actual depth (m); d_{apparent} is the apparent depth (m); n is the liquid refractive index relative to air approximation (dimensionless); P is the lens optical power (D = m^{-1}); f is the signed focal length (m).
Conditions. The apparent-depth approximation assumes viewing from air near the normal; the Rayleigh rule applies to particles much smaller than wavelength.

Why it holds. Reciprocals make smaller focal-length magnitudes correspond to larger power magnitudes. Dioptres use metres in the focal-length reciprocal, so centimetres must be converted first.

Method 11.

1. Select the relevant model: small-particle scattering, near-normal depth or lens power.
2. For depth viewed from air divide real depth by the liquid index.
3. For power convert focal length to metres before taking its reciprocal.
4. Retain the focal-length sign so converging power is positive and diverging power negative.

Recognising it. A supplied focal length and a request for optical power call for a signed reciprocal in metre units.

Pr 43. A pool 2 m deep in water of index 1.33 appears about 1.50 m deep from air near the normal.

Solution. For the pool, near-normal apparent depth is $2/1.33 \approx 1.50$ m, so the bottom ap-

pears shallower. The index does not change the real 2 m depth; it changes the directions of the emerging rays. For the separate converging lens, $f = +0.25$ m. Its power is $1/0.25 = +4\ \mathrm{m}^{-1}$, written +4 dioptres.

Common error. A focal length of 25 cm is not 25 m; using it directly gives the wrong lens power.

In the examination. A reciprocal of a focal length in centimetres does not directly give dioptres; convert to metres first.

Pr 44. A diverging lens has focal length -50 cm. What is its power?

Pr 45. A diverging lens has focal length -50 cm. Find its optical power.

Pr 46. Which diverging lens has greater power magnitude: focal length -0.25 m or -0.50 m?

(a) They match because both diverge

(b) The -0.25 m lens

(c) The -0.50 m lens

12 Recognize advanced optics without long derivations

Aims. Recognize what a quarter-wave plate changes about polarized light; Use Brewster angle and plane-wave relations at recognition level.
Before this section: Linear and circular polarization; phase as position within an oscillation cycle; refractive index.

The wave equation describes a disturbance propagating in space and time. Complex notation is a bookkeeping method for amplitude and phase; the measured field is real. Maxwell laws connect electric and magnetic fields. Birefringent materials have different refractive behaviour for different polarizations. At Brewster incidence the reflected component can be linearly polarized.

Idea 12. Brewster incidence is a special transparent-interface geometry. In ordinary transparent nonmagnetic media, the reflected light at that angle lacks the component polarized in the incidence plane. This concerns polarization of the reflected component, not complete reflection of all incoming energy.

Fact 14.

$$\tan \theta_B = n_2/n_1$$
$$E = cB \text{ (plane wave in vacuum)}$$

where θ_B is the Brewster incidence angle from the normal (degrees or radians); n_1, n_2 is the incident and transmitted medium indices (dimensionless); E is the plane-wave electric-field magnitude (V/m); B is the plane-wave magnetic-field magnitude (T).
Conditions. The wave plate is operating at its intended wavelength; Brewster law here uses ordinary transparent nonmagnetic media.

Why it holds. The refractive-index ratio sets the tangent of the Brewster angle from the normal. Brewster incidence and total internal reflection describe different conditions and effects.

Method 12.

1. Identify whether the question concerns phase delay, interface polarization or vacuum wave fields.
2. For a quarter-wave plate, compare the two orthogonal component amplitudes and their phase difference.
3. For Brewster incidence use the index ratio with inverse tangent.
4. For a plane vacuum wave connect field magnitudes using E = cB without a long derivation.

Recognising it. Use Brewster angle when an interface question links refractive indices and linearly polarized reflection.

Pr 47. A quarter-wave plate can turn suitable linearly polarized light into circular polarization by introducing a quarter-cycle phase difference between equal orthogonal components.

Solution. Resolve suitable incident linear polarization along the plate principal axes. Equal component amplitudes are required for the stated circular-polarization result. A quarter-wave plate adds a quarter-cycle, or $\pi/2$, relative phase delay. Equal amplitudes with this delay give circular polarization; unequal amplitudes generally give elliptical polarization.

Common error. These recognition notes do not replace a lecturer-assigned derivation; return to the source if such derivations are explicitly assessed.

In the examination. For a recognition answer name the condition and consequence; deriving Maxwell equations is unnecessary for these exercises.

Pr 48. Is a quarter-cycle phase delay alone enough to guarantee circular polarization?

Pr 49. For light from air into glass of index 1.5, estimate Brewster angle.

Pr 50. For unpolarized light at Brewster incidence in ordinary transparent nonmagnetic media:

(a) All incident light must be reflected

(b) Total internal reflection must occur

(c) The reflected light is linearly polarized

Hints

Use one hint at a time, then return to the problem.

Pr 3 (p. 65) (1) Identify the complete cycle count and total time. Which ratio counts cycles per second? (2) Divide 50 cycles by 10 s for frequency. (3) Use total time divided by cycle count for period, or take the reciprocal of frequency. Check hertz and seconds respectively.

Pr 7 (p. 66) (1) Use the original frequency and wavelength to establish speed first. (2) Retain that speed because the medium remains nondispersive and unchanged. (3) Divide the established speed by 12 Hz for the new wavelength; do not reuse the old wavelength with the new frequency.

Pr 9 (p. 67) (1) The separation is one quarter of a full spatial cycle. (2) Therefore $\Delta\phi = 2\pi \times (1/4) = \pi/2$.

Pr 12 (p. 68) (1) The new distance is three times the original. Ask how a sphere area changes when radius triples. (2) Square the distance factor for the inverse-square intensity change. (3) Apply that factor to 18 $\mathrm{W/m^2}$ and retain intensity units; this is not a total-power calculation.

Pr 16 (p. 70) (1) Draw the fixed endpoints and the single-loop fundamental. (2) The 1 m string contains half a wavelength. Find the full wavelength before using the 40 m/s speed. (3) Divide speed by wavelength for the fundamental frequency, then use the harmonic relationship for the second harmonic.

Pr 20 (p. 71) (1) The supplied frequency ratio allows comparison without calculating Planck constant. (2) Photon energy follows the same ratio as frequency. (3) Vacuum wavelength follows the reciprocal frequency ratio. State which light has the larger energy and which has the shorter wavelength.

Pr 24 (p. 73) (1) Confirm that light travels from glass index 1.5 into air index 1.0. (2) Divide the lower index by the higher index to obtain the critical-angle sine. (3) Take inverse sine and report an angle from the normal. Check calculator degree mode if using degrees.

Pr 28 (p. 74) (1) Assign positive focal length to the converging lens and positive distance to the real object. (2) Subtract the reciprocal of 24 cm from the reciprocal of 12 cm, then invert for image distance. (3) Calculate signed magnification and interpret distance sign, orientation and size separately.

Pr 32 (p. 76) (1) Identify the objective focal length and eyepiece focal length. (2) Divide 100 cm by 5 cm for initial magnification magnitude. (3) Replace only the denominator with 10 cm and compare ratios. Keep image orientation separate from the requested magnitude.

Pr 36 (p. 77) (1) Distinguish slit separation from slit width; this is a double-slit spacing question. (2) Convert 500 nm and 0.50 mm to metres before combining them with 2 m. (3) Multiply wavelength by screen distance and divide by slit separation. Convert the final spacing to millimetres if helpful.

Pr 38 (p. 78) (1) Round-trip path change: $2 \times 250 = 500\,\mathrm{nm}$. (2) This equals one wavelength.

Pr 41 (p. 79) (1) Start with the unpolarized 80 $\mathrm{W/m^2}$ and apply the first-polarizer rule. (2) Use that transmitted intensity as the analyzer input. (3) Multiply by the squared cosine of 45 degrees. Keep the two stages visible rather than applying an extra half factor.

Pr 45 (p. 81) (1) Retain the negative focal-length sign because the lens diverges. (2) Convert -50 cm into metres. (3) Take the signed reciprocal and report dioptres. A positive result would conflict with the stated lens type.

Pr 49 (p. 82) (1) Choose the Brewster tangent relation, not the critical-angle sine relation. (2) Place glass index 1.5 over air index approximately 1.0. (3) Take inverse tangent of that ratio and measure the result from the normal. Check degree mode if reporting degrees.

Answers

Pr 2 (p. 65) No. Displacement and restoring acceleration are zero; speed is maximum.

Pr 3 (p. 65) Frequency = 50/10 = 5 Hz. Period = 10/50 = 0.20 s. Two marks each.

Pr 4 (p. 65) (a) Correct. Twice the time per cycle gives half as many cycles each second.

Pr 6 (p. 66) No. It is angular frequency; divide it by 2π to get 10 Hz.

Pr 7 (p. 66) Speed = $8 \times 0.75 = 6$ m/s. New wavelength = 6/12 = 0.50 m. Two marks each.

Pr 8 (p. 66) (b) Correct. At fixed frequency, lower speed means less distance travelled per cycle.

Pr 9 (p. 67) $\pi/2$ radians, or 90°. They are neither in phase nor exactly opposite in phase.

Pr 11 (p. 68) 2 W/m^2: divide by $3^2 = 9$.

Pr 12 (p. 68) Inverse-square spreading gives 18/9 = 2 W per square metre. The distance triples, so the area over which power spreads increases ninefold.

Pr 13 (p. 68) (c) Correct. The unchanged source power is shared over each complete sphere.

Pr 15 (p. 70) It is twice the fundamental frequency in the ideal uniform-string model.

Pr 16 (p. 70) Fundamental wavelength = 2 m; frequency = 40/2 = 20 Hz. Second harmonic = 40 Hz. Two marks per frequency with reasoning.

Pr 17 (p. 70) (a) Correct. One loop between neighbouring nodes occupies half a wavelength.

Pr 19 (p. 71) A has twice the photon energy and half the vacuum wavelength.

Pr 20 (p. 71) A has twice the photon energy and half the vacuum wavelength. Two marks for each relationship.

Pr 21 (p. 71) (b) Correct. Unchanged frequency gives unchanged photon energy.

Pr 23 (p. 73) No. Total internal reflection requires travel from higher to lower refractive index.

Pr 24 (p. 73) Sine of critical angle = $\frac{1}{1}.5 = 0.6667$, so critical angle = 41.8 degrees. Total internal reflection occurs above this angle. Credit condition, equation, value and interpretation.

Pr 25 (p. 73) (c) Correct. The transmitted angle from the normal is 90 degrees.

Pr 27 (p. 74) It is upright and twice the object height; the sign convention and lens setup determine its image position.

Pr 28 (p. 74) $1/\text{di} = 1/12 - 1/24 = 1/24$, so di = 24 cm. m = −1. Image is real, inverted and equal size. One mark each for distance, magnification, real and inverted/equal size.

Pr 29 (p. 74) (a) Correct. A screen intercepts the rays where they form the image.

Pr 31 (p. 76) It halves the magnification magnitude.

Pr 32 (p. 76) Initial magnitude = 100/5 = 20. New magnitude = 100/10 = 10. Doubling eyepiece focal length halves angular magnification.

Pr 33 (p. 76) (b) Correct. The eyepiece focal length is the denominator of the ratio.

Pr 35 (p. 77) It halves.

Pr 36 (p. 77) Convert wavelength to 0.0000005 m and separation to 0.0005 m. Spacing = $(0.0000005 \times 2)/0.0005 = 0.002$ m = 2 mm. Credit both conversions, calculation and result.

Pr 37 (p. 77) (c) Correct. Wavelength and slit separation set directions; screen distance converts direction to position.

Pr 38 (p. 78) The path changes by 500 nm, so one fringe passes a fixed observation point.

Pr 40 (p. 79) Zero intensity in the ideal model, because $\cos^2 90° = 0$.

Pr 41 (p. 79) First polarizer gives 40. Analyzer gives 40 times cosine squared 45 degrees = 20 W per square metre. Two marks for each stage.

Pr 42 (p. 79) (a) Correct. The input is already aligned, so the squared cosine of zero is one.

Pr 44 (p. 81) -2 D, since -50 cm $= -0.50$ m.

Pr 45 (p. 81) Convert to -0.50 m. Power=$1/(-0.50)$=-2 dioptres. Credit conversion, formula, sign and unit.

Pr 46 (p. 81) (b) Correct. The smaller focal-length magnitude has the larger reciprocal magnitude.

Pr 48 (p. 82) No. The two orthogonal components must also have equal amplitudes.

Pr 49 (p. 82) Tangent of Brewster angle=1.5/1.0. The angle is arctan(1.5)=56.3 degrees. Credit correct ratio, inverse tangent, result and interpretation as incidence angle.

Pr 50 (p. 82) (c) Correct. The reflected component lacks one linear polarization component.

Optics and Waves, Test A

30 marks. Suggested time 30 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. Which wave can propagate through vacuum?

   (a) A water wave
   (b) Light
   (c) A stretched-string wave
   (d) Sound in air

2. With wave speed fixed, doubling frequency makes wavelength:

   (a) Half as large
   (b) Unchanged
   (c) Four times as large
   (d) Twice as large

3. Group velocity describes the motion of:

   (a) The wave-packet envelope
   (b) The source itself
   (c) One particle only
   (d) A stationary node

4. Two tones at 440 and 443 Hz produce a beat frequency of:

   (a) 1320 Hz
   (b) 883 Hz
   (c) 3 Hz
   (d) 441.5 Hz

5. At the same medium, which has greater photon energy?

   (a) Lower-frequency light
   (b) Both always equal
   (c) Higher-frequency light
   (d) Longer-wavelength light

6. Light enters glass from air. Which quantity remains unchanged?

   (a) Ray direction in every case
   (b) Speed
   (c) Wavelength
   (d) Frequency

Part II. Show your reasoning.

1. A source makes 50 complete oscillations in 10 s. Find frequency and period.

2. A wave has frequency 8 Hz and wavelength 0.75 m. Find speed. If frequency becomes 12 Hz in the same nondispersive medium, find wavelength.

3. At distance r an isotropic source has intensity 18 W per square metre. Find intensity at 3r, ignoring absorption.

4. A 1 m string fixed at both ends supports waves at 40 m/s. Find its fundamental frequency and second harmonic.

5. Light A has twice the frequency of light B. Compare photon energy and vacuum wavelength.

6. Find the critical angle for glass of index 1.5 into air of index 1.0.

Optics and Waves, Test B

30 marks. Suggested time 30 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. An image with magnification -2 is:

   (a) Upright and smaller
   (b) Inverted and twice as tall
   (c) Virtual in every case
   (d) Upright and twice as tall

2. In a simple astronomical telescope, the objective usually has:

   (a) A longer focal length than the eyepiece
   (b) Zero focal length
   (c) The same role as a polarizer
   (d) No effect on the image

3. If double-slit separation increases while wavelength and screen distance stay fixed, fringe spacing:

   (a) Becomes negative
   (b) Increases
   (c) Stays fixed
   (d) Decreases

4. Two ideal crossed polarizers transmit from an initially unpolarized beam:

   (a) One quarter
   (b) Zero
   (c) All its intensity
   (d) Half

5. The unit of optical lens power is:

   (a) Hertz
   (b) Watt
   (c) Dioptre
   (d) Newton

6. A quarter-wave plate acts primarily by changing the relative:

   (a) Charge of light
   (b) Speed of the source
   (c) Mass of photons
   (d) Phase of orthogonal components

Part II. Show your reasoning.

1. A converging lens has focal length 12 cm and an object at 24 cm. Find image distance, magnification and image type.

2. A telescope objective has focal length 100 cm and eyepiece 5 cm. Find angular magnification magnitude. What happens if the eyepiece is replaced by 10 cm?

3. In a double-slit experiment, wavelength is 500 nm, screen distance 2 m and slit separation 0.50 mm. Find fringe spacing.

4. Unpolarized intensity 80 W per square metre passes through a polarizer and an analyzer at 45 degrees. Find the final intensity.

5. A diverging lens has focal length -50 cm. Find its optical power.

6. For light from air into glass of index 1.5, estimate Brewster angle.

Optics and Waves, Test A: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (b) Light is electromagnetic and does not require a material medium.

Written part. Frequency = 50/10 = 5 Hz. Period = 10/50 = 0.20 s. Two marks each. (Section: p. 64.)

Question 2. (a) Wavelength equals speed divided by frequency.

Written part. Speed = 8 × 0.75 = 6 m/s. New wavelength = 6/12 = 0.50 m. Two marks each. (Section: p. 65.)

Question 3. (a) The group is the envelope formed by combining nearby frequencies.

Written part. Inverse-square spreading gives 18/9 = 2 W per square metre. The distance triples, so the area over which power spreads increases ninefold. (Section: p. 67.)

Question 4. (c) Beat frequency is the absolute difference.

Written part. Fundamental wavelength = 2 m; frequency = 40/2 = 20 Hz. Second harmonic = 40 Hz. Two marks per frequency with reasoning. (Section: p. 69.)

Question 5. (c) Photon energy is proportional to frequency.

Written part. A has twice the photon energy and half the vacuum wavelength. Two marks for each relationship. (Section: p. 70.)

Question 6. (d) The boundary does not change the frequency of the incident oscillation.

Written part. Sine of critical angle = $\frac{1}{1}.5 = 0.6667$, so critical angle = 41.8 degrees. Total internal reflection occurs above this angle. Credit condition, equation, value and interpretation. (Section: p. 71.)

Optics and Waves, Test B: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (b) The minus sign indicates inversion and magnitude 2 gives the size ratio.

Written part. $1/\mathrm{di} = 1/12 - 1/24 = 1/24$, so di = 24 cm. m = −1. Image is real, inverted and equal size. One mark each for distance, magnification, real and inverted/equal size. (Section: p. 73.)

Question 2. (a) The long-focus objective and short-focus eyepiece give a large angular-magnification ratio.

Written part. Initial magnitude = 100/5 = 20. New magnitude = 100/10 = 10. Doubling eyepiece focal length halves angular magnification. (Section: p. 75.)

Question 3. (d) Fringe spacing is inversely proportional to slit separation.

Written part. Convert wavelength to 0.0000005 m and separation to 0.0005 m. Spacing = $(0.0000005 \times 2)/0.0005 = 0.002$ m = 2 mm. Credit both conversions, calculation and result. (Section: p. 76.)

Question 4. (b) Their axes differ by 90 degrees and $\cos^2 90^\circ$ is zero.

Written part. First polarizer gives 40. Analyzer gives 40 times cosine squared 45 degrees = 20 W per square metre. Two marks for each stage. (Section: p. 78.)

Question 5. (c) Lens power is inverse focal length in metres, expressed in dioptres.

Written part. Convert to −0.50 m. Power=1/(−0.50)=−2 dioptres. Credit conversion, formula, sign and unit. (Section: p. 80.)

Question 6. (d) A controlled phase delay changes the polarization state.

Written part. Tangent of Brewster angle=1.5/1.0. The angle is arctan(1.5)=56.3 degrees. Credit correct ratio, inverse tangent, result and interpretation as incidence angle. (Section: p. 81.)

Facts at a glance

1. Wave language and simple harmonic motion p. 64

$$T = 1/f$$
$$f = N/t$$
$$a = -\omega^2 x$$

2. Read a travelling wave without deriving it p. 65

$$v = f\lambda$$
$$k = 2\pi/\lambda$$
$$\omega = 2\pi f$$
$$y = A\sin(kx - \omega t + \phi)$$

3. Wavefronts, spreading and group velocity p. 67

$$I \propto A^2$$
$$I_{\text{spherical}} \propto 1/r^2$$

4. Superposition, standing waves and beats p. 69

$$f_{\text{beat}} = |f_1 - f_2|$$
$$\lambda_n = 2L/n \text{ (string fixed at both ends)}$$

5. Light, photons and the electromagnetic picture p. 70

$$E_\gamma = hf = hc/\lambda_0$$
$$\lambda_0 = \text{vacuum wavelength}$$
$$c = 3.00 \times 10^8 \text{ m/s}$$

6. Reflection, refraction and total internal reflection p. 71

$$\theta_i = \theta_r$$
$$n = c/v$$
$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$
$$\sin\theta_c = n_2/n_1$$

7. Mirrors, lenses and image signs p. 73

$$1/f = 1/d_o + 1/d_i$$
$$m = -d_i/d_o$$

8. Optical systems, prisms and instruments p. 75

$$m_{\text{total}} = m_1 m_2$$
$$|M_{\text{telescope}}| = f_o/f_e \text{ (normal adjustment)}$$

9. Interference and diffraction p. 76

$$\Delta y = \lambda L / d$$
$$d \sin\theta = m\lambda \text{ (double-slit bright)}$$
$$a \sin\theta = m\lambda \text{ (single-slit dark, } m \neq 0\text{)}$$

10. Polarization and Malus law p. 78

$$I = I_0 \cos^2\theta$$
$$I_{\text{after first}} = I_{\text{unpolarized}}/2$$

11. Scattering, apparent depth and lens power p. 80

$$I_{\text{Rayleigh}} \propto 1/\lambda^4$$
$$d_{\text{apparent}} \approx d_{\text{real}}/n$$
$$P_{\text{lens}} = 1/f \text{ (f in metres)}$$

12. Recognize advanced optics without long derivations p. 81

$$\tan\theta_B = n_2/n_1$$
$$E = cB \text{ (plane wave in vacuum)}$$

Notation

Symbols are listed by section. The same letter may name different quantities in different sections; read the section's Fact.

1. Wave language and simple harmonic motion

N	number of complete cycles (dimensionless)
t	total elapsed time (s)
f	cycles per second (Hz)
T	time for one cycle (s)
ω	angular frequency (rad/s)

2. Read a travelling wave without deriving it

A	amplitude or maximum displacement (m)
k	wave number (rad/m)
ω	angular frequency (rad/s)
λ	wavelength (m)
f	frequency (Hz)

3. Wavefronts, spreading and group velocity

I	energy-transfer rate per unit area (W/m^2)
A	wave amplitude (depends on disturbance)
r	distance from point source (m)

4. Superposition, standing waves and beats

f_{beat}	number of loudness cycles per second (Hz)
L	vibrating string length (m)
n	harmonic number, 1, 2, 3, ... (dimensionless)
λ_n	wavelength of harmonic n (m)

5. Light, photons and the electromagnetic picture

E_γ	energy of one photon (J)
h	Planck constant (J s)
f	light frequency (Hz)
λ	vacuum wavelength (m)

6. Reflection, refraction and total internal reflection

n	refractive index (dimensionless)
c	vacuum light speed (m/s)
v	light speed in medium (m/s)
θ_c	critical incidence angle (degrees or radians)

7. Mirrors, lenses and image signs

f	signed focal length (cm or m consistently)
d_o	object distance (same length unit as f)
d_i	signed image distance (same length unit as f)
m	transverse magnification (dimensionless)

8. Optical systems, prisms and instruments

f_o	objective focal length (cm or m consistently)
f_e	eyepiece focal length (same length unit as fo)
M	angular magnification (dimensionless)
m_1, m_2	transverse magnifications of successive stages (dimensionless)

9. Interference and diffraction

Δy	double-slit fringe spacing (m)
L	slit-to-screen distance (m)
d	separation of double slits (m)
a	single-slit width (m)
λ	wavelength (m)

10. Polarization and Malus law

I_0	intensity incident on the analyzer (W/m^2 or stated units)
I	intensity transmitted by analyzer (same as I0)
θ	angle between incident polarization and analyzer axis (degrees or radians)

11. Scattering, apparent depth and lens power

d_{real}	actual depth (m)
d_{apparent}	apparent depth (m)
n	liquid refractive index relative to air approximation (dimensionless)
P	lens optical power ($D = m^{-1}$)
f	signed focal length (m)

12. Recognize advanced optics without long derivations

θ_B	Brewster incidence angle from the normal (degrees or radians)
n_1, n_2	incident and transmitted medium indices (dimensionless)
E	plane-wave electric-field magnitude (V/m)
B	plane-wave magnetic-field magnitude (T)

Sources

The introduction and four current sections: wave motion; electromagnetic theory, photons and light; geometrical optics; physical optics. Derivations are replaced by plain-language interpretation and short exam calculations.

The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. Check lecturer-specific derivations and conventions. The practice tests are original study material, not predictions of an examination.

Calculus I

MATH 203: derivative rules, implicit/logarithmic/exponential derivatives, L Hopital, graphing and extrema, optimization, Rolle and mean-value theorems, integration and the fundamental theorem. The full lecture syllabus is the main source.

1 Limits, continuity and the derivative definition

Aims. Distinguish a limit from a value at the point; Find a polynomial derivative from the difference quotient.
Before this section: Expanding a square; factoring and cancelling a nonzero common factor.

A limit describes nearby values, not necessarily the function value at the point. Continuity requires the function value and the two-sided limit to exist and agree. The derivative is the limiting slope of secant lines and gives instantaneous rate of change.

Idea 1. A secant line measures change across separated inputs. As those inputs approach one another, its slope can approach a single instantaneous slope: the derivative. The increment must remain nonzero while it is a denominator, even though the limiting process eventually removes it.

Fact 1.

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

where h is the small nonzero input increment before the limit (same unit as x); $f'(x)$ is the instantaneous rate of change (output unit per input unit).

Why it holds. Factoring and cancelling the increment can remove the apparent zero denominator. The nearby simplified slopes determine the limit, rather than an undefined quotient at zero increment.

Method 1.

1. Write f(x+h) and subtract the complete original f(x).
2. Expand brackets carefully and collect terms.
3. Factor out h and cancel only while h is nonzero.
4. Take the limit h approaching zero in the simplified expression.

Recognising it. If the question says use the definition, construct the difference quotient instead of citing the power rule.

Pr 1. For $f(x) = x^2$, the difference quotient becomes $\frac{2xh+h^2}{h} = 2x + h$.

Solution. For $f(x) = x^2$, the incremented value is $(x+h)^2 = x^2 + 2xh + h^2$. Subtracting f(x) leaves $2xh + h^2$. Dividing by nonzero h gives $2x + h$. As h approaches zero, the secant slope approaches 2x, so $f'(x) = 2x$.

Common error. Zero over zero is an indeterminate form, not a final answer of zero.

In the examination. Place brackets around f(x+h) and f(x) before subtraction, especially when either contains negative terms.

Pr 2. Why should h not be set to zero before simplifying the difference quotient?

Pr 3. Use the definition to find the derivative of f(x)=3x squared.

Pr 4. Why can the increment be cancelled before taking the derivative limit?

(a) It is nonzero while approaching zero

(b) It equals zero at every stage

(c) Any denominator may always be cancelled

2 Basic derivatives and tangent lines

Aims. Differentiate sums and powers correctly; Build a tangent line from the original point and derivative slope.
Before this section: Power rule; substituting an x value; point-slope form of a line.

Differentiate a sum term by term. Constants disappear. The derivative gives the slope at a point, but the tangent line must also pass through the point on the original function. Rewrite roots and reciprocal powers before applying the power rule.

Idea 2. A derivative supplies a slope, not a whole line. Many parallel lines share that slope, so a tangent must also be anchored at its point on the original curve. Calculate the curve height from the original function and the slope from its derivative.

Fact 2.

$$\frac{d}{dx}x^n = nx^{n-1}$$
$$y - f(a) = f'(a)(x - a)$$

where a is the input coordinate of tangent point (input unit); $f(a)$ is the output coordinate of tangent point (output unit); $f'(a)$ is the tangent slope (output unit per input unit).

Why it holds. The original function gives the tangent point. The derivative gives the local slope, and a point plus a finite slope determines the line.

Method 2.

1. Differentiate the original function term by term.
2. Substitute x = a into the derivative for the slope.
3. Substitute x = a into the original function for the point.
4. Use y - f(a) = f′(a)(x-a), then simplify only if requested.

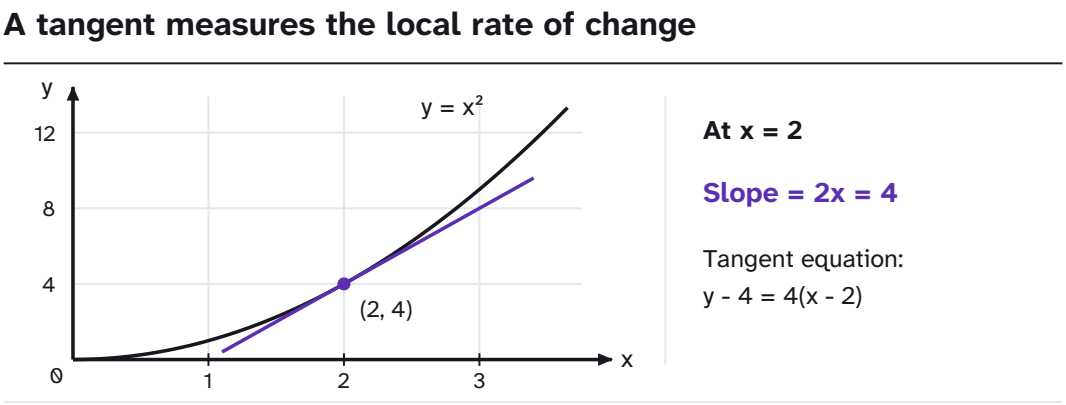


Figure 4. Figure 04. The tangent to $y = x^2$ at $(2, 4)$ has slope 4 and equation $y - 4 = 4(x - 2)$. Its plotted line touches the curve at the marked point. Read slope from coordinates, not the visual angle.

Recognising it. A tangent-equation request needs both a point and a slope, even when differentiation itself is simple.

Pr 5. For $f(x) = x^2 + 1$ at $x = 2$, the point is $(2,5)$ and slope is 4.

Solution. For $f(x) = x^2 + 1$, $f'(x) = 2x$ because the constant contributes zero. At $x = 2$ the slope is 4, while the original function gives $y = 5$. The tangent must therefore pass through $(2,5)$ with slope 4. Its equation is $y - 5 = 4(x-2)$, which expands to $y = 4x - 3$.

Common error. Do not use $f'(a)$ as the y-coordinate.

In the examination. Keep the unsimplified point-slope equation visible: it shows both required ingredients for method marks.

Pr 6. Does substituting $x = 2$ into f' give the y coordinate of the tangent point?

Pr 7. Find the tangent to $f(x)=x$ cubed at $x = 2$.

Pr 8. Two points have the same finite derivative value. Their tangent lines must be:

(a) Exactly the same line in every case

(b) Parallel or coincident

(c) Perpendicular

3 Product and quotient rules

Aims. Differentiate a product using both changing factors; Preserve term order and the squared denominator in the quotient rule.
Before this section: Basic polynomial and sine derivatives; multiplication and fractions.

A product has two changing factors, so both contribute to the derivative. A quotient rule numerator must preserve its order; reversing it changes the sign. Do not differentiate numerator and denominator separately to differentiate a fraction.

Idea 3. When two factors change, a product changes through both of them. A quotient also has two changing parts, but its denominator contribution enters with a subtraction. Writing the full rule before simplifying preserves the separate contributions and protects the sign.

Fact 3.

$$(uv)' = u'v + uv'$$
$$(u/v)' = (u'v - uv')/v^2$$

where u, v is the the two factor functions (problem dependent); u', v' is the their derivatives with respect to x (corresponding output per input unit).

Why it holds. The product rule includes one contribution from each changing factor. Reversing the quotient-rule subtraction reverses the final derivative sign.

Method 3.

1. Identify whether the expression is a product or quotient at the outermost level.
2. Name the component functions u and v and calculate their derivatives.
3. Insert all terms into the appropriate rule before simplifying.
4. For a quotient retain the original domain restriction and check the numerator subtraction order.

Recognising it. Inspect the outer operation: multiplication or division of functions signals the corresponding rule.

Pr 9. For $y = x^2 \sin x$, derivative is $2x \sin x + x^2 \cos x$.

Solution. For y = x^2 sin x, choose u = x^2 and v = sin x. Their derivatives are u′ = 2x and v′ = cos x. The product rule keeps each factor once unchanged: u′v + uv′. Thus y′ = 2x sin x + x^2 cos x; multiplying the two derivatives alone would miss both contributions.

Common error. The quotient denominator is the square of the original denominator.

In the examination. Write the entire quotient-rule numerator in brackets before expanding; the minus sign applies to the whole second product.

Pr 10. For $y = (x+1)/(x-1)$, what is the derivative for $x \neq 1$?

Pr 11. Differentiate $y=(x+1)/(x-1)$, for x not equal to 1.

Pr 12. Why is separately differentiating top and bottom generally wrong for the derivative of u/v?

(a) The denominator must always be constant

(b) Fractions cannot be differentiated

(c) It omits the complete effect of denominator change

4 Chain rule and trigonometric functions

Aims. Identify inner and outer operations in a composite function; Include the inner derivative in trigonometric and power derivatives.
Before this section: Power rule; sine and cosine derivatives; radians.

A composite function has an outer operation acting on an inner expression. Differentiate the outside while preserving the inside, then multiply by the derivative of the inside. Trigonometric derivative formulas use radians.

Idea 4. A composite function is a machine inside a machine. The outer function responds to its inner input, while that inner input changes with the original variable. The chain rule multiplies these rates to describe the complete change.

Fact 4.

$$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$$
$$(\sin x)' = \cos x$$
$$(\cos x)' = -\sin x$$

where $u = g(x)$ is the inner expression (problem dependent); $g'(x)$ is the inner derivative (inner-expression unit per input unit).

Why it holds. An outer derivative alone measures change per unit of inner expression. The inner derivative converts that rate into change per unit of the original variable.

Method 4.

1. Identify the last operation applied to the expression: this is the outer function.
2. Differentiate that outer function while leaving the inner expression in place.
3. Multiply by the derivative of the inner expression.
4. Check the sign for cosine derivatives and use radians for trigonometric rules.

Recognising it. An operation surrounding a nontrivial expression, such as cosine surrounding a square, is a chain-rule cue.

Pr 13. For $y = (3x + 1)^4$, derivative $= 4(3x + 1)^3 \cdot 3 = 12(3x + 1)^3$.

Solution. In y = $(3x+1)^4$, the inner expression is 3x+1 and the outer operation is raising to the fourth power. The outer derivative gives $4(3x+1)^3$. The inner derivative is 3, so it must multiply the outer result. Combining the constants gives $y' = 12(3x+1)^3$.

Common error. Forgetting the inner derivative is the most common lost mark.

In the examination. Before finishing, circle the inner expression and verify that its derivative appears as a factor.

Pr 14. What is the derivative of $\cos(x^2)$?

Pr 15. Differentiate $y=\cos(x^2)$.

Pr 16. In differentiating $\sin(7x)$, what does the extra factor 7 represent?

(a) The rate of change of the inner expression

(b) A second sine factor

(c) A degrees-to-radians conversion

Fact 5. Four more trigonometric derivatives. The supplied final asks for a cotangent derivative. Learn these rules together with sine, cosine and the chain rule.

$$\frac{d}{dx}\tan u = \sec^2 u\, u'$$

$$\frac{d}{dx}\cot u = -\csc^2 u\, u'$$

$$\frac{d}{dx}\sec u = \sec u \tan u\, u'$$

$$\frac{d}{dx}\csc u = -\csc u \cot u\, u'$$

1. Use radians. Here u is a function of x, and u' means its derivative.
2. Apply the outside rule, then multiply by the inside derivative.
3. Use a rule only where the original function is defined. Remember: $\sec u = 1/\cos u$, $\csc u = 1/\sin u$, and $\cot u = \cos u/\sin u$.

Pr 17. Differentiate $y = \cot(5x)$.

5 Implicit differentiation

Aims. Differentiate y terms while treating y as a function of x; Collect and isolate all terms containing y'.
Before this section: Chain and product rules; algebraic rearrangement.

When x and y appear in one equation, y is treated as a function of x. Differentiating a y term therefore introduces y'. Collect all y-prime terms and solve algebraically. This can describe curves that are not single-valued functions globally.

Idea 5. An implicit equation links two changing coordinates without isolating one globally. As one coordinate changes, the other adjusts to stay on the curve. Differentiation records this linked movement, so differentiating a function of the dependent coordinate requires its own rate of change.

Fact 6.

$$\frac{d}{dx}(y^2) = 2yy'$$
$$\frac{d}{dx}(xy) = y + xy'$$

where y' is the dy/dx, the local slope of the curve (y unit per x unit).

Why it holds. A y-expression is a composite function of x, so chain-rule factors appear. Collecting derivative terms reveals a shared factor that can be isolated to find the slope.

Method 5.

1. Differentiate both sides with respect to x.
2. Attach y' when differentiating a function of y, and use the product rule for xy.
3. Move all terms containing y' to one side.
4. Factor out y', divide by its coefficient and state where that division is valid.

Recognising it. Use implicit differentiation when x and y are mixed and explicit rearrangement would be awkward or lose branches.

Pr 18. From $x^2 + y^2 = 25$, differentiation gives $2x + 2yy' = 0$.

Solution. The curve $x^2 + y^2 = 25$ need not be solved globally for a single y value. Differentiating x^2 gives 2x; differentiating y^2 gives 2y y' by the chain rule. The constant 25 differentiates to zero, so $2x + 2y\,y' = 0$. Rearranging gives $y' = -x/y$ wherever $y \neq 0$; division by y requires this restriction.

Common error. Do not treat y as a constant or omit the product rule on xy.

In the examination. For an xy term write $y + xy'$, not simply y or xy'; both factors depend on x.

Pr 19. Why does the derivative of y^2 contain y'?

Pr 20. Find dy/dx for xy+y squared=6.

Pr 21. When y depends on x, the derivative of xy is:

(a) xy'

(b) $y + xy'$

(c) y

6 Logarithmic, exponential and inverse-trig derivatives

Aims. Differentiate exponential and logarithmic composites; Use logarithmic differentiation for a positive variable base and exponent.

Before this section: Chain rule; natural logarithms; product rule; positive-domain restrictions.

Exponential functions reproduce themselves up to an inner derivative. Natural logarithms give a reciprocal factor. Logarithmic differentiation helps when variable powers or products are awkward. Inverse-trigonometric functions are different from reciprocal trigonometric functions.

Idea 6. A variable power changes through both its base and its exponent. A logarithm moves the exponent into a product, where familiar differentiation rules track those two changes. The final step restores the original function after differentiating the logarithmic equation.

Fact 7.

$$(e^u)' = e^u u'$$
$$(\ln|u|)' = u'/u$$
$$(\arctan x)' = 1/(1+x^2)$$

where e is the base of the natural exponential (dimensionless); $\ln u$ is the natural logarithm of a positive dimensionless argument (dimensionless).

Why it holds. Logarithms turn a positive power expression into a product involving its exponent. The derivative of the logarithm includes the original derivative divided by the original function, so that function must be restored.

Method 6.

1. Identify the outer function and check its real domain.
2. For a logarithm use inner derivative divided by the inner expression.
3. For an exponential retain the exponential and multiply by the inner derivative.
4. For y = x^x with x > 0, take logs first, differentiate implicitly, then restore y.

Recognising it. A variable in both base and exponent suggests logarithmic differentiation, after checking a suitable positive domain.

Pr 22. For $y = \ln(x^2 + 1)$, derivative is $\frac{2x}{x^2+1}$.

Solution. For ln(x^2+1), the inner derivative is 2x and the inner expression is always positive. The logarithmic chain rule therefore gives 2x/(x^2+1). For the separate function e^(3x), the exponential is retained unchanged. Multiplication by the inner derivative 3 gives 3e^(3x).

Common error. The derivative of ln u is u prime/u, not 1/ln u.

In the examination. The power rule nx^(n−1) assumes a constant exponent; x^x needs another method.

Pr 23. For x > 0, what is d(x^x)/dx?

Pr 24. Differentiate y=x to the power x for x greater than zero.

Pr 25. Why is the constant-exponent power rule insufficient for x^x?

(a) The expression is constant

(b) Logarithms are invalid for positive x

(c) The exponent changes as well as the base

Fact 8. Other exponential, logarithm and inverse-trig rules. The supplied final includes 3^x and $\log_3 x$. Their derivatives need a factor involving $\ln 3$.

$$\frac{d}{dx}a^u = a^u \ln(a)\, u'$$

$$\frac{d}{dx}\log_a u = \frac{u'}{u \ln a}$$

$$\frac{d}{dx}\arcsin u = \frac{u'}{\sqrt{1-u^2}}$$

$$\frac{d}{dx}\arccos u = -\frac{u'}{\sqrt{1-u^2}}$$

1. The base a is a fixed positive number, with $a \neq 1$. For the logarithm, require $u > 0$.
2. For the displayed arcsine and arccosine derivatives, require $|u| < 1$. Angles are in radians.
3. Find the inside derivative u' first. Inverse sine means arcsin; it is different from 1/ sin.

Pr 26. Differentiate 3^x and $\log_3 x$.

7 L Hopital and indeterminate forms

Aims. Check for a permitted indeterminate quotient before using L Hopital; Evaluate a simple limit by differentiating numerator and denominator.
Before this section: Limits; exponential and trigonometric derivatives; chain rule.

L Hopital rule can evaluate a quotient limit with the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$ under its differentiability and limit conditions. Differentiate numerator and denominator separately for this limit procedure; this is not the quotient rule. Other indeterminate forms must first be rewritten.

Idea 7. An indeterminate quotient describes competition between quantities approaching zero or growing without bound. When its conditions hold, L Hopital compares their rates of change. It is a limit procedure, so it does not replace the quotient rule for the derivative of a fraction.

Fact 9.

$$\lim \frac{f(x)}{g(x)} = \lim \frac{f'(x)}{g'(x)}$$

when the rule applies

where $0/0$ is the indeterminate limiting form, not an arithmetic value (not a measured quantity); ∞/∞ is the indeterminate quotient form (not a measured quantity).
Conditions. The numerator and denominator satisfy the differentiability and denominator-derivative conditions near the limiting point.

Why it holds. The initial limiting form must be checked before applying the rule. Differentiating a numerator containing an inner expression still requires the chain rule.

Method 7.

1. Substitute the limiting input to identify the quotient form.
2. Confirm $0/0$ or infinity/infinity and the rule conditions before differentiating.
3. Differentiate numerator and denominator separately, not with the quotient rule.
4. Evaluate the new limit; if still indeterminate, check conditions again before repeating.

Recognising it. Check the quotient form first; an ordinary fraction alone is not a reason to use L Hopital.

Pr 27. As x approaches zero, $\frac{e^x-1}{x}$ has form $\frac{0}{0}$.

Solution. As x approaches zero, e^x - 1 approaches zero and x approaches zero. The quotient therefore has the required $0/0$ form. Differentiate numerator and denominator separately to obtain e^x/1. Its limit is e^0 = 1, so the original limit is 1.

Common error. Do not use the rule on a quotient whose substitution gives an ordinary finite value.

In the examination. Show the initial indeterminate form; applying the rule without checking its conditions can invalidate the method.

Pr 28. What is the limit of $\sin(3x)/x$ as x approaches zero?

Pr 29. Evaluate the limit of $\sin(3x)/x$ as x approaches zero.

Pr 30. A numerator approaches 2 and its denominator approaches 4. The appropriate first method is:

(a) Use the ordinary quotient limit

(b) Apply L Hopital because it is a fraction

(c) Declare the form indeterminate

8 Increasing, decreasing and concavity

Aims. Use a first-derivative sign chart to find monotonic intervals; Distinguish a stationary point from a proven extremum or inflection.
Before this section: Factoring quadratics; sign tests; first and second derivatives.

The sign of the first derivative controls increasing and decreasing intervals. The sign of the second controls concavity. A critical number lies in the domain where the first derivative is zero or undefined. An inflection point requires a concavity change, not merely a zero second derivative.

Idea 8. The derivative records the local slope of the graph. A negative derivative makes the graph fall as input increases; a positive derivative makes it rise. A change from falling to rising identifies a minimum, whereas a zero derivative without a sign change may not be an extremum.

Fact 10.

$$f' > 0 : \text{ increasing;}$$
$$f'' > 0 : \text{ concave up}$$

where f' is the first derivative, controlling increase or decrease (output per input unit); f'' is the second derivative, controlling concavity (output per input unit squared).

Why it holds. A zero derivative gives a candidate rather than a complete extremum proof. Signs on both sides tell whether increasing or decreasing behaviour actually changes.

Method 8.

1. Find where f′ is zero or undefined within the original domain.
2. Split the domain at those critical numbers and test the sign on each interval.
3. Use sign changes to classify local maxima and minima.
4. Use f′′ signs separately for concavity and require a concavity change for an inflection point.

Recognising it. Use a derivative sign chart for monotonic intervals and local extrema; use a separate concavity test for inflections.

Pr 31. For $f = x^3 - 3x$, derivative is $3(x-1)(x+1)$.

Solution. For f(x) = x³ - 3x, f′ = 3x² - 3 = 3(x−1)(x+1). The critical numbers are −1 and 1. Testing x = −2, 0 and 2 gives derivative signs positive, negative and positive. The function increases on (-∞,−1) and (1,∞), and decreases on (−1,1); the sign changes identify a local maximum at −1 and minimum at 1.

Common error. A derivative equal to zero does not automatically mean a maximum.

In the examination. Give open intervals for increasing and decreasing behaviour, then state extrema as points if asked.

Pr 32. Is f′′(c) = 0 alone enough to prove an inflection point?

Pr 33. For $f(x) = x^2 - 4x$, find increasing and decreasing intervals and the minimum point.

Pr 34. A first derivative changes from negative to positive at an interior critical point. This gives:

(a) An inflection point guaranteed

(b) A local minimum

(c) A local maximum

9 Graphing, asymptotes and absolute extrema

Aims. Find absolute extrema by comparing all candidates on a closed interval; Distinguish a removable hole from a vertical asymptote.
Before this section: Domain restrictions; critical numbers; evaluating a function at endpoints.

Graphing combines domain, intercepts, derivative signs and limiting behaviour. A cancelled factor can create a hole rather than an asymptote. On a closed interval a continuous function attains absolute extrema; compare endpoint and interior critical values.

Idea 9. An absolute extremum compares the entire allowed graph. On a closed interval, an endpoint can be highest or lowest without having a horizontal tangent. Comparing a short list of original function values turns this global problem into a finite check.

Fact 11.

Candidates = endpoints plus interior critical numbers

where $[a, b]$ is the closed interval including both endpoints (input unit); $f(c)$ is the function value at a candidate point (output unit).

Why it holds. Interior critical numbers supply possible internal extrema. Endpoints must also be considered because the allowed domain ends there.

Method 9.

1. Check the domain and whether the function is continuous on the stated closed interval.
2. Find all interior critical numbers.
3. Evaluate the original function at those numbers and at both endpoints.
4. Compare the values and report each extremum value together with where it occurs.

Recognising it. Absolute maximum or minimum on a closed interval calls for endpoint-plus-critical-value comparison.

Pr 35. For x^2 on $[-1, 2]$, candidates are -1,0,2.

Solution. For f(x) = x^2 on [−1,2], the derivative 2x is zero at x = 0. The candidate inputs are the two endpoints −1 and 2 and the interior critical number 0. The original function values are f(−1)=1, f(0)=0 and f(2)=4. The smallest is 0 at x=0; the largest is 4 at x=2.

Common error. Do not omit endpoints or compare derivative values instead of function values.

In the examination. Use the original function for the comparison table; comparing derivative values does not locate the highest graph point.

Pr 36. Can an endpoint be the absolute maximum even when its derivative is not zero?

Pr 37. Find absolute maximum and minimum of $f(x) = x^2$−2x on [0,3].

Pr 38. A continuous function is strictly increasing on a closed interval. Its absolute maximum occurs:

(a) Only where its derivative is zero

(b) Nowhere because it keeps increasing

(c) At the right endpoint

10 Optimization and mean-value theorems

Aims. Translate a simple optimization constraint into one variable; Check mean-value-theorem assumptions and solve for the interior input.

Before this section: Area and perimeter; derivatives; continuity and differentiability.

Optimization translates a word problem into a one-variable function and an allowed domain. Rolle theorem requires equal endpoint values as well as continuity and interior differentiability. The mean-value theorem matches some instantaneous slope to the average slope.

Idea 10. The mean-value theorem links average change across an interval with instantaneous change somewhere inside. A secant joins endpoint heights, and at least one interior tangent must be parallel under the required conditions. The theorem matches slopes rather than automatically selecting a midpoint.

FACT 12.

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$
$$a < c < b$$

where x is the one rectangle side length (length unit); A is the rectangle area (square length unit); c is the interior input guaranteed by the mean-value theorem (input unit).

WHY IT HOLDS. Continuity and interior differentiability exclude breaks and corners that could defeat the guarantee. The matching input lies strictly inside the interval but need not be its midpoint.

METHOD 10.

1. For optimization name the objective, write the constraint and identify the feasible domain.
2. Eliminate one variable, differentiate and find interior critical candidates.
3. Check endpoints or derivative signs to justify the optimum.
4. For a theorem question verify its assumptions before setting instantaneous slope equal to average slope.

RECOGNISING IT. Use the theorem when a function and interval are supplied and an interior slope-matching input is requested.

Pr 39. A rectangle with perimeter 20 has area $x(10 - x)$.

Solution. Perimeter 20 gives 2x + 2y = 20, so the other side is y = 10 - x. Area becomes A(x) = x(10-x) with $0 < x < 10$ for a nondegenerate rectangle. Its derivative is $10 - 2x$, which vanishes at x = 5. The second derivative is -2, confirming a maximum; both sides are 5 and area is 25 square units.

Common error. A candidate outside the physical domain is not a valid optimization answer.

In the examination. A critical number alone is not a maximum proof; include a sign change, second derivative or candidate comparison.

Pr 40. For $f(x)=x^2$ on [1,3], what c satisfies the mean-value theorem?

Pr 41. Use the mean-value theorem for $f(x) = x^2$ on [1,3] to find c.

Pr 42. Must the input guaranteed by the mean-value theorem be the midpoint?

(a) No; it solves the slope-matching equation

(b) Yes; every function uses the midpoint

(c) It must be an endpoint

11 Antiderivatives and substitution

Aims. Choose a substitution whose differential matches the integrand; Return the antiderivative to x and include the integration constant.

Before this section: Chain rule; power integration; constant factors.

An indefinite integral represents a family of antiderivatives, so include an arbitrary constant. Substitution reverses the chain rule: select an inner expression whose derivative is present. Differentiate your answer as a fast correctness check.

Idea 11. Substitution recognizes the chain rule running backward. A temporary variable replaces an inner expression, while its differential accounts for a matching factor elsewhere. A leftover constant is harmless if tracked carefully rather than dropped.

Fact 13.

$$\int x^n\,dx = x^{n+1}/(n+1) + C \ (n \neq -1)$$

$$\int dx/x = \ln|x| + C$$

where u is the chosen inner expression (problem dependent); du is the differential of the inner expression (same unit as u); C is the arbitrary additive integration constant (antiderivative unit).

Why it holds. A useful substitution replaces the nested expression and its differential factor together. Differentiating the final result checks both the power and its multiplier.

Method 11.

1. Choose an inner expression whose derivative is also present as a factor.
2. Write u and du and replace every part of the integral consistently.
3. Integrate in u, preserving any numerical factors.
4. Substitute back to x, add C and differentiate once to check.

Recognising it. Look for an inner expression accompanied by its derivative or a constant multiple of that derivative.

Pr 43. For the integral of $2x(x^2+1)^3$, set $u = x^2+1$ and $du = 2x\,dx$.

Solution. For the integral of $2x(x^2+1)^3$, choose $u = x^2+1$. Then $du = 2x\,dx$, so the complete integral becomes the integral of $u^3\,du$. The antiderivative is $u^4/4 + C$. Substitute back to get $(x^2+1)^4/4 + C$; differentiation restores $2x(x^2+1)^3$.

Common error. The power rule exception is $n=-1$; do not divide by zero.

In the examination. Do not leave a mixture of u and x in the substituted integral; it hides missing differential factors.

Pr 44. What is the integral of $6x(x^2+2)^2$ dx?

Pr 45. Evaluate the integral of $6x(x^2+2)$ squared with respect to x.

Pr 46. Which substitution most directly simplifies $6x(x^2+2)^2$?

(a) Rename the whole integrand without computing a differential

(b) $u = x^2 + 2$

(c) $u = 6x$

Fact 14. Integrate sine, cosine and an exponential. These short rules reverse the chain rule. A sine integral appears in the supplied final.

$$\int \sin(ax+b)\,dx = -\frac{\cos(ax+b)}{a} + C$$

$$\int \cos(ax+b)\,dx = \frac{\sin(ax+b)}{a} + C$$

$$\int e^{ax+b}\,dx = \frac{e^{ax+b}}{a} + C$$

1. Here a and b are constants and $a \neq 0$. Trigonometric angles use radians.
2. Divide by the coefficient a; differentiation would multiply by it.
3. Add C for an indefinite integral. Differentiate your answer to check it.

Pr 47. Find $\int \sin(3x)\,dx$.

12 Definite integrals, signed area and the fundamental theorem

Aims. Distinguish signed integral from total geometrical area; Apply the fundamental theorem, including a variable-upper-limit chain factor.
Before this section: Antiderivatives; positive and negative regions; substituting bounds.

A definite integral gives signed accumulation: contributions below the axis are negative. Geometrical area requires positive contributions from each region. The fundamental theorem connects integration and differentiation; for a variable upper limit, include its derivative.

Idea 12. A definite integral combines signed contributions over fixed bounds. An antiderivative provides a shortcut through its change between endpoints. If the integrand is nonnegative throughout, the signed accumulation is also geometric area; otherwise positive and negative regions can cancel.

Fact 15.

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

$$\frac{d}{dx}\int_a^{g(x)} f(t)\,dt = f(g(x))g'(x)$$

where a, b is the lower and upper integration bounds (input unit); F is the an antiderivative of f (f unit times input unit); t is the dummy integration variable (input unit).

Why it holds. The fundamental theorem requires upper antiderivative value minus lower value. Substitution changes the endpoint labels unless the antiderivative is returned to the original variable before evaluation.

Method 12.

1. Identify whether the question asks for signed accumulation or total area.
2. Find an antiderivative and calculate upper value minus lower value.
3. For total area split at sign changes and make each region contribution positive.
4. For differentiation with upper bound g(x), evaluate f at g(x) and multiply by g′(x).

Recognising it. With fixed limits, keep endpoint values consistent with the integration variable and evaluate their difference.

Pr 48. The $\int_{-1}^{1} x\,dx$ is zero by cancellation, but total geometrical area is 1.

Solution. For y = x from -1 to 1, the region below the axis contributes negative signed area. An antiderivative is $x^2/2$; evaluation at 1 and -1 gives $1/2 - 1/2 = 0$. Geometrically each half is a triangle with area 1/2. Total geometrical area is $1/2 + 1/2 = 1$, while the signed integral remains zero.

Common error. Never keep x-limits on an integral that has been fully changed to u.

In the examination. For a definite integral evaluate both bounds; no arbitrary +C is needed in the final numerical answer.

Pr 49. Does a zero definite integral prove the function is zero throughout the interval?

Pr 50. Evaluate the integral from 0 to 1 of $2x(x^2+1)$ dx.

Pr 51. After changing variable in a definite integral, what should happen to the bounds?

(a) Always keep the old bound numbers with the new variable

(b) Remove them permanently

(c) Transform them, or return to the original variable before evaluation

FACT 16. Read a Riemann sum as rectangles. The course syllabus includes sigma notation and area as a limit. The symbol $\sum$ simply means add the terms.

$$\Delta x = \frac{b-a}{n}$$

$$\text{rectangle sum} = \sum_{i=1}^{n} f(x_i^*)\,\Delta x$$

1. Split the interval from a to b into n equal widths.
2. Choose a sample point x_i^* in each small interval. Its function value is that rectangle's signed height.
3. Add height times width. More, narrower rectangles approach the definite integral when the function is continuous.
4. Below-axis heights contribute negative signed area.

Pr 52. Use two right-end rectangles to estimate $\int_0^1 x\,dx$.

Hints

Use one hint at a time, then return to the problem.

Pr 3 (p. 100) (1) Write $f(x+h) = 3(x+h)^2$, then subtract the complete original function. (2) Expand and factor h from the numerator of the difference quotient. (3) Cancel nonzero h, then let it approach zero. Keep the coefficient 3 throughout.

Pr 7 (p. 101) (1) Evaluate x^3 at $x = 2$ to find the point on the original curve. (2) Differentiate x^3 and evaluate the derivative at the same input. (3) Use the point and slope in $y - y_0 = m(x - x_0)$, then check that the point satisfies your line.

Pr 11 (p. 103) (1) Choose $u = x + 1$ and $v = x - 1$, then find both derivatives. (2) Form $(u'v - uv')/v^2$ with brackets around both numerator terms. (3) Simplify $(x-1) - (x+1)$ carefully and preserve $x \neq 1$.

Pr 15 (p. 104) (1) Identify cosine as the outer function and x^2 as the inner expression. (2) Differentiate cosine while leaving x^2 inside, and retain the negative sign. (3) Multiply by the derivative of x^2. Check that both rates appear.

Pr 17 (p. 104) (1) The inside is $u = 5x$, so $u' = 5$. (2) The cotangent rule gives $y' = -\csc^2(5x) \times 5$.

Pr 20 (p. 106) (1) Differentiate both sides of $xy + y^2 = 6$, using the product rule for xy. (2) Include y' when differentiating y^2, and collect all derivative terms. (3) Factor out y', move the remaining term and divide by its coefficient where nonzero.

Pr 24 (p. 107) (1) For $x > 0$, set $y = x^x$ and take natural logarithms. (2) Rewrite the right side as $x \ln x$; use the product rule while the left derivative is y'/y. (3) Multiply by y, then replace it with the original x^x.

Pr 26 (p. 107) (1) In both cases the inside expression is $u = x$, so $u' = 1$. (2) For the exponential, multiply by $\ln 3$. For the logarithm, divide by $\ln 3$.

Pr 29 (p. 109) (1) Check the numerator and denominator as $x \to 0$ and identify the form. (2) Differentiate $\sin(3x)$ with its inner derivative and differentiate the denominator separately. (3) Evaluate the resulting derivative quotient at the limiting input; do not use a quotient-rule numerator.

Pr 33 (p. 110) (1) Differentiate $x^2 - 4x$ and find where its derivative is zero. (2) Test derivative signs using one input on each side of that critical number. (3) Report the resulting intervals and evaluate the original function at the critical input to give the minimum point.

Pr 37 (p. 111) (1) Find the derivative of $x^2 - 2x$ and its critical input inside $[0, 3]$. (2) List that input and both endpoints as candidates. (3) Evaluate the original function at each candidate; report both the largest and smallest values and where each occurs.

Pr 41 (p. 112) (1) Check that x^2 is continuous on $[1, 3]$ and differentiable inside. (2) Compute the endpoint secant slope from function values and interval length. (3) Set $f'(c)$ equal to that slope, solve and verify $1 < c < 3$.

Pr 45 (p. 114) (1) Choose $u = x^2 + 2$, calculate du and compare it with $6x\,dx$. (2) Rewrite the full integral in u, retaining any constant multiplier. (3) Integrate the power, return to x, include the arbitrary constant and differentiate to check.

Pr 47 (p. 114) (1) The sine antiderivative is negative cosine. (2) The inside derivative is 3, so divide by 3. (3) Check: $\dfrac{d}{dx}[-\cos(3x)/3] = \sin(3x)$.

Pr 50 (p. 115) (1) Notice $x^2 + 1$ and its derivative factor $2x\,dx$. (2) Substitute with transformed bounds, or find an antiderivative and return it to x. (3) Evaluate upper minus lower between 0 and 1. Since the integrand is nonnegative there, a negative result would indicate an error.

Pr 52 (p. 116) (1) Each width is $\Delta x = (1-0)/2 = 1/2$. (2) The right endpoints are $1/2$ and 1, giving heights $1/2$ and 1. (3) The estimate is $(1/2)(1/2) + (1)(1/2) = 3/4$.

Answers

Pr 2 (p. 100) It would produce division by zero; the limit concerns nonzero h approaching zero.

Pr 3 (p. 100) [3(x+h) squared −3x squared]/h = [6xh+3h squared]/h = 6x+3h. Taking the limit gives 6x. One mark each for setup, expansion, cancellation and limit.

Pr 4 (p. 100) (a) Correct. Simplification is valid at nonzero increments, and those nearby values determine the limit.

Pr 6 (p. 101) No. It gives the slope; use f(2) for the y coordinate.

Pr 7 (p. 101) f(2)=8; derivative=3x squared; slope at 2 is 12. Tangent y−8=12(x−2), equivalently y=12x-16. One mark per step.

Pr 8 (p. 101) (b) Correct. Equal slopes give parallel lines, which can coincide in a special case.

Pr 10 (p. 103) $-2/(x-1)^2$: the numerator is (x−1) − (x+1) = −2.

Pr 11 (p. 103) Derivative = [(x−1)-(x+1)]/(x−1) squared = −2/(x−1) squared. Award marks for quotient rule, numerator order, denominator square and simplification.

Pr 12 (p. 103) (c) Correct. The quotient rule accounts for both changes and includes the squared denominator.

Pr 14 (p. 104) $-2x \sin(x^2)$, combining the cosine derivative with the inner derivative 2x.

Pr 15 (p. 104) Outer derivative is -sin(x^2); inner derivative is 2x. Multiply to obtain $-2x \sin(x^2)$. Credit both layers, sign and product.

Pr 16 (p. 104) (a) Correct. The inner expression changes seven units per unit increase in x.

Pr 17 (p. 104) $y' = -5\csc^2(5x)$. The minus sign and the factor 5 both matter.

Pr 19 (p. 106) Because y depends on x; the chain rule multiplies 2y by dy/dx.

Pr 20 (p. 106) Differentiate: y+x y'+2y y'=0. Collect: (x+2y)y'=-y. Hence y'=-y/(x+2y), where the denominator is nonzero. Four marks across these steps.

Pr 21 (p. 106) (b) Correct. Both factors change, so the product rule gives both terms.

Pr 23 (p. 107) x^x(ln x + 1). Take ln y = x ln x, then y′/y = ln x + 1.

Pr 24 (p. 107) Take logarithms: ln y=x ln x. Differentiate: y'/y=ln x+1. Multiply by y: y'= x to the x × (ln x+1). Four marks for these stages.

Pr 25 (p. 107) (c) Correct. Treating the exponent as fixed misses part of the change.

Pr 26 (p. 107) $(3^x)' = 3^x \ln 3$, and $(\log_3 x)' = 1/(x \ln 3)$ for $x > 0$.

Pr 28 (p. 109) 3, since the derivative quotient is 3 cos(3x)/1 and cos 0 = 1.

Pr 29 (p. 109) The form is $\frac{0}{0}$. L Hopital gives limit 3 cos(3x)/1 = 3. Alternatively use the standard sine limit. Credit form, derivative, evaluation and answer.

Pr 30 (p. 109) (a) Correct. A nonzero finite denominator limit gives a directly determined quotient.

Pr 32 (p. 110) No. Concavity must change across c.

Pr 33 (p. 110) Derivative = 2x-4, zero at 2. It is negative for x<2 and positive for x>2. Minimum is (2,−4). One mark each for derivative, intervals and point.

Pr 34 (p. 110) (b) Correct. The graph falls toward the point and rises away.

Pr 36 (p. 111) Yes. Absolute extrema on a closed interval must include endpoint checks.

Pr 37 (p. 111) Critical point x=1. Values: f(0)=0, f(1)=−1, f(3)=3. Minimum −1 at x=1; maximum 3 at $x = 3$. Credit candidates, evaluations and both extrema.

Pr 38 (p. 111) (c) Correct. The last allowed input has the largest output.

Pr 40 (p. 112) c=2: the average slope is (9−1)/(3−1)=4 and f′(c)=2c=4.

Pr 41 (p. 112) The polynomial is continuous and differentiable. Average slope = (9−1)/(3−1)=4. Solve 2c=4 to get c=2, which is inside (1,3). Four marks for hypotheses, slope, solution and interval check.

Pr 42 (p. 112) (a) Correct. A midpoint works for some functions but is not the general theorem conclusion.

Pr 44 (p. 114) $(x^2+2)^3 + C$. With $u=x^2+2$, the integral becomes 3 times the integral of u^2 du.

Pr 45 (p. 114) Set $u=x^2+2$, $du = 2x\,dx$. Integral becomes 3 times integral u squared du = u cubed+C = (x^2+2) cubed+C. Credit substitution, factor, integration and constant.

Pr 46 (p. 114) (b) Correct. Its differential matches the remaining x factor up to a constant.

Pr 47 (p. 114) $-\frac{1}{3}\cos(3x) + C$.

Pr 49 (p. 115) No. Positive and negative contributions can cancel.

Pr 50 (p. 115) Set $u = x^2 + 1$; limits become 1 and 2. Integral of u du from 1 to 2 = (4−1)/2 = 3/2. Credit substitution, new limits, antiderivative and subtraction.

Pr 51 (p. 115) (c) Correct. The endpoints must still describe the same original inputs.

Pr 52 (p. 116) Estimate: 3/4. Exact integral: 1/2. The right-end estimate is too large because the function increases.

Calculus I, Test A

30 marks. Suggested time 30 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. The derivative of x^2 at $x = 3$ is:

   (a) 3
   (b) 6
   (c) 9
   (d) 0

2. The derivative of a constant 7 is:

   (a) 7
   (b) 1
   (c) x
   (d) 0

3. The derivative of xe^x is:

   (a) $e^x(1 + x)$
   (b) 0
   (c) e^x
   (d) xe^x

4. The derivative of $\sin(5x)$ is:

   (a) $-5\sin(5x)$
   (b) $\sin(5)$
   (c) $\cos(5x)$
   (d) $5\cos(5x)$

5. For $x^2 + y^2 = 25$, the slope at $(3,4)$ is:

   (a) 0
   (b) $\frac{3}{4}$
   (c) $-\frac{3}{4}$
   (d) $\frac{4}{3}$

6. The derivative of $\arctan x$ is:

   (a) $\arctan(1)$
   (b) $\frac{1}{1+x^2}$
   (c) $\frac{1}{\cos^2 x}$
   (d) $-\frac{1}{1+x^2}$

Part II. Show your reasoning.

1. Use the definition to find the derivative of f(x)=3x squared.
2. Find the tangent to f(x)=x cubed at $x = 2$.
3. Differentiate y=(x+1)/(x−1), for x not equal to 1.
4. Differentiate y=cos(x^2).
5. Find dy/dx for xy+y squared=6.
6. Differentiate y=x to the power x for x greater than zero.

Calculus I, Test B

30 marks. Suggested time 30 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. Before applying L Hopital directly, the quotient should have form:

   (a) $\frac{0}{0}$ or $\frac{\infty}{\infty}$
   (b) $\frac{1}{1}$ only
   (c) $\frac{0}{5}$ only
   (d) Any fraction

2. If f' is negative on an interval, f is:

   (a) Decreasing
   (b) Constant
   (c) Always concave down
   (d) Increasing

3. To find absolute extrema on a closed interval, check:

   (a) Only where f' is zero
   (b) Only endpoints
   (c) Endpoints and interior critical numbers
   (d) Only where f is zero

4. Rolle theorem additionally requires:

   (a) An infinite interval
   (b) Equal endpoint function values
   (c) A discontinuity
   (d) A nonzero derivative everywhere

5. An indefinite integral must generally include:

   (a) Only a numerical answer
   (b) A fixed upper limit
   (c) No variable
   (d) An arbitrary constant

6. The $\int_{-2}^{2} x\,dx$ equals:

   (a) 8
   (b) -4
   (c) 0
   (d) 4

Part II. Show your reasoning.

1. Evaluate the limit of $\sin(3x)/x$ as x approaches zero.
2. For $f(x) = x^2 - 4x$, find increasing and decreasing intervals and the minimum point.
3. Find absolute maximum and minimum of $f(x) = x^2 - 2x$ on $[0,3]$.
4. Use the mean-value theorem for $f(x) = x^2$ on $[1,3]$ to find c.
5. Evaluate the integral of $6x(x^2+2)$ squared with respect to x.
6. Evaluate the integral from 0 to 1 of $2x(x^2+1)$ dx.

Calculus I, Test A: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (b) The derivative is 2x, so its value at 3 is 6.

Written part. [3(x+h) squared −3x squared]/h = [6xh+3h squared]/h = 6x+3h. Taking the limit gives 6x. One mark each for setup, expansion, cancellation and limit. (Section: p. 99.)

Question 2. (d) A constant function has zero rate of change.

Written part. f(2)=8; derivative=3x squared; slope at 2 is 12. Tangent y−8=12(x−2), equivalently y=12x-16. One mark per step. (Section: p. 100.)

Question 3. (a) The product rule differentiates each factor in turn.

Written part. Derivative = [(x−1)-(x+1)]/(x−1) squared = −2/(x−1) squared. Award marks for quotient rule, numerator order, denominator square and simplification. (Section: p. 102.)

Question 4. (d) The inner derivative contributes a factor of 5.

Written part. Outer derivative is -sin(x^2); inner derivative is 2x. Multiply to obtain −2x sin(x^2). Credit both layers, sign and product. (Section: p. 103.)

Question 5. (c) Implicit differentiation gives slope $-\frac{x}{y}$.

Written part. Differentiate: y+x y'+2y y'=0. Collect: (x+2y)y'=-y. Hence y'=-y/(x+2y), where the denominator is nonzero. Four marks across these steps. (Section: p. 105.)

Question 6. (b) This is the standard inverse-tangent derivative.

Written part. Take logarithms: ln y=x ln x. Differentiate: y'/y=ln x+1. Multiply by y: y'=x to the x × (ln x+1). Four marks for these stages. (Section: p. 106.)

Calculus I, Test B: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (a) The rule concerns specified indeterminate quotient forms.

Written part. The form is $\frac{0}{0}$. L Hopital gives limit $3\cos(3x)/1 = 3$. Alternatively use the standard sine limit. Credit form, derivative, evaluation and answer. (Section: p. 108.)

Question 2. (a) Negative slope means the function decreases as x increases.

Written part. Derivative = 2x-4, zero at 2. It is negative for x<2 and positive for x>2. Minimum is $(2,-4)$. One mark each for derivative, intervals and point. (Section: p. 109.)

Question 3. (c) The largest or smallest value may occur at either kind of candidate.

Written part. Critical point x=1. Values: $f(0)=0$, $f(1)=-1$, $f(3)=3$. Minimum -1 at x=1; maximum 3 at $x = 3$. Credit candidates, evaluations and both extrema. (Section: p. 110.)

Question 4. (b) Equal endpoint heights make the average slope zero.

Written part. The polynomial is continuous and differentiable. Average slope = $(9-1)/(3-1)=4$. Solve 2c=4 to get c=2, which is inside $(1,3)$. Four marks for hypotheses, slope, solution and interval check. (Section: p. 111.)

Question 5. (d) Differentiation loses constants, so all antiderivatives differ by a constant on an interval.

Written part. Set u=x^2+2, $du = 2x\,dx$. Integral becomes 3 times integral u squared du = u cubed+C = (x^2+2) cubed+C. Credit substitution, factor, integration and constant. (Section: p. 113.)

Question 6. (c) The odd integrand has equal and opposite signed areas.

Written part. Set $u = x^2+1$; limits become 1 and 2. Integral of u du from 1 to 2 = $(4-1)/2$ = 3/2. Credit substitution, new limits, antiderivative and subtraction. (Section: p. 114.)

Facts at a glance

1. Limits, continuity and the derivative definition p. 99

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

2. Basic derivatives and tangent lines p. 100

$$\frac{d}{dx} x^n = n x^{n-1}$$
$$y - f(a) = f'(a)(x - a)$$

3. Product and quotient rules p. 102

$$(uv)' = u'v + uv'$$
$$(u/v)' = (u'v - uv')/v^2$$

4. Chain rule and trigonometric functions p. 103

$$\frac{d}{dx} f(g(x)) = f'(g(x))g'(x)$$
$$(\sin x)' = \cos x$$
$$(\cos x)' = -\sin x$$

5. Implicit differentiation p. 105

$$\frac{d}{dx}(y^2) = 2yy'$$
$$\frac{d}{dx}(xy) = y + xy'$$

6. Logarithmic, exponential and inverse-trig derivatives p. 106

$$(e^u)' = e^u u'$$
$$(\ln |u|)' = u'/u$$
$$(\arctan x)' = 1/(1 + x^2)$$

7. L Hopital and indeterminate forms p. 108

$$\lim \frac{f(x)}{g(x)} = \lim \frac{f'(x)}{g'(x)}$$
when the rule applies

8. Increasing, decreasing and concavity p. 109

$$f' > 0 : \text{ increasing;}$$
$$f'' > 0 : \text{ concave up}$$

9. Graphing, asymptotes and absolute extrema p. 110

Candidates = endpoints plus interior critical numbers

10. Optimization and mean-value theorems p. 111

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$
$$a < c < b$$

11. Antiderivatives and substitution p. 113

$$\int x^n \, dx = x^{n+1}/(n+1) + C \ (n \neq -1)$$
$$\int dx/x = \ln|x| + C$$

12. Definite integrals, signed area and the fundamental theorem p. 114

$$\int_a^b f(x) \, dx = F(b) - F(a)$$
$$\frac{d}{dx} \int_a^{g(x)} f(t) \, dt = f(g(x))g'(x)$$

Notation

Symbols are listed by section. The same letter may name different quantities in different sections; read the section's Fact.

1. Limits, continuity and the derivative definition

h	small nonzero input increment before the limit (same unit as x)
$f'(x)$	instantaneous rate of change (output unit per input unit)

2. Basic derivatives and tangent lines

a	input coordinate of tangent point (input unit)
$f(a)$	output coordinate of tangent point (output unit)
$f'(a)$	tangent slope (output unit per input unit)

3. Product and quotient rules

u, v	the two factor functions (problem dependent)
u', v'	their derivatives with respect to x (corresponding output per input unit)

4. Chain rule and trigonometric functions

$u = g(x)$	inner expression (problem dependent)
$g'(x)$	inner derivative (inner-expression unit per input unit)

5. Implicit differentiation

y'	dy/dx, the local slope of the curve (y unit per x unit)

6. Logarithmic, exponential and inverse-trig derivatives

e	base of the natural exponential (dimensionless)
$\ln u$	natural logarithm of a positive dimensionless argument (dimensionless)

7. L Hopital and indeterminate forms

$0/0$	indeterminate limiting form, not an arithmetic value (not a measured quantity)
∞/∞	indeterminate quotient form (not a measured quantity)

8. Increasing, decreasing and concavity

f'	first derivative, controlling increase or decrease (output per input unit)
f''	second derivative, controlling concavity (output per input unit squared)

9. Graphing, asymptotes and absolute extrema

$[a, b]$	closed interval including both endpoints (input unit)
$f(c)$	function value at a candidate point (output unit)

10. Optimization and mean-value theorems

x	one rectangle side length (length unit)
A	rectangle area (square length unit)
c	interior input guaranteed by the mean-value theorem (input unit)

11. Antiderivatives and substitution

u	chosen inner expression (problem dependent)
du	differential of the inner expression (same unit as u)
C	arbitrary additive integration constant (antiderivative unit)

12. Definite integrals, signed area and the fundamental theorem

a, b	lower and upper integration bounds (input unit)
F	an antiderivative of f (f unit times input unit)
t	dummy integration variable (input unit)

Sources

MATH 203: derivative rules, implicit/logarithmic/exponential derivatives, L Hopital, graphing and extrema, optimization, Rolle and mean-value theorems, integration and the fundamental theorem. The full lecture syllabus is the main source.

The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. Check lecturer-specific derivations and conventions. The practice tests are original study material, not predictions of an examination.

Mechanics and Heat Lab

All distinct experiments in the folder: pendulum, free fall, static and dynamic springs, force table, cooling, Boyle law, expansion, specific heat, latent heat and Joule equivalent. Duplicate versions are consolidated.

1 Earn marks from tables, graphs and uncertainty

Aims. Calculate a fitted-line slope with the correct axis units; Distinguish percentage error from measurement uncertainty.
Before this section: Coordinate differences; division; raw and derived measurements.

A lab result is a measured quantity with units and a justified method. Record raw observations before calculating derived columns. A slope is change in the vertical variable divided by change in the horizontal variable. Its units are essential and determine how to recover the desired constant.

Idea 1. An experimental result combines a number, units and a defensible measurement method. Repeated readings reveal spread, but they may all share the same calibration offset. Percentage error compares a result with a reference; it does not by itself explain whether the discrepancy comes from random variation or systematic bias.

Fact 1.

$$m = \Delta y / \Delta x$$

$$\%\ \text{error} = 100|x_{\text{measured}} - x_{\text{reference}}|/|x_{\text{reference}}|$$

where m is the slope of the selected vertical variable against horizontal variable (vertical unit / horizontal unit); x_{measured} is the experimental estimate (same as reference); $x_{\text{reference}}$ is the accepted comparison value (same as measurement).

Why it holds. Repeating a measurement can reduce random scatter in the mean. A shared calibration offset remains in repeated readings unless it is identified and corrected.

Method 1.

1. Record raw readings with units before calculating derived quantities.
2. Label axes with quantities and units and draw a suitable best-fit line.
3. Choose two widely separated points on that fitted line and calculate rise over run.
4. Use axis units to interpret the slope; compare with a reference using the stated percentage-error formula.

Recognising it. Use percentage error for a reference comparison, and discuss instrument or method bias separately from random spread.

Pr 1. A fitted line rises by 0.8 square seconds over 0.2 m.

Solution. The example has vertical change 0.8 s^2 and horizontal change 0.2 m. Slope is rise over run: $0.8/0.2 = 4$ s^2/m. The reciprocal, 0.25 m/s^2, belongs to a graph with reversed axes and is a different slope. A slope should come from the fitted line, not necessarily from two adjacent raw observations.

Common error. Do not force a line through the origin unless the model and measurement offsets justify it.

In the examination. Write slope units next to the numerical calculation; they help reveal an inverted axis choice.

Pr 2. Can percentage error alone describe repeatability of a measurement?

Pr 3. A measured g is 9.6 m/s^2 and reference g is 9.8 m/s^2. Calculate percentage error and name one possible systematic error.

Pr 4. A ruler has a fixed zero offset. Taking many readings and averaging them will:

(a) Retain the systematic offset unless corrected

(b) Automatically remove the offset

(c) Prove the ruler is accurate if the readings agree

Fact 2. Uncertainty is different from percentage error. An uncertainty describes measurement quality. Percentage error compares a result with a reference value.

$$\text{reported result} = \text{value} \pm \text{uncertainty}$$

$$\%\ \text{error} = 100\frac{|x_{\text{measured}} - x_{\text{reference}}|}{|x_{\text{reference}}|}$$

1. Use the uncertainty method specified by your experiment or instrument; do not invent a precision from extra calculator digits.
2. Give the uncertainty the same unit as the value. Round the value to the same decimal place as the reported uncertainty.
3. Repeated measurements help estimate and reduce the effect of random scatter. They do not remove a consistent zero offset.
4. An uncertainty range is not a guarantee that every possible error is inside it.

Pr 5. A result is (9.6 ± 0.2) m/s^2, and the reference is 9.8 m/s^2. State the given uncertainty and calculate percentage error.

2 Simple pendulum

Aims. Measure pendulum length and period consistently; Recover g from one trial or a T^2-versus-L graph.
Before this section: Complete oscillation; squaring; slope interpretation.

For small angles a pendulum period depends mainly on length and gravity, not bob mass. Measure length from pivot to the centre of the bob. Timing several complete oscillations reduces the relative impact of reaction time. Keep the starting angle small and avoid pushing the bob.

Idea 2. A pendulum period is the time for a complete return, not merely a trip to the other side. Timing many cycles spreads the start-stop reaction uncertainty over a longer total duration. Its length is a geometric quantity from pivot to bob centre, and a consistent length definition is as important as careful timing.

Fact 3.

$$T = t/N$$
$$T^2 = (4\pi^2/g)L$$
$$g = 4\pi^2/m \text{ for } T^2 \text{ versus } L$$

where L is the pivot-to-bob-centre length (m); t is the time for N complete oscillations (s); T is the period of one oscillation (s); g is the gravitational acceleration (m/s^2).
Conditions. The simple model assumes small angular displacement, negligible damping and a light inextensible suspension.

Why it holds. Dividing a multi-cycle time by cycle count gives the mean single-cycle period. For small angles, period depends on length and gravity rather than bob mass.

Method 2.

1. Measure length from pivot to the bob centre and release from a small angle without a push.
2. Time several complete oscillations and divide total time by their count.
3. Calculate T^2 and repeat at several lengths if graphing.
4. For T^2 on the vertical axis and L horizontally, use $g = 4\pi^2/\text{slope}$.

Recognising it. A pendulum time for several cycles first requires conversion to period before applying the gravity relation.

Pr 6. For L=1 m and T=2 s, $g = 4\pi^2 \times 1/4 = 9.87$ m/s^2.

Solution. For the example L = 1 m and T = 2 s. Square the measured period: $T^2 = 4$ s^2. Rearranging $T^2 = 4\pi^2 L/g$ gives $g = 4\pi^2 L/T^2$. Substitution gives $g = 4\pi^2 \times 1/4 \approx 9.87$ m/s^2.

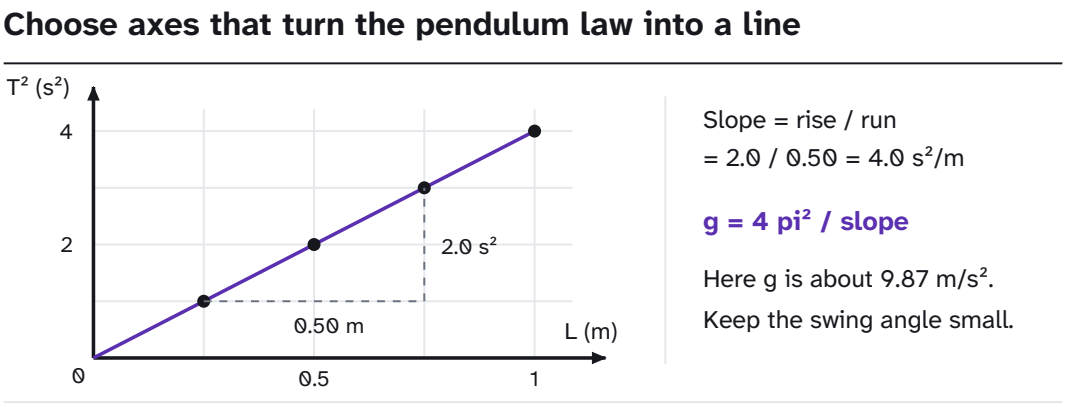


Figure 5. Figure 05. For small pendulum oscillations, a plot of T^2 against L has slope $4\pi^2/g$. The illustrated slope is 4.0 s^2/m, giving g about 9.87 m/s^2. Points are ideal model values, not experimental data.

Common error. Using total timing t instead of the single-cycle period T makes g dramatically wrong.

In the examination. State which variables are on which axes before converting a slope into g.

Pr 7. A bob crosses equilibrium twice in one full cycle. Should each crossing be counted as a cycle?

Pr 8. A pendulum takes 30 s for 15 cycles at length 0.99 m. Calculate period and g.

Pr 9. The pendulum bob is replaced by a heavier bob at the same effective length and small angle. The ideal period:

(a) Halves whenever mass doubles

(b) Stays approximately the same

(c) Increases in proportion to mass

3 Free fall and axis choice

Aims. Linearize free fall with a time-squared graph; Select the correct g conversion for either axis orientation.
Before this section: Constant acceleration from rest; square of time; fitted-line slope.

For release from rest with negligible air drag, fall distance grows with time squared. The course sheets mention both graph orientations; the equation must decide the slope conversion. A light sensor or electronic timer avoids large manual timing errors.

Idea 3. Free fall from rest is quadratic in time, so plotting distance directly against time gives a curve. Squaring time creates a straight-line relationship. Which variable is vertical then determines whether slope represents half of gravity or the reciprocal form; the graph label is part of the calculation.

Fact 4.

$$h = \tfrac{1}{2}gt^2;$$
$$m_{h \text{ vs } t^2} = g/2;$$
$$m_{t^2 \text{ vs } h} = 2/g$$

where h is the vertical fall distance (m); t is the fall time (s); g is the gravitational acceleration (m/s^2); m is the graph slope (depends on axes).
Conditions. The object starts from rest and air resistance is negligible over the measured fall.

Why it holds. For height plotted vertically against time squared, slope has acceleration units. Reversing the axes reverses the ideal slope and changes the conversion needed for gravity.

Method 3.

1. Release from rest and measure fall time with the stated apparatus.
2. Calculate t^2 for each height.
3. State the graph axes and compare $h = (g/2)t^2$ with $y = mx$.
4. For h against t^2 use $g = 2m$; for t^2 against h use $g = 2/m$.

Recognising it. When a free-fall graph is given, read the axes before recalling any slope-to-gravity rule.

Pr 10. A plot of h against t^2 has slope 4.9 m per square second, so g=9.8 m/s^2.

Solution. With h on the vertical axis, the fitted slope is 4.9 m/s^2. The equation $h = (g/2)t^2$ shows this slope is half g, so $g = 9.8$ m/s^2. Reversing the axes makes slope $2/g$, about 0.204 s^2/m. The reciprocal slopes describe the same ideal relationship; neither slope can be interpreted without its axes.

Common error. Do not use g=2 times slope for a t-squared-versus-height graph.

In the examination. Do not memorize slope = g/2 without the graph orientation; use the displayed equation to decide.

Pr 11. A t^2-versus-h slope is 0.20 s^2/m. What is g?

Pr 12. The slope of t^2 against height is 0.20 square seconds per metre. Find g.

Pr 13. For release from rest without drag, why is height plotted against time squared?

(a) It removes gravity from the experiment

(b) It makes height proportional to time

(c) It makes the ideal relationship linear

4 Static Hooke law

Aims. Distinguish spring extension from loaded length; Find spring stiffness from a load or a force-extension slope.
Before this section: Weight mg; length differences; centimetres to metres.

Within the proportional range, spring extension is proportional to the applied load. Extension is loaded length minus unloaded length. The restoring spring force points opposite the extension, while the applied load magnitude is k times extension. An elastic spring returns to its original length after unloading, but that alone does not prove a straight-line force-extension relation.

Idea 4. A spring experiment measures extension relative to the unloaded state. The hanging mass supplies a weight, which is a force rather than a length. Within the proportional range, plotting load against extension reveals stiffness: a steep line means more force is needed for each extra metre of stretch.

Fact 5.

$$F = k\Delta L$$
$$F = mg$$
$$k = m_{F \text{ vs } \Delta L}$$

where F is the applied load magnitude (N); ΔL is the loaded minus unloaded spring length (m); k is the spring stiffness (N/m); m is the total suspended mass (kg).
Conditions. Use Hooke law only over the proportional linear range; elastic recovery alone does not guarantee strict proportionality.

Why it holds. Using total loaded length would include an irrelevant original-length offset. Weight must be calculated from mass before dividing by extension to obtain stiffness.

Method 4.

1. Record the unloaded spring length first.
2. Add the load safely and calculate extension as loaded minus unloaded length.

3. Convert extension to metres and load mass to force using mg.
4. Find k from F/ΔL or the slope of F against ΔL; check the graph remains linear.

Recognising it. For static spring problems, convert mass to force and loaded-minus-unloaded length to extension.

Pr 14. A 2 N load produces 0.04 m extension, so k=50 N/m.

Solution. The applied load is 2 N and the extension is 0.04 m, not the total spring length. Using $F = k\Delta L$ gives $k = 2/0.04 = 50$ N/m. This means an additional 1 m extension would require 50 N only if the same linear model remained valid. If unloading leaves a changed length, the spring has not returned elastically; avoid extrapolating the original model.

Common error. Centimetres must be converted to metres before reporting k in N/m.

In the examination. Include the hanger in total suspended mass if the apparatus instructions require it; otherwise the force scale can acquire an offset.

Pr 15. A 0.30 kg total load extends the spring 6 cm with $g = 10$ m/s^2. What is k?

Pr 16. A 0.30 kg mass extends a spring by 6 cm. Use g=10 m/s^2. Find k.

Pr 17. A spring original length is 10 cm and its loaded length is 14 cm. Which length belongs in the Hooke-law extension term?

(a) The 4 cm difference

(b) The 14 cm loaded length

(c) The 10 cm original length

5 Dynamic spring and gravity

Aims. Relate mass-spring period to load and spring stiffness; Use graph axes to distinguish a k experiment from a g experiment.
Before this section: Static spring extension; period measurement; slope units.

An oscillating mass-spring system has a period that increases with mass and decreases with stiffness. Combining static extension with dynamic period allows gravity to be estimated. Small oscillations and the same effective load model are needed for consistency.

Idea 5. Static stretch and dynamic oscillation probe the same spring in different ways. A larger oscillating mass produces a longer period, while a stiffer spring produces a shorter one. Squaring period makes its relationship with mass linear, so the graph slope contains the stiffness information.

Fact 6.

$$T^2 = 4\pi^2 m/k$$
$$T^2 = (4\pi^2/g)\Delta L$$

where T is the oscillation period (s); m is the oscillating mass in the stated model (kg); k is the spring stiffness (N/m); ΔL is the static equilibrium extension (m); g is the gravitational acceleration (m/s^2).
Conditions. Small oscillations stay in the spring proportional range; the static and dynamic models must treat hanger and effective spring mass consistently.

Why it holds. For period squared against mass, a larger slope corresponds to a smaller spring stiffness. A period-squared-versus-extension graph instead contains gravity, so the two slope interpretations cannot be exchanged.

Method 5.

1. Measure small vertical oscillations about equilibrium and obtain period from several cycles.
2. For T^2 against mass compare the slope with $4\pi^2$/k.
3. For T^2 against static extension compare the slope with $4\pi^2$/g.
4. Apply the matching rearrangement and retain the effective-mass assumptions used by the apparatus.

Recognising it. Inspect whether the horizontal axis is mass or static extension before converting a spring-oscillation slope.

Pr 18. A T-squared-versus-extension slope of 4.0 square seconds per metre gives g= $4\pi^2/4.0$=9.87 m/s^2.

Solution. The example plots T^2 vertically against static extension horizontally. Its slope is 4.0 s^2/m, corresponding to $4\pi^2$/g. Rearrangement gives g = $4\pi^2/4.0 \approx 9.87$ m/s^2. A T^2-versus-mass slope would instead determine k; substituting it here would mix two different experiments.

Common error. The symbols for mass and graph slope can both be m; label them explicitly to avoid confusion.

In the examination. Use slope units to identify the requested constant: s^2/kg for stiffness, s^2/m for gravity in these two plots.

Pr 19. A T^2-versus-mass slope is 0.80 s^2/kg. What is k?

Pr 20. A T-squared-versus-mass graph has slope 0.80 square seconds per kg. Find spring constant.

Pr 21. Two springs give period-squared-versus-mass lines. The stiffer spring has:

(a) The same slope regardless of stiffness

(b) The smaller slope

(c) The larger slope

6 Force table and equilibrium

Aims. Calculate the equilibrant from perpendicular forces; Explain how the force-table ring indicates approximate equilibrium.
Before this section: Vector components; Pythagoras; angles from positive x.

The force table demonstrates vector addition. The resultant of two forces is their vector sum. The equilibrant has equal magnitude and opposite direction. A centred ring that does not press on the pin indicates approximate balance, subject to pulley friction.

Idea 6. The force table balances vector directions, not just force magnitudes. Two perpendicular pulls create a resultant pointing between them; a third force balances that resultant only if it points exactly opposite. A centred ring provides experimental evidence of approximate balance when it is not held by the pin.

Fact 7.

$$\sum F_x = 0$$
$$\sum F_y = 0$$
$$F_i = m_i g$$

where F_i is the tension force from hanging mass in the ideal setup (N); m_i is the total hanging mass for a string (kg); R is the resultant force magnitude (N).
Conditions. Mass-to-tension conversion neglects pulley friction and string mass; actual balance is approximate.

Why it holds. Equal resultant and equilibrant magnitudes are not enough without opposite directions. A force pushing the ring against the central pin can be balanced partly by the pin, hiding string-force imbalance.

Method 6.

1. Convert each hanging mass to a force using the chosen g.
2. Resolve all forces into signed horizontal and vertical components.
3. Find the resultant, then reverse both components for the equilibrant.
4. Adjust the balancing force and check the ring is centred without pressing on the pin.

Recognising it. When equilibrium is requested, first find the resultant, then reverse its components for the equilibrant.

Pr 22. For 3 N east and 4 N north, the resultant is 5 N northeast.

Solution. The 3 N east and 4 N north forces give resultant components (+3,+4) N. Its magnitude is $\sqrt{(3^2+4^2)} = 5$ N. Its precise direction is 53.1° north of east; northeast here describes its quadrant. The equilibrant has components (−3,−4) N and magnitude 5 N, directed 53.1° south of west.

Common error. The equilibrant is not the resultant in the same direction.

In the examination. If an angle is requested, northeast alone is insufficient; give a numerical angle and reference axis.

Pr 23. Does the equilibrant have the same direction as the resultant?

Pr 24. Two forces are 6 N east and 8 N north. Find equilibrant magnitude and direction.

Pr 25. Why should the force-table ring be free of contact with the central pin?

(a) The pin removes all pulley friction

(b) A ring touching a pin has no weight

(c) The pin could supply an extra balancing force

7 Newton cooling law

Aims. Calculate temperature excess above the surroundings; Interpret an exponential cooling trend without confusing it with Celsius temperature.
Before this section: Subtracting temperatures; halving; logarithm recognition.

A hot object approaches room temperature as its temperature excess decays. The simple model assumes approximately constant surroundings and an effective cooling constant over the measured range. The temperature excess, not the absolute temperature, is the decaying quantity.

Idea 7. Cooling depends on how much hotter the object is than its surroundings. An object at 50°C in a 20°C room has an excess of 30°C, not 50°C. The simple exponential model shrinks this excess, then room temperature must be added back to recover the actual object temperature.

Fact 8.

$$T - T_r = (T_0 - T_r)e^{-kt}$$
$$\ln[(T - T_r)/(T_0 - T_r)] = -kt$$

where T is the object temperature (°C or K consistently); T_r is the room temperature (same unit as T); k is the effective cooling constant (inverse time); t is the elapsed time (s or min consistently with k).
Conditions. Room temperature is approximately constant and the effective cooling constant applies over the measured range.

Why it holds. Equal temperature excesses in otherwise equivalent conditions give equal cooling-rate magnitudes in the model. As excess falls, the cooling rate becomes smaller and the object approaches room temperature.

Method 7.

1. Measure room temperature and record it as the reference.
2. Subtract room temperature from each object temperature.

3. Compare excesses over time or use the stated logarithmic plot.
4. Add the room temperature back when converting a final excess into an object temperature.

Recognising it. A statement that temperature excess halves means halve object-minus-room temperature, not the Celsius reading itself.

Pr 26. If room temperature is 20 degrees C and the object cools from 80 to 50, the excess falls from 60 to 30 degrees, so it halves.

Solution. The room is at 20°C, so the initial 80°C object has excess $80 - 20 = 60$°C. At 50°C its excess is $50 - 20 = 30$°C. The excess has halved, even though 50 is not half of 80. This distinction explains why the model approaches room temperature rather than cooling toward 0°C.

Common error. The object approaches room temperature, not zero degrees C.

In the examination. When logging data, use a dimensionless excess ratio such as (T-Tr)/(T0-Tr), with temperatures consistently measured.

Pr 27. At room temperature 25°C, an object starts at 85°C and its excess halves. What is its new temperature?

Pr 28. Room temperature is 25 degrees C. A body starts at 85 degrees C and its temperature excess halves in 5 minutes. What is its temperature then?

Pr 29. A 60°C object is placed in a 20°C room or a 40°C room under otherwise identical conditions. It initially cools faster in:

(a) The 20°C room

(b) The 40°C room

(c) Both rooms equally because the object starts at 60°C

8 Boyle law

Aims. Use absolute pressure in a fixed-temperature gas comparison; Predict the inverse pressure-volume trend and check thermal settling.

Before this section: Inverse proportion; multiplying and dividing; consistent pressure and volume units.

At fixed temperature and fixed amount of gas, absolute pressure is inversely proportional to volume. A narrow uniform tube may use gas-column length as a proxy for volume. Let the gas settle thermally after changing pressure.

Idea 8. Boyle law follows the pressure-volume trade-off for a fixed amount of gas at constant temperature. The gas can occupy less space only while exerting greater absolute pressure. Rapid compression can warm the gas, so thermal settling is needed before treating readings as one constant-temperature comparison.

Fact 9.

$$P_1V_1 = P_2V_2$$
$$P \propto 1/V$$

where P is the absolute gas pressure (Pa or consistent pressure unit); V is the gas volume (m^3 or consistent volume unit).
Conditions. Gas amount and temperature remain constant; the ideal Boyle-law approximation is suitable for the measured range.

Why it holds. The pressure-volume product stays constant only while amount and temperature stay fixed. Using gauge pressure instead of absolute pressure changes the comparison unless converted appropriately.

Method 8.

1. Confirm the gas amount stays fixed and readings are taken at the same temperature.
2. Use absolute pressure rather than gauge pressure.
3. Write P1V1 = P2V2 and isolate the requested volume or pressure.
4. Check that larger pressure gives smaller volume and allow time for thermal settling between trials.

Recognising it. Use Boyle law when gas amount is fixed and two states are explicitly at the same temperature.

Pr 30. A gas at 100 kPa and 60 cubic centimetres has volume 30 cubic centimetres at 200 kPa, if temperature stays fixed.

Solution. Initially the gas has pressure 100 kPa and volume 60 cm^3. At fixed temperature and amount, their product stays constant. Doubling pressure to 200 kPa requires half the volume: V2 = 100 × 60/200 = 30 cm^3. Keeping kPa and cm^3 is valid for this ratio because the corresponding units match on both sides.

Common error. Compressing quickly can warm the gas and violate the constant-temperature assumption.

In the examination. Do not convert a gauge reading to absolute pressure by ignoring atmospheric pressure; use the apparatus definition.

Pr 31. If volume is compressed quickly, why wait before recording a Boyle-law reading?

Pr 32. A trapped gas has volume 40 cubic centimetres at absolute pressure 120 kPa. Find its volume at 160 kPa at the same temperature.

Pr 33. A gas is compressed rapidly and its temperature rises. Can the original constant-temperature Boyle comparison be applied immediately?

(a) Only if gauge pressure replaces absolute pressure

(b) No; the temperature condition is not yet satisfied

(c) Yes; any gas compression follows the same constant product

9 Linear thermal expansion

Aims. Calculate a linear expansion coefficient with compatible length units; Extract the coefficient from a length-change-versus-temperature graph.
Before this section: Small length changes; temperature differences; mm-to-m conversion.

Heating changes the length of a rod slightly. The coefficient of expansion describes fractional length change per degree. A dial gauge often reads millimetres while the rod length is recorded in metres, so consistent units are critical.

Idea 9. An expansion coefficient describes fractional length change per degree, so it should not depend on choosing millimetres or metres for the apparatus readout. Problems arise when the tiny displacement is entered in millimetres while the original rod remains in metres. Consistent length units must precede division.

Fact 10.

$$\Delta L = \alpha L_0 \Delta T$$
$$\alpha = m_{\Delta L \text{ vs } \Delta T} / L_0$$

where α is the linear expansion coefficient (K^{-1}); L_0 is the initial rod length (m); ΔL is the length increase (m); ΔT is the temperature increase (K).
Conditions. The temperature range is small enough to treat α as approximately constant and the rod temperature as sufficiently uniform.

Why it holds. The same displacement has different numerical values in metres and millimetres. Dividing length change by original length removes the length unit, leaving inverse temperature.

Method 9.

1. Record original rod length and the initial and final temperatures.
2. Convert gauge displacement to the same length unit as the rod.
3. Rearrange $\Delta L = \alpha L0 \Delta T$ to solve for α.
4. For a graph of ΔL against ΔT divide its slope by L0.

Recognising it. When a dial displacement and rod length use different units, convert them before calculating the coefficient.

Pr 34. A 1 m rod expands 0.5 mm over 50 K, giving alpha = $0.0005/(1 \times 50)$=0.00001 per K.

Solution. The 1 m rod expands by 0.5 mm, which is 0.0005 m. The temperature rise is 50 K. Divide extension by original length and temperature rise: $\alpha = 0.0005/(1 \times 50)$. The result is 0.00001 K^{-1}, or 1.0×10^{-5} K^{-1}.

Common error. A graph against absolute temperature may have an intercept; do not confuse it with a change-of-temperature plot.

In the examination. Write the displacement conversion explicitly; the small numerical size of α is physically expected.

Pr 35. What conversion error occurs if 0.5 mm is entered as 0.5 m?

Pr 36. A 0.50 m rod expands 0.30 mm after warming by 40 K. Find its expansion coefficient.

Pr 37. A student enters a millimetre displacement number as metres while keeping rod length in metres. The coefficient is:

(a) 1000 × too small

(b) Unaffected because coefficients have no length unit

(c) 1000 × too large

10 Specific heat, latent heat and Joule equivalent

Aims. Separate sensible heating, phase-change energy and electrical input; Keep the joule-per-calorie conversion dimensional.
Before this section: Energy balance; electrical power; grams-to-kilograms conversion.

Mixing calorimetry balances heat lost by the hot sample against heat gained by water and calorimeter. Latent heat accounts for a phase change. Electrical heating provides IVt joules; comparing it with heat measured in calories gives joules per calorie, not a dimensionless ratio.

Idea 10. Electrical heating supplies energy over a measured time. If a sample already at its melting point uses all that input only for melting, dividing by melted mass gives latent heat per kilogram. If temperature changes or losses also occur, the same division would misattribute energy to melting.

Fact 11.

$$m_s c_s (T_s - T_f) = (m_w c_w + C_{\text{cal}})(T_f - T_w)$$
$$Q = mL$$
$$E = IVt$$

where Q is the thermal energy transferred (J); L is the specific latent heat (J/kg); I is the heater current (A); V is the heater voltage (V); t is the heating duration (s).
Conditions. The latent-heat example assumes the sample starts at its melting point and all stated energy is used for melting.

Why it holds. Voltage times current gives power, so multiplying by time gives total supplied energy. Latent heat is energy per mass, not total energy and not a temperature rise.

Method 10.

1. Identify whether temperature changes, phase changes or both occur.

2. Calculate electrical input as IVt when heater readings are supplied.
3. Use mcΔT for temperature change and mL for a phase change.
4. If comparing joules with calories, divide joules by calories and state J/cal.

Recognising it. Use the electrical-input and latent-heat relations together only after identifying where the supplied energy goes.

Pr 38. If 500 J melts 0.002 kg at its melting point with no other heating, latent heat is 250000 J/kg.

Solution. The sample receives 500 J only for melting and the melted mass is 0.002 kg. Rearranging $Q = mL$ gives $L = 500/0.002 = 250000$ J/kg. In the separate conversion example, 418.4 J is compared with 100 cal. Their ratio is $418.4/100 = 4.184$ J/cal; the unit is essential because the two energy scales differ.

Common error. Ignoring calorimeter heat capacity or heat loss can bias the inferred material property.

In the examination. State where the energy goes before dividing by mass; not all heating energy necessarily produces the phase change.

Pr 39. If both electrical input and thermal gain are already in joules, is their ratio a J/cal conversion?

Pr 40. A heater at 6 V carries 2 A for 100 s. If all energy melts a sample already at its melting point and the melted mass is 0.004 kg, find supplied energy and latent heat.

Pr 41. Why must the sample already be at its melting point in the simple electrical melting calculation?

(a) Otherwise some input may first warm the sample

(b) Otherwise electrical energy cannot enter matter

(c) Because latent heat means temperature rise per second

11 Calorimetry: determine a solid specific heat

Aims. Balance heat lost by a solid against heat gained by water and calorimeter; Solve for specific heat while using the correct temperature changes.

Before this section: Heat capacity; mass in kg; a common final equilibrium temperature.

A hot solid placed in cooler water transfers energy to the water and calorimeter. At equilibrium they share a final temperature. The ideal model assumes negligible heat loss to surroundings and no phase change. The calorimeter contribution must be included if it is specified.

Idea 11. Calorimetry matches energy leaving a hot sample with energy entering cooler water and, if relevant, its container. Each body has its own temperature change despite reaching one shared final temperature. Writing those two changes explicitly is often the most important step.

Fact 12.

$$c_s = \frac{(m_w c_w + C_{\text{cal}})(T_f - T_w)}{m_s(T_s - T_f)}$$

where c_s is the solid specific heat capacity (J/(kg K)); m_s, m_w is the solid and water masses (kg); C_{cal} is the calorimeter heat capacity (J/K); T_f is the final equilibrium temperature (°C or K).
Conditions. The example neglects surroundings losses and calorimeter heat capacity and assumes no phase change.

Why it holds. The solid temperature drop determines how much energy it loses for each unit of its specific heat. The water temperature rise and known specific heat determine the energy gained in the ideal balance.

Method 11.

1. Identify the initially hot body, initially cold water and common final temperature.
2. Write heat lost by the solid equal to heat gained by water plus calorimeter.
3. Use positive temperature drops and rises on opposite sides.
4. Solve for cs and state the heat-loss and calorimeter assumptions.

Recognising it. Use a heat-lost equals heat-gained balance when a hot solid and cooler water reach one final temperature.

Pr 42. With a negligible calorimeter, 0.10 kg solid cooling from 80 to 30 degrees C heats 0.20 kg water from 25 to 30 degrees C.

Solution. Water warms from 25°C to 30°C, a rise of 5 K. Its heat gain is 0.20 × 4200 × 5 = 4200 J. The 0.10 kg solid cools from 80°C to 30°C, a drop of 50 K. With negligible calorimeter heat capacity and losses, 0.10 × cs × 50 = 4200, giving cs = 840 J/(kg K).

Common error. Using the same temperature change for hot solid and water is generally wrong.

In the examination. Use the calorimeter heat-capacity term when supplied; discarding it undercounts heat gained by the cold side.

Pr 43. Why is the solid temperature change not the same as the water temperature change?

Pr 44. A 0.20 kg solid cools from 90 to 40 degrees C and warms 0.10 kg water from 30 to 40 degrees C. Ignore calorimeter heat capacity and losses; use water c=4200 J/(kg K). Find the solid specific heat.

Pr 45. If a calorimeter warms but its heat gain is ignored, what happens to the inferred solid specific heat under this calculation?

(a) It cannot change because water mass is unchanged

(b) It is underestimated

(c) It is overestimated

12 Latent heat and electrical-to-thermal conversion

Aims. List all energy stages for ice melting and subsequent warming; Interpret how lost heat biases a measured electrical-to-calorie ratio.
Before this section: Specific heat versus latent heat; phase-change temperatures; units of energy.

A phase-change experiment separates melting from subsequent warming of the melted material. In the Joule-equivalent experiment electrical energy is compared with heat gained by water and calorimeter. If both quantities are already in joules, their ratio tests energy accounting; it is not a joule-per-calorie conversion.

Idea 12. A joule-per-calorie experiment compares the same intended energy transfer measured on two unit scales. If some supplied energy escapes without appearing in measured heat gain, the denominator becomes too small. Distinguishing units from efficiency prevents treating every energy ratio as a unit conversion.

Fact 13.

$$Q_{\text{ice}} = mL_f + mc_w(T_f - 0°\text{C})$$
$$J_{\text{eq}} = E_{\text{joules}}/Q_{\text{calories}}$$

where m is the ice mass (kg); L_f is the specific latent heat of fusion (J/kg); c_w is the liquid water specific heat (J/(kg K)); J_{eq} is the joules per calorie conversion estimate (J/cal).
Conditions. The ice formulas use an ordinary-pressure melting point of approximately 0°C and account for the stated final phase.

Why it holds. Joules divided by calories has a conversion unit. Joules divided by joules is dimensionless and describes energy accounting rather than a joule-per-calorie conversion.

Method 12.

1. Identify whether ice starts below or at its melting point.
2. Include warming of ice if needed, then melting, then warming of meltwater to the final temperature.
3. For conversion, keep electrical energy in joules and measured thermal gain in calories.
4. Assess a loss by asking whether the recorded heat gain is too small or too large for the supplied energy.

Recognising it. Check the denominator units before interpreting any electrical-to-thermal energy ratio.

Pr 46. If ice begins below zero degrees C, an additional warming-of-ice term is required before melting.

Solution. Ice below 0°C first needs energy to warm to 0°C; that stage uses ice specific heat. At 0°C, energy mLf melts it without a temperature rise in the ideal fixed-pressure model. If the final state is liquid above 0°C, add mcw(Tf - 0°C) for the meltwater. When ice al-

ready starts at 0°C the first stage is absent; the melting and later warming stages remain if required.

Common error. Do not divide joules by joules and label the dimensionless answer J/cal.

In the examination. Do not use the water specific heat for warming solid ice; the two phases have different heat capacities.

Pr 47. If 836.8 J supplies a recorded heat gain of 200 cal, what is E/Q, and how would unrecorded heat loss bias it?

Pr 48. An electrical experiment supplies 836.8 J and records heat gain of 200 cal. Find the conversion factor. If unaccounted heat escapes, does E/Q measured tend to be too high or too low?

Pr 49. Both electrical input and measured thermal gain are stated in joules. Their ratio has:

(a) Units of J/cal automatically

(b) Units of watts

(c) No unit, so it is not directly a J/cal conversion

Hints

Use one hint at a time, then return to the problem.

Pr 3 (p. 133) (1) Compare the measured and reference values using an absolute difference; the denominator must be the reference. (2) Divide that difference by 9.8 m/s^2 and multiply by 100 percent. (3) For the second part name a specific fixed bias, such as a timing calibration offset, and explain why it would affect readings consistently.

Pr 5 (p. 133) (1) The stated uncertainty is $0.2\,m/s^2$. (2) Percentage error is $100|9.6 - 9.8|/9.8 \approx 2.04\%$.

Pr 8 (p. 135) (1) Use the 30 s reading as the total for 15 complete cycles, not the single period. (2) Divide total time by cycle count and square the resulting period. (3) Use length 0.99 m in the rearranged pendulum relation for gravity. Check that the squared period is in seconds squared.

Pr 12 (p. 137) (1) The stated graph puts time squared vertically and height horizontally. Write the fall equation with that vertical quantity isolated. (2) Compare it with a straight-line equation to identify the slope as the reciprocal-gravity form. (3) Use the slope 0.20 s^2/m to solve for gravity. Verify that the resulting units are m/s^2.

Pr 16 (p. 138) (1) The 0.30 kg mass supplies a weight; the 6 cm measurement is its extension. (2) Convert extension to metres and multiply mass by the specified gravity. (3) Divide the applied force by extension to obtain stiffness. Retain N/m and assume the spring remains in its proportional range.

Pr 20 (p. 139) (1) The horizontal axis is mass, so choose the stiffness relation rather than the extension-based gravity relation. (2) Write the slope coefficient of mass in the squared-period equation. (3) Rearrange to make stiffness the subject and substitute 0.80 s^2/kg. Check that the result has N/m units.

Pr 24 (p. 141) (1) Draw 6 N east and 8 N north as perpendicular components. (2) Find their resultant magnitude and its reference angle from the component ratio. (3) Reverse both components to obtain the equilibrant. Give a numerical angle with a compass reference, not merely southwest.

Pr 28 (p. 142) (1) Subtract room temperature 25°C from the initial 85°C reading to obtain excess. (2) Halve that excess for the stated 5-minute interval. (3) Add the unchanged room temperature back. Check that the result lies between initial object temperature and room temperature.

Pr 32 (p. 143) (1) Check that the supplied 120 and 160 kPa are absolute pressures and temperatures match. (2) Set the initial pressure-volume product equal to the final product, then isolate final volume. (3) Use 120 kPa, 40 cm^3 and 160 kPa consistently. The larger final pressure should give a smaller final volume.

Pr 36 (p. 145) (1) The 0.30 mm reading is a length increase, not final rod length. (2) Convert it to metres, then divide by original length 0.50 m and temperature rise 40 K. (3) Report an inverse-kelvin coefficient. Its small magnitude is expected for a typical solid.

Pr 40 (p. 146) (1) Find heater power from 6 V and 2 A before using the 100 s duration. (2) Multiply power by duration to obtain input energy in joules. (3) Because all input melts 0.004 kg at the melting point, divide that energy by melted mass for latent heat. Keep the two answers separately labelled.

Pr 44 (p. 147) (1) Write the solid temperature drop and water temperature rise as separate positive differences. (2) Calculate water heat gain using 0.10 kg and its given specific heat; ignore the calorimeter only because the question says so. (3) Equate that heat to solid mass 0.20 kg times unknown specific heat times its cooling drop, then isolate specific heat.

Pr 48 (p. 149) (1) Use electrical energy as the numerator and thermal gain in calories as the denominator. (2) Divide 836.8 J by 200 cal and retain the unit J/cal. (3) For the loss question, keep input fixed and ask what happens to the ratio if the measured heat-gain denominator becomes too small.

Answers

Pr 2 (p. 133) No. It compares with a reference; repeated-reading spread or an uncertainty estimate is needed to discuss repeatability.

Pr 3 (p. 133) Percentage error = $100 \times 0.2/9.8 = 2.04$ percent. A consistent height-scale zero offset is one possible systematic error. Two marks for the calculation and two for a specific error with its consistency explained.

Pr 4 (p. 133) (a) Correct. Every reading carries the same bias, so averaging alone preserves it.

Pr 5 (p. 133) Uncertainty: $\pm 0.2\ \mathrm{m/s^2}$. Percentage error: about 2.04%. They answer different questions.

Pr 7 (p. 135) No. Count a return to the same position moving in the same direction as one full cycle.

Pr 8 (p. 135) T=30/15=2 s. g=$4\pi^2 \times 0.99/4$=9.77 $\mathrm{m/s^2}$. Two marks per calculation.

Pr 9 (p. 135) (b) Correct. Bob mass cancels in the small-angle simple-pendulum model.

Pr 11 (p. 137) 10 $\mathrm{m/s^2}$, using g = 2/0.20.

Pr 12 (p. 137) g = 2/0.20 = 10 $\mathrm{m/s^2}$. Four marks for axis identification, rearrangement, substitution and unit.

Pr 13 (p. 137) (c) Correct. Distance from rest is proportional to time squared.

Pr 15 (p. 138) 50 N/m: the force is 3 N and extension is 0.06 m.

Pr 16 (p. 138) Force = 3 N; extension = 0.06 m; k=3/0.06=50 N/m. Award marks for force, conversion, ratio and unit.

Pr 17 (p. 138) (a) Correct. Extension is the change from the unloaded length.

Pr 19 (p. 139) About 49.35 N/m, from $4\pi^2/0.80$.

Pr 20 (p. 139) Slope = $4\pi^2/k$, so k=$4\pi^2$/0.80=49.35 N/m. Credit the selected axes, rearrangement, numerical value and units.

Pr 21 (p. 139) (b) Correct. Stiffness appears inversely in this slope.

Pr 23 (p. 141) No. It has equal magnitude and opposite direction.

Pr 24 (p. 141) Resultant magnitude is 10 N at 53.1 degrees north of east. Equilibrant is 10 N at 233.1 degrees from positive x, or 53.1 degrees south of west. Two marks for magnitude and two for direction.

Pr 25 (p. 141) (c) Correct. Contact can hide an imbalance among the intended string forces.

Pr 27 (p. 142) 55°C: initial excess 60°C becomes 30°C, then add 25°C.

Pr 28 (p. 142) Initial excess = 60 degrees C. Halved excess = 30 degrees C. Add room temperature: 25+30=55 degrees C. Credit both differences and final interpretation.

Pr 29 (p. 142) (a) Correct. Its initial temperature excess is larger in the colder room.

Pr 31 (p. 143) Compression can change temperature; waiting helps the gas return to the intended common temperature.

Pr 32 (p. 143) V2 = 120 × 40/160 = 30 cubic centimetres. Credit controlled conditions, equation, rearrangement and unit.

Pr 33 (p. 143) (b) Correct. Wait for the specified thermal condition or use an appropriate different model.

Pr 35 (p. 145) The calculated expansion coefficient becomes 1000 × too large.

Pr 36 (p. 145) Convert expansion to 0.00030 m. Alpha=0.00030/(0.50 × 40)=0.000015 per K. Credit conversion, formula, substitution and result.

Pr 37 (p. 145) (c) Correct. The entered displacement is 1000 times larger than the true metre value.

Pr 39 (p. 146) No. It is dimensionless and tests energy accounting or an efficiency interpretation under the model.

Pr 40 (p. 146) Energy=6 × 2 × 100=1200 J. Latent heat=$\frac{1200}{0.004}$ = 300000 J/kg. Two marks per calculation; the no-loss and initial-phase-temperature assumptions matter.

Pr 41 (p. 146) (a) Correct. Pre-melting warming requires a separate energy term.

Pr 43 (p. 147) They have different starting temperatures, though they reach the same final equilibrium temperature.

Pr 44 (p. 147) Water gains 0.10 × 4200 × 10=4200 J. Solid loses 0.20 × cs × 50=10cs. Thus cs=420 J/(kg K). Two marks for each side and result.

Pr 45 (p. 147) (b) Correct. Ignoring the container undercounts energy lost by the solid, giving too small a specific heat.

Pr 47 (p. 149) 4.184 J/cal. If heat escapes and measured Q is too small, E/Q tends to be too high.

Pr 48 (p. 149) Factor=836.8/200=4.184 J/cal. Heat escaping reduces measured Q for the supplied E, tending to make the ratio too high. Two marks for each part.

Pr 49 (p. 149) (c) Correct. Like energy units cancel.

Mechanics and Heat Lab, Test A

30 marks. Suggested time 30 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. If y is force in N and x is extension in m, slope units are:

   (a) N/m
   (b) m/N
   (c) N × m
   (d) No units

2. Pendulum length should be measured to:

   (a) The top of the table
   (b) The lowest swing point
   (c) The bottom of the bob
   (d) The centre of the bob

3. For t^2 vertically versus h horizontally, g equals:

   (a) Slope squared
   (b) Slope/2
   (c) 2 times slope
   (d) 2/slope

4. The spring constant is the slope of:

   (a) Time versus temperature
   (b) Extension versus force
   (c) Force versus extension
   (d) Mass versus time

5. For fixed k, multiplying the oscillating mass by four makes the period:

   (a) Four times
   (b) Half
   (c) Twice
   (d) Unchanged

6. The equilibrant of a 7 N eastward resultant is:

   (a) 7 N east
   (b) 14 N west
   (c) Zero
   (d) 7 N west

Part II. Show your reasoning.

1. A measured g is 9.6 m/s^2 and reference g is 9.8 m/s^2. Calculate percentage error and name one possible systematic error.

2. A pendulum takes 30 s for 15 cycles at length 0.99 m. Calculate period and g.

3. The slope of t^2 against height is 0.20 square seconds per metre. Find g.

4. A 0.30 kg mass extends a spring by 6 cm. Use g=10 m/s^2. Find k.

5. A T-squared-versus-mass graph has slope 0.80 square seconds per kg. Find spring constant.

6. Two forces are 6 N east and 8 N north. Find equilibrant magnitude and direction.

Mechanics and Heat Lab, Test B

30 marks. Suggested time 30 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. In the simple cooling model, the cooling rate is proportional to:

   (a) Mass only
   (b) Zero
   (c) Temperature excess above surroundings
   (d) Absolute time only

2. Boyle law requires:

   (a) Fixed temperature and gas amount
   (b) Changing gas amount
   (c) Fixed pressure
   (d) Fixed volume

3. Two rods are made of the same material and have the same temperature rise. A rod twice as long has expansion:

   (a) Twice as large
   (b) Unchanged
   (c) Four times as large
   (d) Half as large

4. Electrical energy supplied by constant voltage and current for time t is:

   (a) V/I
   (b) IVt
   (c) I/t
   (d) Vt/I

5. In an ideal mixing experiment, the final equilibrium temperature lies:

   (a) Below both
   (b) Between the initial temperatures
   (c) Always at zero
   (d) Above both initial temperatures

6. To determine joules per calorie, divide electrical energy in joules by heat measured in:

   (a) Watts
   (b) Calories
   (c) Kelvin
   (d) Joules

Part II. Show your reasoning.

1. Room temperature is 25 degrees C. A body starts at 85 degrees C and its temperature excess halves in 5 minutes. What is its temperature then?

2. A trapped gas has volume 40 cubic centimetres at absolute pressure 120 kPa. Find its volume at 160 kPa at the same temperature.

3. A 0.50 m rod expands 0.30 mm after warming by 40 K. Find its expansion coefficient.

4. A heater at 6 V carries 2 A for 100 s. If all energy melts a sample already at its melting point and the melted mass is 0.004 kg, find supplied energy and latent heat.

5. A 0.20 kg solid cools from 90 to 40 degrees C and warms 0.10 kg water from 30 to 40 degrees C. Ignore calorimeter heat capacity and losses; use water c=4200 J/(kg K). Find the solid specific heat.

6. An electrical experiment supplies 836.8 J and records heat gain of 200 cal. Find the conversion factor. If unaccounted heat escapes, does E/Q measured tend to be too high or too low?

Mechanics and Heat Lab, Test A: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (a) Slope units are vertical units divided by horizontal units.

Written part. Percentage error = $100 \times 0.2/9.8 = 2.04$ percent. A consistent height-scale zero offset is one possible systematic error. Two marks for the calculation and two for a specific error with its consistency explained. (Section: p. 132.)

Question 2. (d) The ideal mass is represented at the bob centre.

Written part. T=30/15=2 s. g=$4\pi^2 \times 0.99/4$=9.77 m/s^2. Two marks per calculation. (Section: p. 134.)

Question 3. (d) Rearranging h = gt squared/2 gives $t^2 = (2/g)$h.

Written part. g = 2/0.20 = 10 m/s^2. Four marks for axis identification, rearrangement, substitution and unit. (Section: p. 136.)

Question 4. (c) Force = k times extension matches y = mx.

Written part. Force = 3 N; extension = 0.06 m; k=3/0.06=50 N/m. Award marks for force, conversion, ratio and unit. (Section: p. 137.)

Question 5. (c) Period is proportional to the square root of mass.

Written part. Slope = $4\pi^2$/k, so k=$4\pi^2$/0.80=49.35 N/m. Credit the selected axes, rearrangement, numerical value and units. (Section: p. 138.)

Question 6. (d) Equal magnitude and opposite direction produce zero net force.

Written part. Resultant magnitude is 10 N at 53.1 degrees north of east. Equilibrant is 10 N at 233.1 degrees from positive x, or 53.1 degrees south of west. Two marks for magnitude and two for direction. (Section: p. 140.)

Mechanics and Heat Lab, Test B: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (c) Newton cooling uses the difference between object and surrounding temperatures.

Written part. Initial excess = 60 degrees C. Halved excess = 30 degrees C. Add room temperature: 25+30=55 degrees C. Credit both differences and final interpretation. (Section: p. 141.)

Question 2. (a) Only pressure and volume are varied in the ideal Boyle experiment.

Written part. V2 = 120 × 40/160 = 30 cubic centimetres. Credit controlled conditions, equation, rearrangement and unit. (Section: p. 142.)

Question 3. (a) Expansion is proportional to original length.

Written part. Convert expansion to 0.00030 m. Alpha=0.00030/(0.50 × 40)=0.000015 per K. Credit conversion, formula, substitution and result. (Section: p. 144.)

Question 4. (b) Electrical power is VI; multiplying by time gives energy.

Written part. Energy=6 × 2 × 100=1200 J. Latent heat=$\frac{1200}{0.004}$ = 300000 J/kg. Two marks per calculation; the no-loss and initial-phase-temperature assumptions matter. (Section: p. 145.)

Question 5. (b) Energy flows from the hotter body to the cooler one until equilibrium.

Written part. Water gains 0.10 × 4200 × 10=4200 J. Solid loses 0.20 × cs × 50=10cs. Thus cs=420 J/(kg K). Two marks for each side and result. (Section: p. 146.)

Question 6. (b) The denominator determines the per-calorie unit.

Written part. Factor=836.8/200=4.184 J/cal. Heat escaping reduces measured Q for the supplied E, tending to make the ratio too high. Two marks for each part. (Section: p. 148.)

Facts at a glance

1. Earn marks from tables, graphs and uncertainty p. 132

$$m = \Delta y / \Delta x$$
$$\%\ \text{error} = 100 |x_{\text{measured}} - x_{\text{reference}}| / |x_{\text{reference}}|$$

2. Simple pendulum p. 134

$$T = t/N$$
$$T^2 = (4\pi^2/g)L$$
$$g = 4\pi^2/m \text{ for } T^2 \text{ versus } L$$

3. Free fall and axis choice p. 136

$$h = \tfrac{1}{2} g t^2;$$
$$m_{h \text{ vs } t^2} = g/2;$$
$$m_{t^2 \text{ vs } h} = 2/g$$

4. Static Hooke law p. 137

$$F = k\Delta L$$
$$F = mg$$
$$k = m_{F \text{ vs } \Delta L}$$

5. Dynamic spring and gravity p. 138

$$T^2 = 4\pi^2 m/k$$
$$T^2 = (4\pi^2/g)\Delta L$$

6. Force table and equilibrium p. 140

$$\sum F_x = 0$$
$$\sum F_y = 0$$
$$F_i = m_i g$$

7. Newton cooling law p. 141

$$T - T_r = (T_0 - T_r)e^{-kt}$$
$$\ln[(T - T_r)/(T_0 - T_r)] = -kt$$

8. Boyle law p. 142

$$P_1 V_1 = P_2 V_2$$
$$P \propto 1/V$$

9. Linear thermal expansion p. 144

$$\Delta L = \alpha L_0 \Delta T$$
$$\alpha = m_{\Delta L \text{ vs } \Delta T} / L_0$$

10. Specific heat, latent heat and Joule equivalent p. 145

$$m_s c_s (T_s - T_f) = (m_w c_w + C_{\text{cal}})(T_f - T_w)$$
$$Q = mL$$
$$E = IVt$$

11. Calorimetry: determine a solid specific heat p. 146

$$c_s = \frac{(m_w c_w + C_{\text{cal}})(T_f - T_w)}{m_s (T_s - T_f)}$$

12. Latent heat and electrical-to-thermal conversion p. 148

$$Q_{\text{ice}} = mL_f + mc_w (T_f - 0°\text{C})$$
$$J_{\text{eq}} = E_{\text{joules}} / Q_{\text{calories}}$$

Notation

Symbols are listed by section. The same letter may name different quantities in different sections; read the section's Fact.

1. Earn marks from tables, graphs and uncertainty

m	slope of the selected vertical variable against horizontal variable (vertical unit / horizontal unit)
x_{measured}	experimental estimate (same as reference)
$x_{\text{reference}}$	accepted comparison value (same as measurement)

2. Simple pendulum

L	pivot-to-bob-centre length (m)
t	time for N complete oscillations (s)
T	period of one oscillation (s)
g	gravitational acceleration (m/s^2)

3. Free fall and axis choice

h	vertical fall distance (m)
t	fall time (s)
g	gravitational acceleration (m/s^2)
m	graph slope (depends on axes)

4. Static Hooke law

F	applied load magnitude (N)
ΔL	loaded minus unloaded spring length (m)
k	spring stiffness (N/m)
m	total suspended mass (kg)

5. Dynamic spring and gravity

T	oscillation period (s)
m	oscillating mass in the stated model (kg)
k	spring stiffness (N/m)
ΔL	static equilibrium extension (m)
g	gravitational acceleration (m/s^2)

6. Force table and equilibrium

F_i	tension force from hanging mass in the ideal setup (N)
m_i	total hanging mass for a string (kg)
R	resultant force magnitude (N)

7. Newton cooling law

T	object temperature (°C or K consistently)
T_r	room temperature (same unit as T)
k	effective cooling constant (inverse time)
t	elapsed time (s or min consistently with k)

8. Boyle law

P	absolute gas pressure (Pa or consistent pressure unit)
V	gas volume (m^3 or consistent volume unit)

9. Linear thermal expansion

α	linear expansion coefficient (K^{-1})
L_0	initial rod length (m)
ΔL	length increase (m)
ΔT	temperature increase (K)

10. Specific heat, latent heat and Joule equivalent

Q	thermal energy transferred (J)
L	specific latent heat (J/kg)
I	heater current (A)
V	heater voltage (V)
t	heating duration (s)

11. Calorimetry: determine a solid specific heat

c_s	solid specific heat capacity (J/(kg K))
m_s, m_w	solid and water masses (kg)
C_{cal}	calorimeter heat capacity (J/K)
T_f	final equilibrium temperature (°C or K)

12. Latent heat and electrical-to-thermal conversion

m	ice mass (kg)
L_f	specific latent heat of fusion (J/kg)
c_w	liquid water specific heat (J/(kg K))
J_{eq}	joules per calorie conversion estimate (J/cal)

Sources

All distinct experiments in the folder: pendulum, free fall, static and dynamic springs, force table, cooling, Boyle law, expansion, specific heat, latent heat and Joule equivalent. Duplicate versions are consolidated.

The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. Check lecturer-specific derivations and conventions. The practice tests are original study material, not predictions of an examination.

Instruments and Lab Safety

SCI 101 lecture chapters 1-3: accident prevention, chemical hazards, laboratory techniques, emergency principles and biosafety. This is exam study material; real laboratory work follows the current instructor procedure and chemical-specific safety information.

1 Hazard, risk and prevention

Aims. Distinguish a hazard from the risk of harm in a particular situation; Choose a control that interrupts the actual exposure route.
Before this section: Hazard means potential for harm; exposure means contact with the harmful agent.

A hazard is a source of possible harm. Risk depends on exposure and the chance and severity of harm. Accident prevention is shared: prepare, follow approved procedures and report unsafe conditions. A familiar chemical or routine experiment can still present hazards.

Idea 1. A substance hazard is its potential to cause harm; the practical risk depends on how it is used and who can be exposed. Identifying the chemical is therefore only the start. A useful assessment links the harmful property to an activity, an exposure route and a suitable control.

Fact 1.

Good answer: hazard → exposure route → control

Why it holds. The same hazard can create different risks under different exposure conditions. A control is relevant only if it reduces the source, exposure likelihood or harmful consequence.

Method 1.

1. Name the harmful property and the activity creating exposure.
2. Identify who could be exposed, by which route and under what conditions.
3. Select a control that reduces the source or interrupts the route.
4. Explain the remaining need for the approved procedure and required personal protection.

Recognising it. For an unfamiliar reagent, preparation means identifying hazards and the approved method before beginning work.

Pr 1. A volatile harmful liquid on an open bench creates an inhalation hazard.

Solution. The example liquid is volatile and harmful, so vapour can leave an open container. A person nearby may inhale that vapour; this connects the chemical hazard to an exposure route. An approved fume-hood procedure controls vapour at its source when

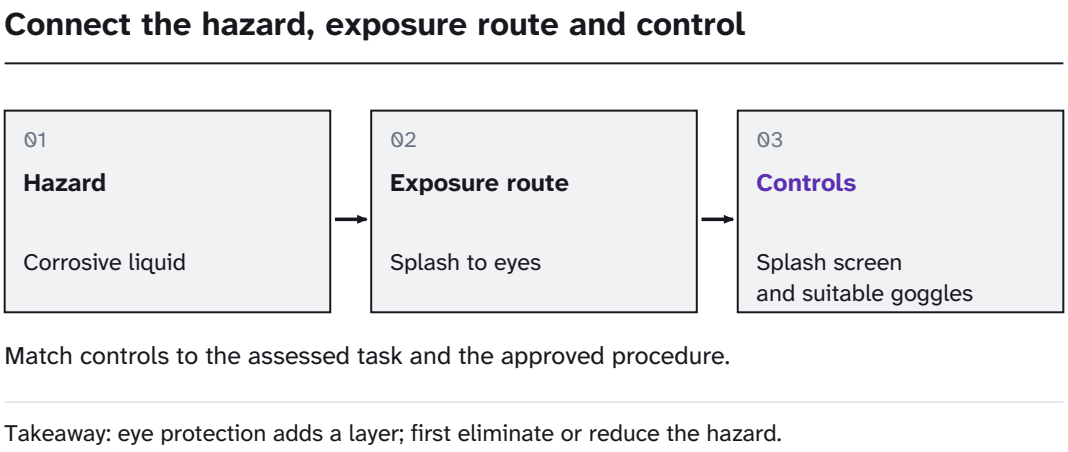


Figure 6. Figure 06. A useful safety answer connects a specific hazard to a route of exposure and suitable controls. A splash screen and appropriate goggles can limit splash exposure; they do not replace reducing or removing the hazard where possible.

suitable for that substance and operation. Goggles protect eyes from appropriate hazards but do not remove vapour from inhaled air.

Common error. Do not write only be careful; name a concrete control.

In the examination. For a scenario answer write a linked chain: hazard, exposure route, suitable control and reason.

Pr 2. Why is naming only a hazardous chemical insufficient to describe a risk scenario?

Pr 3. A student uses an unfamiliar reagent without reading its safety information. Identify the problem and two appropriate preparatory actions.

Pr 4. The same harmful volatile liquid is sealed in approved storage or poured openly during a task. Why can risk differ?

(a) The opportunity for exposure differs

(b) Its chemical identity necessarily changes

(c) A sealed container makes the liquid chemically harmless

2 Personal protective equipment and contamination

Aims. Trace how contaminated gloves transfer material to clean surfaces; Select PPE for the task and remove it without spreading contamination.

Before this section: Contamination transfer; clean and contaminated work zones; task-specific protection.

Splash goggles protect against chemical splashes; ordinary spectacles do not provide equivalent coverage. Gloves must be compatible with the chemical and inspected. No glove protects indefinitely. Contaminated gloves can carry chemicals to phones, door handles and notebooks.

Idea 2. PPE is selected for the route and type of exposure, not simply worn as a general safety symbol. Ordinary prescription glasses leave openings around the eyes, while a coat mainly protects covered skin and clothing. Protection for a splash task must match the actual eye and body hazards specified by the method.

Fact 2.

Select PPE for the specific hazard and task

Why it holds. A barrier cannot protect an area it does not cover adequately. Different materials and tasks can require different PPE, so everyday glasses are not automatically splash-rated eye protection.

Method 2.

1. Identify the contaminant and which surfaces or body areas need protection.
2. Use PPE specified for the material and task rather than assuming one glove type protects against everything.
3. Keep contaminated gloves away from phones, door handles and other clean shared items.
4. Remove and dispose of PPE according to the approved procedure, then perform required hand hygiene.

Recognising it. When asked whether clothing is sufficient, identify the exposed area and required protection for the particular task.

Pr 5. A student touches a phone with contaminated gloves.

Solution. The student glove is already contaminated before touching the phone. Contact transfers material from glove to phone, creating a new contaminated surface. Later bare-hand contact can move that material to skin, food or the face. Removing gloves correctly before touching clean items breaks that transfer chain; the phone may still need the approved decontamination procedure.

Common error. More layers of unsuitable gloves do not establish chemical compatibility.

In the examination. Explain what is transferred and to where; simply saying wear gloves does not solve a contamination scenario.

Pr 6. Do gloves make it acceptable to touch your face during laboratory work?

Pr 7. Explain why ordinary prescription glasses and a lab coat alone are insufficient for a splash-risk task.

Pr 8. Why should glove material be selected for the chemical and task?

(a) Thicker-looking gloves always remove all exposure routes

(b) Different glove materials resist different chemicals differently

(c) Every glove blocks every chemical equally

3 Exposure routes and toxic effects

Aims. Identify inhalation, ingestion, skin or eye contact and injection exposures; Distinguish route of exposure from the timing of a toxic effect.

Before this section: Contact with an agent can occur by several routes; visible vapour is not required.

Chemicals can enter by inhalation, ingestion, skin or eye absorption, or injection through contaminated sharps. Acute exposure is short-term; chronic exposure is repeated or long-term. Effects can be local or systemic, immediate or delayed, reversible or irreversible.

Idea 3. Exposure route describes how material reaches the body, not whether the person notices it. Vapour can be inhaled, while a contaminated sharp can introduce material through skin. Odour is a sensory property with limits, so absence of smell cannot establish absence of a hazardous exposure.

Fact 3.

Dose, duration, route and substance all influence harm

Why it holds. Different routes require different preventive controls. Some hazards have little warning odour, and smell may not reliably indicate a safe concentration.

Method 3.

1. Read the activity and locate the point where the agent could enter or contact the body.
2. Name the route using the specific evidence in the scenario.
3. Distinguish a rapid acute effect from an effect associated with longer or repeated exposure.
4. Match prevention to the identified route and follow the material-specific procedure.

Recognising it. Classify the physical entry mechanism first, then explain why sensory impressions cannot replace safety information.

Pr 9. Eating in a laboratory can lead to ingestion of contamination.

Solution. Eating in the laboratory can move contamination from hands or surfaces to food. Swallowing that material is ingestion, even when the contaminant is not visible. A contaminated needle can breach the skin and introduce material directly: this is injection exposure. The absence of visible vapour does not remove either risk because neither route depends on inhalation.

Common error. A lack of smell does not prove safety.

In the examination. Name both the route and the evidence, for example ingestion because contaminated food is swallowed.

Pr 10. A droplet splashes into an eye. Is the route inhalation?

Pr 11. Classify inhaling vapour and a contaminated needle puncture by exposure route. Explain why no smell does not prove no hazard.

Pr 12. A hazardous substance has no noticeable odour during use. What can be concluded?

(a) There is no inhalation hazard

(b) PPE and handling controls are unnecessary

(c) Odour alone cannot establish safety

4 Labels, SDS and chemical properties

Aims. Use a label and SDS to connect chemical properties with controls; Distinguish corrosive and oxidizing hazards in an explanation.
Before this section: Chemical identity; label symbols; an SDS gives substance-specific hazard and handling information.

Read the identity and hazard information before use. Safety data sheets describe hazards, handling, storage, exposure controls, first aid and other substance-specific information. Flammable, oxidizing, corrosive and toxic refer to different hazards and can apply to one substance simultaneously.

Idea 4. One chemical can have several hazards that require separate reasoning. Flammability concerns ignition and combustion, whereas corrosivity concerns damaging contact. An answer that names one generic precaution may miss the second hazard, even when it sounds broadly cautious.

Fact 4.

$$\text{Identity} + \text{hazard} + \text{controls} + \text{incompatibilities}$$

Why it holds. An ignition control addresses a fire pathway but does not by itself prevent corrosive contact. Splash/contact controls address exposed tissues but do not make a flammable material nonflammable.

Method 4.

1. Confirm the chemical identity from its label before selecting a procedure.
2. Read relevant SDS information for hazards, incompatibilities and required handling controls.
3. Explain how the named property causes harm in the scenario.
4. Use the approved storage or handling arrangement rather than inferring compatibility from appearance.

Recognising it. When a label carries more than one hazard, give a distinct risk-and-control link for each.

Pr 13. A corrosive liquid can damage tissue, while an oxidizer may intensify fire even when it is not itself a fuel.

Solution. A corrosive liquid can damage living tissue or certain materials by contact. An oxidizer can support or intensify combustion even when it is not itself the fuel. The mechanisms differ, so the same general statement handle carefully does not explain either control. A good answer links splash/contact controls to corrosivity and incompatibility or fire-prevention controls to oxidizing behaviour as specified by the SDS.

Common error. Do not infer that all colourless liquids are harmless or that one symbol captures every hazard.

In the examination. Pair each property with its consequence; listing pictogram names alone gives less evidence of understanding.

Pr 14. Does a chemical being nonflammable prove it cannot intensify a fire?

Pr 15. A bottle has both flammable and corrosive hazards. State two distinct risks and a control principle for each.

Pr 16. A bottle is both flammable and corrosive. Which statement is best?

(a) Both fire and damaging-contact risks need suitable controls

(b) Removing ignition sources alone resolves every hazard

(c) Eye protection alone makes ignition impossible

5 Glassware, measurement and safe apparatus

Aims. Select apparatus according to measurement purpose and safe operating limits; Explain why volumetric preparation and heating require different vessels.
Before this section: Volume; calibration mark; damaged glassware can fail during use.

Inspect glassware for chips and cracks and apparatus for damaged wiring before use. A beaker is useful for holding and mixing; a graduated cylinder gives more useful volume estimates; a volumetric pipette or flask is designed for a specified accurate volume. Read liquid levels at eye height using the appropriate meniscus.

Idea 5. Accurate liquid handling depends on both choosing the right instrument and using it as calibrated. A fixed-volume volumetric pipette is intended to deliver one specified volume under its stated conditions. Safe pipetting and proper reading matter as much as the printed volume.

Fact 5.

Choose the instrument for required precision, not convenience

Why it holds. A vessel marked for approximate volume may not deliver the required fixed volume accurately. The calibration method determines how the instrument should be filled, read and drained.

Method 5.

1. State whether the task requires approximate volume, accurate delivery or preparation to a final volume.
2. Choose the apparatus designed for that purpose and check its condition.
3. Follow the specified filling, meniscus and operating procedure.
4. Before heating, verify that the vessel and setup are approved for the process.

Recognising it. For an accurate fixed transfer, distinguish a volumetric pipette from a flask that contains a final prepared volume.

Pr 17. A volumetric flask prepares a solution to its marked final volume.

Solution. The volumetric flask is calibrated to contain a stated final volume at its reference conditions. The solution is brought to the mark as part of an approved preparation procedure. Its main role is volume accuracy, not serving as a directly heated reaction vessel. For heating a reaction, use the vessel and support specified by the laboratory method rather than substituting the flask.

Common error. Extra decimal places do not make an imprecise instrument accurate.

In the examination. In apparatus questions state the purpose and limitation together: prepares final volume, unsuitable for directly heating the reaction.

Pr 18. Does having a calibration mark mean a flask is suitable for direct heating?

Pr 19. Name a suitable device to transfer an accurate fixed 10 mL volume and explain two good practices.

Pr 20. A volumetric flask and a volumetric pipette both show 10 mL. Why are they not automatically interchangeable?

(a) The same printed number guarantees identical delivery

(b) They are designed for different volume-handling purposes

(c) Only a flask ever has calibration

6 Hoods, heating, cooling and pressure

Aims. Explain why heating a closed unsuitable vessel can create a pressure hazard; Recognize when an apparatus setup requires instructor review.

Before this section: Heating can increase vapour production and gas pressure; equipment has specific operating limits.

A fume hood controls airborne chemical exposure when used correctly. It is not a storage cabinet or a waste-evaporation device. Heating and cooling systems can introduce burns, pressure changes and material failure. Approved setups account for both temperature and pressure.

Idea 6. A heated closed vessel can store pressure as well as thermal energy. A volatile liquid supplies vapour, and heating can increase that vapour pressure. If the vessel or closure is unsuitable, failure can release hot hazardous contents suddenly; a hood does not turn it into pressure-rated equipment.

Fact 6.

Control vapour, ignition, pressure and thermal contact separately

Why it holds. Heating an enclosed volatile liquid can raise internal pressure. Container failure can eject hot liquid or vapour, adding exposure and burn consequences.

Method 6.

1. Identify heat source, volatile materials and any sealed volume.
2. Check the approved procedure and apparatus pressure and temperature limits.
3. Use the specified containment, heating and ventilation arrangements.
4. If the setup differs or becomes unsafe, stop and alert the responsible instructor instead of modifying it ad hoc.

Recognising it. A tightly sealed volatile-liquid heating setup requires checking the approved apparatus rather than improvising a modification.

Pr 21. A volatile solvent warmed in an unsuitable sealed container can build pressure.

Solution. Warming the volatile solvent increases vapour formation. In an unsuitable sealed container, expanding gas and vapour can increase internal pressure. The container or closure may fail because it was not designed for that operation. The example therefore calls for stopping and consulting the approved setup; improvising a vent is not a validated correction.

Common error. A hood does not remove every fire, pressure or contact hazard.

In the examination. Explain the mechanism, pressure buildup from heating, before giving the required stop-and-consult response.

Pr 22. Does placing an unsuitable sealed heated vessel in a fume hood solve its pressure hazard?

Pr 23. Give two reasons why heating a tightly sealed vessel containing a volatile liquid is hazardous.

Pr 24. Does placing an unsuitable sealed heated vessel in a fume hood make the pressure setup acceptable?

(a) Yes; all hoods prevent pressure buildup

(b) Yes; a hood removes liquid volatility

(c) No; ventilation does not make a vessel pressure-rated

7 Waste, housekeeping and emergency priorities

Aims. Use identification and approved procedures to choose a waste or spill response; State safe priorities when a spill material is unknown.

Before this section: Chemical incompatibility; designated waste streams; laboratory emergency instructions.

Waste must follow the laboratory segregation and disposal procedure. Broken glass, sharps and chemical waste are different streams. Good housekeeping keeps exits and emergency equipment accessible. In an incident, alert others and follow the established response; do not improvise a hazardous cleanup.

Idea 7. Emergency equipment is useful only when people can reach and operate it promptly. A storage box can create risk without being chemically hazardous itself because it delays the response to another event. Housekeeping therefore protects the reliability of the laboratory safety system.

Fact 7.

$$\text{Recognize} \rightarrow \text{alert} \rightarrow \text{follow trained response}$$

Why it holds. A blocked eyewash can delay response to an eye exposure. Keeping the access route clear reduces that delay before an emergency occurs.

Method 7.

1. Identify the material only if that can be done safely from available records or labels.
2. Keep others clear and alert the instructor or responsible responder.
3. Use the laboratory response plan for the identified hazard and your training level.
4. For routine waste, use the designated compatible container and labelling procedure.

Recognising it. An obstructed emergency station is a prevention and access problem, even when no spill or exposure is currently happening.

Pr 25. An unknown spill is not an invitation to mix in a neutralizer.

Solution. An unknown spill has uncertain properties, so an assumed neutralizer may be incompatible. Adding another chemical could cause heat, gas formation or another hazardous reaction. Alerting the instructor and keeping people clear limits exposure while the material and response are assessed. Cleanup then follows the approved spill procedure for the identified material and trained personnel.

Common error. Do not recommend that an untrained student fight every fire or clean every spill.

In the examination. In an emergency scenario prioritize alerting and exposure prevention; do not invent a chemical cleanup recipe.

Pr 26. Should an unknown laboratory liquid automatically be washed into a sink?

Pr 27. A box blocks access to the eyewash station. Explain the risk and an appropriate preventive action.

Pr 28. Why can a harmless empty box near emergency equipment still create risk?

(a) It can delay access when the equipment is needed

(b) It must release toxic vapour

(c) Emergency equipment works only when boxes are present

8 Biosafety and the control hierarchy

Aims. Classify controls within the hazard-control hierarchy; Distinguish engineering containment from training, work practices and PPE.
Before this section: A biological hazard may be spread through aerosols, contact or sharps depending on the activity.

Biosafety aims to prevent unintended exposure to biological agents and their release. The general control hierarchy prefers removing a hazard, then using a safer substitute, then engineering controls, administrative controls and work practices, and PPE. The suitable controls depend on the task: removing or replacing the biological material may not be possible. Training and procedures are administrative controls; suitable containment equipment is an engineering control. Safe work practices include hand hygiene and avoiding aerosol generation.

Idea 8. Controls work at different places in the hazard pathway. Equipment containment can isolate the source, procedures guide behaviour, and PPE forms a barrier worn by a person. Classifying a control means explaining its mechanism; it does not mean one category makes every other category unnecessary.

Fact 8.

Eliminate → substitute → engineering → administrative/work practices → PPE

Why it holds. Engineering controls act through equipment or physical containment. Training and handwashing procedures guide work practices, while goggles are worn personal protection.

Method 8.

1. Identify the biological material and activity-specific exposure route.
2. Consider elimination or substitution where feasible before downstream controls.
3. Classify equipment containment as engineering, training and procedures as administrative or work-practice controls, and worn barriers as PPE.
4. Explain how the selected combination addresses the task and follows institutional biosafety procedures.

Recognising it. When asked to classify controls, describe what each does before assigning its hierarchy category.

Pr 29. Training is administrative.

Solution. Training changes how people work, so it is an administrative control. A suitable biological safety cabinet provides engineering containment for its approved uses. Gloves form a worn barrier and are PPE. A safe handling technique is a work practice, generally grouped with administrative controls; the categories describe different protective mechanisms.

Common error. A device with airflow is not automatically suitable for every chemical or biological hazard.

In the examination. Do not label every cabinet or hood as interchangeable containment; state suitable equipment for the defined biological task.

Pr 30. What is the general hierarchy from source-focused to last-line protection?

Pr 31. Classify goggles, safety training, a suitable containment cabinet and correct handwashing.

Pr 32. A suitable containment cabinet and goggles are used together. Why can both be needed?

(a) PPE always eliminates the source hazard

(b) They interrupt exposure through different protective mechanisms

(c) They belong to exactly the same control category

Fact 9. The control hierarchy in the correct order. The hierarchy ranks ways to control a hazard. A list of control types is not automatically in the preferred order.

1. Eliminate: remove the hazard.
2. Substitute: use a safer alternative after checking its risks.
3. Engineering controls: separate people from the hazard using suitable equipment or containment.
4. Administrative controls and work practices: use training, procedures and safe methods.
5. PPE: use suitable worn protection. It adds protection but depends on correct selection and use.
6. In a laboratory, eliminating or replacing the material may not be possible. The assessed task determines the suitable combination of controls.

Pr 33. Classify a suitable containment cabinet, safety training, gloves and correct handwashing.

Hints

Use one hint at a time, then return to the problem.

Pr 3 (p. 166) (1) Identify what the student has not established before starting: hazards and suitable controls. (2) Choose one preparatory action that obtains reagent-specific safety information. (3) Choose another that confirms the approved procedure or required supervision. Explain how each prevents an uninformed exposure.

Pr 7 (p. 168) (1) List which areas a lab coat covers and which areas remain exposed. (2) Consider whether ordinary prescription glasses provide the task-required splash protection around the eyes. (3) Explain the coverage gaps and identify the need for approved splash-rated protection and any other task-specific PPE, following the laboratory method.

Pr 11 (p. 169) (1) For each event, identify how the material crosses into or contacts the body. (2) Airborne vapour enters through breathing; a contaminated puncture breaches skin directly. Name the route for each. (3) For the smell explanation, state that detectability and hazardous exposure are different questions; refer to substance-specific safety information.

Pr 15 (p. 170) (1) Separate the two label properties before suggesting controls. (2) For flammability, explain an ignition/fire risk and a suitable approved ignition-control principle. (3) For corrosivity, explain tissue-damage risk and an approved exposure-prevention principle. Do not treat one control as covering both mechanisms.

Pr 19 (p. 172) (1) The task asks to transfer a fixed accurate volume, not prepare a solution to a final flask mark. (2) Select an appropriate fixed-volume volumetric pipette and approved pipette aid. (3) For good practices, explain correct meniscus reading and use according to its calibration/draining instructions. Mouth pipetting is not acceptable.

Pr 23 (p. 173) (1) Trace how heating affects vapour inside a tightly sealed space. (2) Connect the resulting pressure to possible vessel or closure failure. (3) State a second consequence involving hot released contents, such as burns or hazardous exposure, and relate it to the need for the approved setup.

Pr 27 (p. 174) (1) Identify the purpose of an eyewash and when rapid access matters. (2) Explain how a box could delay a person trying to respond to an eye exposure. (3) Propose promptly clearing and maintaining the route through the responsible housekeeping procedure. Link the action to access rather than claiming the box itself is corrosive.

Pr 31 (p. 175) (1) Sort the examples into equipment, behaviour/procedure and worn protection. (2) Classify the suitable cabinet as engineering containment and goggles as PPE. (3) Place training under administrative controls and correct handwashing under work practices, commonly grouped with administrative controls. Explain the basis rather than just listing labels.

Pr 33 (p. 176) (1) The cabinet is an engineering control. (2) Training is administrative; gloves are PPE. (3) Handwashing is a safe work practice, often grouped with administrative controls.

Answers

Pr 2 (p. 166) Risk also depends on how exposure can occur, how likely it is and how serious its consequences could be.

Pr 3 (p. 166) Problem: unassessed chemical hazards. Actions: consult the reagent SDS and approved procedure; clarify required controls and waste handling with the instructor before use. One mark for hazard, one for exposure reasoning and one per specific action.

Pr 4 (p. 166) (a) Correct. Handling conditions can change exposure while the underlying hazardous property remains.

Pr 6 (p. 168) No. Gloves can carry contamination directly to the face.

Pr 7 (p. 168) Prescription glasses do not seal against splashes; appropriate splash goggles are needed. A coat does not protect hands, so compatible gloves may be required by the assessed procedure. Award one mark per protection gap and suitable control.

Pr 8 (p. 168) (b) Correct. Compatibility and performance depend on material, chemical and conditions.

Pr 10 (p. 169) No. It is eye or mucous-membrane contact.

Pr 11 (p. 169) Vapour: inhalation. Needle: injection. Some harmful substances are odourless, or odour detection is unreliable. One mark per route and two for the explanation.

Pr 12 (p. 169) (c) Correct. Safety depends on identified properties and exposure controls, not smell alone.

Pr 14 (p. 170) No. An oxidizer may intensify combustion without being a fuel.

Pr 15 (p. 170) Fire risk: control ignition sources using the approved setup. Tissue damage from splashes: prevent contact using appropriate containment and compatible protection. One mark per risk and per linked control.

Pr 16 (p. 170) (a) Correct. The properties create separate pathways requiring an appropriate combination of controls.

Pr 18 (p. 172) No. Calibration and suitability for heating are different design requirements.

Pr 19 (p. 172) Use an appropriate 10 mL volumetric pipette with a pipette aid. Read the meniscus at eye level and follow the calibrated delivery procedure. Credit device, aid and two measurement practices.

Pr 20 (p. 172) (b) Correct. A flask contains a prepared volume; a pipette is designed to transfer a calibrated volume according to its type.

Pr 22 (p. 173) No. A hood does not make a vessel pressure-rated or validate an unsafe heating setup.

Pr 23 (p. 173) Vapour generation can raise pressure, and thermal expansion can further stress the vessel. Rupture can release hot liquid or vapour. Credit two mechanisms and their resulting consequences.

Pr 24 (p. 173) (c) Correct. Engineering ventilation and pressure suitability are different requirements.

Pr 26 (p. 174) No. Its identity, compatibility and approved disposal route must be established.

Pr 27 (p. 174) It can delay emergency irrigation after exposure. Keep access clear and report the obstruction so it is corrected promptly under laboratory procedures. Credit emergency delay, possible worsening of injury, removal/reporting and prevention.

Pr 28 (p. 174) (a) Correct. Obstruction creates a response-time hazard even if the box material is harmless.

Pr 30 (p. 175) Elimination, substitution, engineering controls, administrative controls and work practices, then PPE.

Pr 31 (p. 175) Goggles: PPE. Training: administrative control. Suitable containment cabinet: engineering control. Correct handwashing: safe work practice (often grouped with administrative controls). One mark each.

Pr 32 (p. 175) (b) Correct. Containment and worn barriers can form complementary protections specified for the task.

Pr 33 (p. 176) Engineering; administrative; PPE; work practice, respectively. Engineering controls come before administrative controls and PPE in the general hierarchy.

Instruments and Lab Safety, Test A

20 marks. Suggested time 20 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. Which best describes risk?

   (a) The mass of any bottle
   (b) The same as smell
   (c) The label colour alone
   (d) The chance and severity of harm under the exposure conditions

2. Which is appropriate before touching a clean door handle?

   (a) Keep contaminated gloves on
   (b) Wipe gloves on clothing
   (c) Remove contaminated gloves correctly
   (d) Smell the gloves

3. Eating with contaminated hands is mainly which exposure route?

   (a) Inhalation
   (b) Injection
   (c) Noise exposure
   (d) Ingestion

4. An oxidizer is especially associated with:

   (a) Intensifying combustion
   (b) Preventing all fires
   (c) Measuring temperature
   (d) Making all chemicals inert

Part II. Show your reasoning.

1. A student uses an unfamiliar reagent without reading its safety information. Identify the problem and two appropriate preparatory actions.

2. Explain why ordinary prescription glasses and a lab coat alone are insufficient for a splash-risk task.

3. Classify inhaling vapour and a contaminated needle puncture by exposure route. Explain why no smell does not prove no hazard.

4. A bottle has both flammable and corrosive hazards. State two distinct risks and a control principle for each.

Instruments and Lab Safety, Test B

20 marks. Suggested time 20 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. Which is designed to prepare one accurately specified solution volume?

   (a) A beaker
   (b) A watch glass
   (c) A volumetric flask
   (d) A test-tube rack

2. A fume hood should be used as:

   (a) Permanent chemical storage
   (b) A controlled work enclosure for appropriate vapour hazards
   (c) A way to evaporate all waste
   (d) A food cupboard

3. An unknown chemical spill should first prompt a student to:

   (a) Alert the instructor and keep people clear
   (b) Mix it with another chemical
   (c) Ignore it
   (d) Taste it

4. Which is an administrative control?

   (a) Gloves
   (b) Training and written procedures
   (c) A sealed rotor
   (d) Splash goggles

Part II. Show your reasoning.

1. Name a suitable device to transfer an accurate fixed 10 mL volume and explain two good practices.

2. Give two reasons why heating a tightly sealed vessel containing a volatile liquid is hazardous.

3. A box blocks access to the eyewash station. Explain the risk and an appropriate preventive action.

4. Classify goggles, safety training, a suitable containment cabinet and correct handwashing.

Instruments and Lab Safety, Test A: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (d) Risk depends on both the hazard and how people are exposed.

Written part. Problem: unassessed chemical hazards. Actions: consult the reagent SDS and approved procedure; clarify required controls and waste handling with the instructor before use. One mark for hazard, one for exposure reasoning and one per specific action. (Section: p. 165.)

Question 2. (c) This prevents spreading contamination outside the work area.

Written part. Prescription glasses do not seal against splashes; appropriate splash goggles are needed. A coat does not protect hands, so compatible gloves may be required by the assessed procedure. Award one mark per protection gap and suitable control. (Section: p. 167.)

Question 3. (d) Material can pass from the hands to food and the digestive system.

Written part. Vapour: inhalation. Needle: injection. Some harmful substances are odourless, or odour detection is unreliable. One mark per route and two for the explanation. (Section: p. 168.)

Question 4. (a) Oxidizers can support or intensify combustion.

Written part. Fire risk: control ignition sources using the approved setup. Tissue damage from splashes: prevent contact using appropriate containment and compatible protection. One mark per risk and per linked control. (Section: p. 169.)

Instruments and Lab Safety, Test B: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (c) The calibration mark defines the specified contained volume under its calibration conditions.

Written part. Use an appropriate 10 mL volumetric pipette with a pipette aid. Read the meniscus at eye level and follow the calibrated delivery procedure. Credit device, aid and two measurement practices. (Section: p. 171.)

Question 2. (b) It is an engineering control for relevant airborne exposures.

Written part. Vapour generation can raise pressure, and thermal expansion can further stress the vessel. Rupture can release hot liquid or vapour. Credit two mechanisms and their resulting consequences. (Section: p. 172.)

Question 3. (a) Unknown hazards require assessment and the established response.

Written part. It can delay emergency irrigation after exposure. Keep access clear and report the obstruction so it is corrected promptly under laboratory procedures. Credit emergency delay, possible worsening of injury, removal/reporting and prevention. (Section: p. 173.)

Question 4. (b) Administrative controls organize how work is authorized and performed.

Written part. Goggles: PPE. Training: administrative control. Suitable containment cabinet: engineering control. Correct handwashing: safe work practice (often grouped with administrative controls). One mark each. (Section: p. 174.)

Facts at a glance

1. Hazard, risk and prevention p. 165

Good answer: hazard → exposure route → control

2. Personal protective equipment and contamination p. 167

Select PPE for the specific hazard and task

3. Exposure routes and toxic effects p. 168

Dose, duration, route and substance all influence harm

4. Labels, SDS and chemical properties p. 169

Identity + hazard + controls + incompatibilities

5. Glassware, measurement and safe apparatus p. 171

Choose the instrument for required precision, not convenience

6. Hoods, heating, cooling and pressure p. 172

Control vapour, ignition, pressure and thermal contact separately

7. Waste, housekeeping and emergency priorities p. 173

Recognize → alert → follow trained response

8. Biosafety and the control hierarchy p. 174

Eliminate → substitute → engineering → administrative/work practices → PPE

Notation

Symbols are listed by section. The same letter may name different quantities in different sections; read the section's Fact.

Sources

SCI 101 lecture chapters 1-3: accident prevention, chemical hazards, laboratory techniques, emergency principles and biosafety. This is exam study material; real laboratory work follows the current instructor procedure and chemical-specific safety information.

The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. Check lecturer-specific derivations and conventions. The practice tests are original study material, not predictions of an examination.

Humans and the Environment

ENV101 elective: ecosystems, relationships among organisms, pollution, ecological footprint, waste management and sustainable development. The Arabic source lectures are explained here in English.

1 Environment and ecosystem components

Aims. Classify living and nonliving ecosystem components; Explain one interaction linking an organism with an abiotic factor.
Before this section: An ecosystem includes organisms and the physical conditions around them.

An ecosystem combines organisms with the physical and chemical surroundings with which they interact. Biotic factors are living components; abiotic factors include light, temperature, water and minerals. A population is one species in an area; a community includes interacting populations.

Idea 1. An ecosystem includes a living community and the conditions through which it interacts. Sunlight and minerals are not alive, but they affect what organisms can do. An abiotic factor is therefore not an unimportant factor; it is simply a nonliving part of the system.

Fact 1.

$$\text{Organism} \rightarrow \text{population} \rightarrow \text{community} \rightarrow \text{ecosystem}$$

Why it holds. Living status determines the basic biotic/abiotic classification. Nonliving conditions can still strongly affect organism growth, survival and interactions.

Method 1.

1. Name the ecosystem and define the area being considered.
2. Sort its organisms as biotic and its physical or chemical conditions as abiotic.
3. Choose one organism and explain how a named abiotic factor affects it.
4. Use a causal sentence rather than an unconnected list of examples.

Recognising it. For classification, ask whether the item is an organism or a physical/chemical environmental factor.

Pr 1. Fish, algae and bacteria are biotic components of a pond.

Solution. Fish, algae and bacteria are organisms, so they are biotic pond components. Water temperature and dissolved oxygen are abiotic conditions even though living things can affect them. Fish depend on sufficient dissolved oxygen for respiration. The example therefore shows both a classification and an interaction between the living community and its physical environment.

Common error. A habitat is where an organism lives; an ecosystem includes the interacting system.

In the examination. For explain questions add an interaction after classifying: name the factor, the organism and the effect.

Pr 2. Are bacteria abiotic because they are microscopic?

Pr 3. Classify sunlight, grass, soil mineral content and a rabbit as biotic or abiotic.

Pr 4. Soil mineral content affects plant growth. Why is it still abiotic?

(a) It is a nonliving chemical condition

(b) Anything affecting plants is biotic

(c) All material found in soil is living

2 Energy flow, food chains and decomposers

Aims. Assign producer and consumer roles within a stated food chain; Distinguish energy flow from nutrient cycling.
Before this section: Producers capture energy; consumers obtain energy by feeding; decomposers break down organic material.

Producers use light or chemical energy to make organic material from inorganic substances. Consumers obtain organic material by feeding. Decomposers break down dead material and help recycle nutrients. Energy passes through feeding levels and much becomes dispersed heat. Nutrients cycle, but usable energy is not recycled in the same way.

Idea 2. Food-chain energy is a flow through successive roles. Only some energy at one trophic level becomes available to the next, because organisms use energy and not all material is eaten or assimilated. A specified transfer percentage is a model instruction for the calculation, not a universal ecological constant.

Fact 2.

$$\text{Producer} \rightarrow \text{primary consumer} \rightarrow \text{secondary consumer}$$

Why it holds. A percentage must be applied to the energy available at the immediately preceding level. Repeating a fractional transfer reduces energy successively rather than subtracting the same original amount each time.

Method 2.

1. Start the chain with its producer and draw arrows in the direction of food-energy transfer.
2. Assign each consumer level from what that organism eats in the stated chain.
3. Explain that energy is used and dissipated at each transfer.

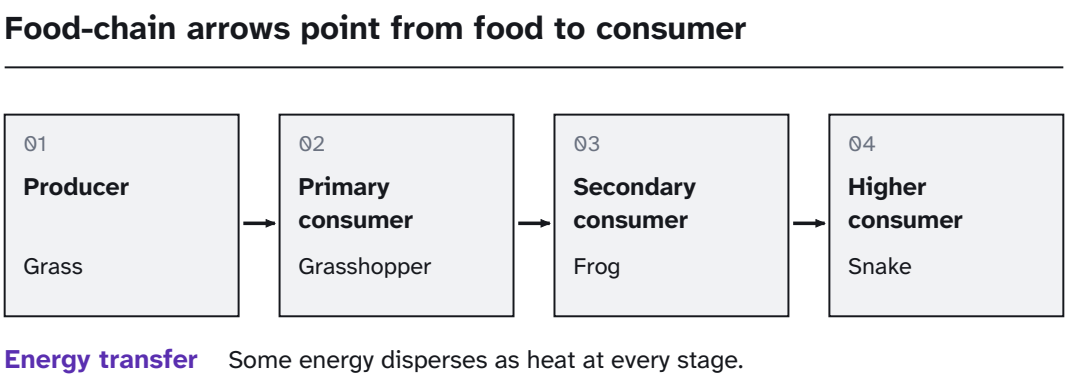


Figure 7. Figure 07. The arrow direction is food to consumer: grass to grasshopper to frog to snake. This is one simplified chain within a food web. Usable energy decreases along the chain; nutrients can be recycled.

4. Describe decomposers returning nutrients to the environment without claiming energy is recycled.

Recognising it. If a question explicitly gives transfer efficiency, apply it once for every trophic-level step.

Pr 5. In grass to grasshopper to frog, grass is the producer, grasshopper the primary consumer and frog the secondary consumer.

Solution. Grass captures light energy and makes organic material, so it is the producer. The grasshopper eats the producer and is the primary consumer. The frog eats that consumer and is the secondary consumer. The arrows grass → grasshopper → frog show transfer toward the eater; dead material from all levels can enter decomposer pathways.

Common error. The 10 percent rule is a rough teaching approximation, not a universal exact constant.

In the examination. Consumer levels are roles in the specified chain; do not assign a permanent level without looking at the feeding relationship.

Pr 6. Does energy cycle back through the ecosystem in the same way as mineral nutrients?

Pr 7. A question specifies 10 percent transfer efficiency at each step. Producers have 10000 energy units. Find energy available to primary and secondary consumers.

Pr 8. A question assumes 10 percent transfer at every step. What fraction of producer energy reaches secondary consumers?

(a) 20 percent

(b) 1 percent

(c) 10 percent

3 Population growth and limiting factors

Aims. Calculate net population change from four demographic processes; Explain how a limiting factor can constrain population growth.

Before this section: Population means members of one species in a defined area and period; basic signed addition.

Population size changes through births, deaths, immigration and emigration. Rapid exponential growth can occur when resources are abundant. Limited food, space and other conditions constrain growth. Carrying capacity describes a population size an environment can sustain under specified conditions.

Idea 3. A population boundary makes gains and losses countable. Births and immigration add members inside the boundary; deaths and emigration remove them. Net change is the balance of those processes, while final population also includes the members already present at the start.

Fact 3.

$$\Delta N = \text{births} + \text{immigration} - \text{deaths} - \text{emigration}$$

Why it holds. Immigration and births have the same positive effect on local count despite different origins. A positive net change must be added to the initial population rather than reported as the final size.

Method 3.

1. Set the population boundary and time interval.
2. Treat births and immigration as additions and deaths and emigration as removals.
3. Add the gains and subtract the losses to obtain net change.
4. If explaining growth limits, link a specific resource or condition to survival or reproduction.

Recognising it. A starting population plus four demographic counts calls for a change calculation followed by a final-size calculation.

Pr 9. If births are 30, deaths 10, immigration 5 and emigration 8, the population increases by 17.

Solution. The example gives 30 births and 5 immigrants, so total additions are 35. It gives 10 deaths and 8 emigrants, so total removals are 18. Net change is $35 - 18 = +17$ individuals. This is the increase during the interval, not the final population; a starting population would be needed to calculate the latter.

Common error. Carrying capacity is not an unchanging property independent of habitat conditions.

In the examination. Label a positive result increase and a negative result decrease; do not report net change as the final population size.

Pr 10. If births equal deaths, must population size stay unchanged?

Pr 11. A population begins with 200 individuals. During a period there are 25 births, 12 deaths, 8 immigrants and 6 emigrants. Find the final size.

Pr 12. Births equal deaths, but immigration exceeds emigration. The local population:

(a) Stays constant

(b) Must decrease

(c) Increases

4 Relationships among organisms

Aims. Identify ecological interactions from their effects on each partner; Support an interaction label with scenario evidence.
Before this section: Benefit, harm and little measurable effect are relative to the organisms in the interaction.

Competition occurs when organisms use a limited shared resource. Predation involves one organism consuming another. In mutualism both benefit; in commensalism one benefits without a substantial effect on the other; in parasitism one benefits at the host expense.

Idea 4. An interaction label summarizes what happens to each partner compared with the relevant situation without the interaction. A benefit for one partner does not automatically mean a benefit for the other. Writing the two effects separately is more reliable than guessing a label from the organisms involved.

Fact 4.

Mutualism $(+,+)$
commensalism $(+,0)$
parasitism $(+,-)$

Why it holds. Mutualism requires a benefit to both partners. A benefit/harm pattern differs from a benefit/no-clear-effect pattern, so the second partner cannot be ignored.

Method 4.

1. Identify the two organisms and the resource or activity linking them.
2. Describe the effect on the first organism and on the second separately.
3. Match the two effects to the appropriate interaction category.
4. Use evidence from the case rather than assuming all close associations are mutualistic.

Recognising it. For relationship scenarios, identify the sign of the effect on each organism before naming the interaction.

Pr 13. A parasite obtaining nutrients while harming its host is a plus-minus interaction.

Solution. The parasite obtains nutrients, so it benefits from the interaction. The host is harmed, giving the benefit/harm pattern commonly written plus/minus. The two competing plants both face restricted access to limited light. Competition is therefore a minus/minus interaction in the comparison with unrestricted resource access, even if one competitor does better than the other.

Common error. A relationship may be context-dependent; answer using the effects given in the question.

In the examination. Write the effect on each partner before naming the interaction; this makes similar scenarios easier to distinguish.

Pr 14. Does one competitor winning mean competition is automatically a benefit/harm interaction?

Pr 15. Identify a relationship where both benefit and another where one benefits while the other is harmed. Explain the signs.

Pr 16. One organism benefits while the other has no clear benefit or harm. Is this necessarily parasitism?

(a) No; parasitism requires harm to the host

(b) Yes; any one-sided benefit is parasitism

(c) Yes; no effect means harm

5 Pollution: sources, pathways and effects

Aims. Build a source-pathway-receptor explanation of pollution harm; Select prevention that interrupts a stated nutrient-runoff pathway.
Before this section: Nutrients support growth; decomposition can consume dissolved oxygen.

Pollution changes environmental conditions in ways that can cause harm or reduce usefulness. Sources can be natural or human-made. Air, water and soil pollution are connected through transport and deposition. A useful answer links a source to a pathway, an effect and a control.

Idea 5. Pollution explanations are strongest when each link shows how one change causes the next. Fertilizer does not have to poison fish directly to harm them: nutrients can change biological growth, and subsequent decomposition can alter oxygen conditions. Prevention can target an earlier link before ecological damage occurs.

Fact 5.

$$\text{Source} \rightarrow \text{pathway} \rightarrow \text{receptor} \rightarrow \text{effect}$$

Why it holds. Runoff transports nutrients from a land source into water. Decomposition of increased organic material can use dissolved oxygen and harm oxygen-dependent

organisms.

Method 5.

1. Name the pollution source and the released material.
2. Trace how the material reaches the affected ecosystem.
3. Explain the intermediate process connecting exposure to the final effect.
4. Choose a preventive measure and state which link it reduces.

Recognising it. A source-to-effect question needs an intermediate pathway and mechanism, plus prevention linked to that chain.

Pr 17. Nutrient-rich runoff can stimulate excessive algal growth.

Solution. Excess nutrients from fertilizer can be carried by runoff into a water body. Nutrient enrichment can stimulate excessive algal growth. When organic material is decomposed, microbial respiration can consume dissolved oxygen. Low oxygen can harm fish; reducing excess fertilizer or intercepting runoff addresses the source or pathway before that effect occurs.

Common error. Do not say algae always produce an immediate oxygen shortage without explaining decomposition and conditions.

In the examination. Use connecting verbs such as carries, stimulates and consumes to make every step in the causal chain explicit.

Pr 18. Why is algae cause fish death an incomplete explanation?

Pr 19. Explain a three-step route from excess fertilizer use to fish harm and name one preventive measure.

Pr 20. Which measure most directly interrupts the pathway from fertilized land to a nearby stream?

(a) Counting fish without changing the source or pathway

(b) Intercepting nutrient-rich runoff before it reaches the stream

(c) Adding more fertilizer whenever algae increase

6 Ecological footprint and biocapacity

Aims. Compare ecological demand with biocapacity on a common basis; Interpret a numerical balance as a deficit or reserve.

Before this section: Demand versus available capacity; subtraction; per-person and total quantities must not be mixed.

Ecological footprint estimates demand for biologically productive area needed to support consumption and absorb certain wastes. Biocapacity represents available biological production capacity. A deficit occurs when demand exceeds capacity. A global hectare is a productivity-adjusted unit, not simply any local hectare.

Idea 6. Footprint and biocapacity compare demand with biological supply within an accounting model. A deficit means the demand figure exceeds the available capacity figure on the same basis. The calculation only makes sense if both quantities describe the same population basis and productivity-adjusted unit.

Fact 6.

$$\text{Ecological deficit} = \text{footprint} - \text{biocapacity}$$
$$\text{if positive}$$

Why it holds. Subtracting capacity from demand tests whether demand exceeds capacity. A positive gap indicates deficit; the reverse comparison would indicate a reserve rather than a negative physical area.

Method 6.

1. Identify footprint as demand and biocapacity as available biological production capacity.
2. Check that both figures use the same units, time basis and population basis.
3. Subtract biocapacity from footprint to test for a deficit.
4. Explain the result within the indicator scope rather than treating it as every environmental impact.

Recognising it. For a footprint balance, check units and per-person versus total basis before subtracting and interpreting.

Pr 21. A population with footprint 3 global hectares per person and biocapacity 2 has a deficit of 1 per person.

Solution. The example footprint is 3 global hectares per person, describing demand. Biocapacity is 2 global hectares per person, describing available capacity on the same basis. Demand exceeds capacity by $3 - 2 = 1$ global hectare per person. The result is an ecological deficit, not an area measured in ordinary unadjusted local hectares.

Common error. Ecological footprint and carbon footprint are related but not identical measures.

In the examination. Write global hectares per person in a per-person result; omitting the population basis changes its meaning.

Pr 22. Can a region total footprint be compared directly with biocapacity per person?

Pr 23. A region has footprint 4.5 and biocapacity 2.8 global hectares per person. Identify and calculate the balance.

Pr 24. A region has footprint lower than its biocapacity on the same basis. This indicates:

(a) An ecological deficit

(b) Proof that every environmental impact is zero

(c) An ecological reserve in this accounting comparison

7 Waste management and the hierarchy

Aims. Distinguish prevention, reuse, recycling, recovery and disposal; Apply the waste hierarchy to a concrete material scenario.
Before this section: Reuse keeps an item in service; recycling reprocesses its material.

Waste management involves preventing, collecting, sorting, treating and disposing of waste. Prevention and reduction usually come before reuse, recycling, recovery and final disposal. Hazardous items require appropriate separate handling; not every household discard belongs in an ordinary bin.

Idea 7. The waste hierarchy separates avoiding a discard from keeping an item in use and from reprocessing its material. These actions can all reduce disposal, but they act at different stages. Correct classification follows what actually happens to the item, not a general impression that the action is environmentally helpful.

Fact 7.

$$\text{Prevent/reduce} \rightarrow \text{reuse} \rightarrow \text{recycle} \rightarrow \text{recover} \rightarrow \text{dispose}$$

Why it holds. Prevention avoids waste generation before later treatment is needed. Reuse keeps the item in service, whereas recycling reprocesses its material into a new product.

Method 7.

1. Identify whether the action avoids waste, keeps the item in use or processes discarded material.
2. Classify the action using the general hierarchy.
3. For a proposed choice consider prevention or reduction first where feasible.
4. Check material hazards and the approved local collection stream before actual disposal.

Recognising it. Inspect the action verb in a scenario: avoid, use again, reprocess or dispose.

Pr 25. Refilling a reusable bottle is reuse.

Solution. Refilling a durable bottle keeps the same item in service, so it is reuse. Processing separated used glass into new glass changes the material into a new product, so it is recycling. Avoiding unnecessary packaging prevents that material becoming waste at all. These actions belong at different stages of the hierarchy and should not all be labelled recycling.

Common error. Recycling does not erase all upstream resource use or justify unlimited consumption.

In the examination. Use the action in the scenario as evidence: avoid means prevention, use again means reuse, reprocess means recycling.

Pr 26. Is using a clean jar again to store suitable items recycling?

Pr 27. Classify avoiding excess packaging, reusing a jar, reprocessing aluminium cans and placing residual waste in an approved landfill.

Pr 28. A metal drink container is melted and made into a new metal product. This is:

(a) Recycling

(b) Reuse of the same item

(c) Waste prevention

8 Sustainable development and integrated decisions

Aims. Assess a project across environmental, social and economic dimensions; State measurable indicators and a trade-off for a sustainability decision.

Before this section: Present and future needs; a benefit in one dimension can have costs in another.

Sustainable development addresses present needs while preserving future ability to meet needs. Environmental, social and economic dimensions interact. A project can improve one dimension while creating costs in another, so evaluation should consider benefits, trade-offs and long-term effects.

Idea 8. Sustainability decisions combine outcomes that do not all move together. An energy-efficiency project may save electricity and money while requiring upfront spending or causing temporary disruption. Indicators make these claims testable; a trade-off explains how a benefit in one area comes with a cost or constraint elsewhere.

Fact 8.

$$\text{Environment} + \text{society} + \text{economy}$$

Why it holds. Environmental, social and economic effects need distinct evidence. Comparing effects over time prevents an upfront cost or one immediate benefit from dominating the whole judgement.

Method 8.

1. Define the project and the needs it is intended to meet.
2. Give one environmental, one social and one economic effect.
3. Select an indicator that could test each effect.
4. Explain one trade-off over the project life rather than declaring the project sustainable from a single benefit.

Recognising it. When evaluating a project, choose one measurable indicator per dimension and connect a concrete benefit with a concrete cost.

Pr 29. A public-transport project can reduce emissions, improve access and lower travel costs, but its construction and operation also require resources and funding.

Solution. The public-transport example can reduce transport emissions if its use changes travel patterns. Improved access to education or work is a social benefit to evaluate. Travel costs, operating costs and funding provide economic questions. Construction also consumes resources and money, so a full assessment compares these costs with benefits over time.

Common error. Do not equate sustainable development with only economic growth or only nature protection.

In the examination. Replace vague words such as greener with a measurable indicator such as electricity use, emissions, access or cost over a stated period.

Pr 30. Is lower operating energy alone enough to prove a project is sustainable?

Pr 31. Give one environmental, one social and one economic indicator for a campus energy-efficiency project, plus one trade-off to evaluate.

Pr 32. Which is a measurable social indicator for a campus energy-efficiency project?

(a) Electricity use in kilowatt-hours alone

(b) Student-reported thermal comfort using a defined survey

(c) The project is very green

Hints

Use one hint at a time, then return to the problem.

Pr 3 (p. 188) (1) Use living organism versus nonliving environmental factor as the test, not whether the item is useful. (2) Consider sunlight and mineral content separately from grass and rabbit. (3) Assign each of the four items a category and give a short reason for any item you initially found ambiguous.

Pr 7 (p. 189) (1) Identify the two steps: producers to primary consumers, then primary to secondary. (2) For the first transfer apply the stated 10 percent to 10000 energy units. (3) For the second transfer apply 10 percent to the first result, not to the original producer amount. Label the two consumer-level energies.

Pr 11 (p. 191) (1) Separate the starting 200 individuals from changes during the period. (2) Add births and immigrants, then subtract deaths and emigrants to obtain net change. (3) Add that net change to 200. Report final size as individuals rather than confusing it with the smaller change value.

Pr 15 (p. 192) (1) Choose one relationship with a benefit for each partner and one with benefit for one and harm for the other. (2) Name the appropriate relationship category for each chosen example. (3) Explain the two signs in words, showing what each partner gains or loses rather than giving an unexplained plus/minus label.

Pr 19 (p. 193) (1) Begin with excess fertilizer and explain how its nutrients reach water. (2) Link nutrient enrichment to excess growth and then decomposition-related oxygen depletion under suitable conditions. (3) Finish with fish harm and one measure that reduces the source or intercepts the pathway. Use connected causal statements.

Pr 23 (p. 195) (1) Identify which of 4.5 and 2.8 describes demand and which describes available capacity. (2) Compare them to decide deficit or reserve before calculating the gap. (3) Subtract the smaller from the larger for its magnitude and retain global hectares per person. State the type of balance explicitly.

Pr 27 (p. 196) (1) Read each action separately rather than giving one category for the whole list. (2) Ask whether the action avoids material use, keeps the item in service, reprocesses material or places a residue in final disposal. (3) Match those distinctions to packaging, jar, cans and landfill in order. Use a short label and explanatory action for each.

Pr 31 (p. 197) (1) List the three dimensions and identify a campus outcome for each. (2) Choose measures such as electricity consumption, occupant comfort and financial savings or payback, giving a clear basis for comparison. (3) Add a trade-off, for example upfront spending versus later savings, and explain what decision needs evaluation rather than merely listing two unrelated facts.

Answers

Pr 2 (p. 188) No. Bacteria are living organisms and therefore biotic components.

Pr 3 (p. 188) Sunlight and mineral content are abiotic. Grass and rabbit are biotic. One mark each.

Pr 4 (p. 188) (a) Correct. Influence on life does not itself make a factor alive.

Pr 6 (p. 189) No. Energy flows through and is eventually dissipated as heat; nutrients can be recycled.

Pr 7 (p. 189) Primary consumers receive 1000 units; secondary consumers receive 100 units. Two marks for each transfer. These values depend on the stated efficiency.

Pr 8 (p. 189) (b) Correct. Two transfers multiply one tenth by one tenth.

Pr 10 (p. 191) No. Immigration and emigration can still produce a net change.

Pr 11 (p. 191) Net change =25-12+8-6=15. Final size=215. Credit signs of all four processes and final total.

Pr 12 (p. 191) (c) Correct. The birth/death balance is zero, leaving a positive migration balance.

Pr 14 (p. 192) No. Both incur the effects of competing for a limited resource; one may still outperform the other.

Pr 15 (p. 192) Mutualism is (+,+): both gain a benefit. Parasitism is (+,-): parasite benefits and host is harmed. One mark for each name and each explanation.

Pr 16 (p. 192) (a) Correct. Benefit with little or no effect describes a different pattern from benefit with harm.

Pr 18 (p. 193) It omits the conditions and intermediate processes, especially decomposition and dissolved-oxygen depletion.

Pr 19 (p. 193) Runoff carries nutrients to water; algal growth increases; decomposition can lower dissolved oxygen and harm fish. Reduce excess nutrient application or intercept runoff. One mark for each causal step and one for prevention.

Pr 20 (p. 193) (b) Correct. This acts on transport between source and receptor.

Pr 22 (p. 195) No. Convert both to totals or both to per-person values first.

Pr 23 (p. 195) It has an ecological deficit of 4.5-2.8=1.7 global hectares per person. Credit comparison, subtraction, units and deficit interpretation.

Pr 24 (p. 195) (c) Correct. Capacity exceeds the counted demand.

Pr 26 (p. 196) It is reuse, because the same item is used again without material reprocessing.

Pr 27 (p. 196) Prevention, reuse, recycling and disposal, respectively. One mark for each classification.

Pr 28 (p. 196) (a) Correct. The discarded material is reprocessed into a new product.

Pr 30 (p. 197) No. Social access, economic viability, construction impacts and long-term effects also require assessment.

Pr 31 (p. 197) Examples: electricity use/emissions; comfort or accessibility; cost savings/payback; upfront cost versus long-term savings. One mark for each relevant, measurable item.

Pr 32 (p. 197) (b) Correct. A defined comfort measure directly examines the learner experience.

Humans and the Environment, Test A

20 marks. Suggested time 20 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. Which is an abiotic factor?

   (a) Algae
   (b) Bacteria
   (c) Water temperature
   (d) Fish

2. Decomposers mainly help by:

   (a) Recycling nutrients from dead matter
   (b) Creating sunlight
   (c) Stopping all energy loss
   (d) Removing all producers

3. Which directly reduces a local population?

   (a) Reproduction
   (b) Births
   (c) Immigration
   (d) Emigration

4. Both organisms benefit in:

   (a) Parasitism
   (b) Mutualism
   (c) Predation
   (d) Competition

Part II. Show your reasoning.

1. Classify sunlight, grass, soil mineral content and a rabbit as biotic or abiotic.

2. A question specifies 10 percent transfer efficiency at each step. Producers have 10000 energy units. Find energy available to primary and secondary consumers.

3. A population begins with 200 individuals. During a period there are 25 births, 12 deaths, 8 immigrants and 6 emigrants. Find the final size.

4. Identify a relationship where both benefit and another where one benefits while the other is harmed. Explain the signs.

Humans and the Environment, Test B

20 marks. Suggested time 20 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. Excess nutrient input into water is most closely associated with:

   (a) Eutrophication
   (b) Lens refraction
   (c) Sterilization
   (d) No biological change

2. A footprint larger than biocapacity indicates:

   (a) Unlimited sustainability
   (b) An ecological reserve
   (c) An ecological deficit
   (d) Zero resource use

3. Which is highest in the general waste hierarchy?

   (a) Disposal after mixing
   (b) Landfill
   (c) Uncontrolled burning
   (d) Waste prevention

4. A complete sustainability assessment considers:

   (a) Only this month
   (b) Environmental, social and economic effects
   (c) Only profit
   (d) Only tree numbers

Part II. Show your reasoning.

1. Explain a three-step route from excess fertilizer use to fish harm and name one preventive measure.

2. A region has footprint 4.5 and biocapacity 2.8 global hectares per person. Identify and calculate the balance.

3. Classify avoiding excess packaging, reusing a jar, reprocessing aluminium cans and placing residual waste in an approved landfill.

4. Give one environmental, one social and one economic indicator for a campus energy-efficiency project, plus one trade-off to evaluate.

Humans and the Environment, Test A: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (c) Temperature is a physical condition rather than a living organism.

Written part. Sunlight and mineral content are abiotic. Grass and rabbit are biotic. One mark each. (Section: p. 187.)

Question 2. (a) They break down organic matter and return nutrients to the system.

Written part. Primary consumers receive 1000 units; secondary consumers receive 100 units. Two marks for each transfer. These values depend on the stated efficiency. (Section: p. 188.)

Question 3. (d) Emigration removes individuals from the local population.

Written part. Net change =25-12+8-6=15. Final size=215. Credit signs of all four processes and final total. (Section: p. 190.)

Question 4. (b) Mutualism is defined by benefits to both participants.

Written part. Mutualism is (+,+): both gain a benefit. Parasitism is (+,-): parasite benefits and host is harmed. One mark for each name and each explanation. (Section: p. 191.)

Humans and the Environment, Test B: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (a) Nutrient enrichment can drive excessive growth and subsequent oxygen depletion.

Written part. Runoff carries nutrients to water; algal growth increases; decomposition can lower dissolved oxygen and harm fish. Reduce excess nutrient application or intercept runoff. One mark for each causal step and one for prevention. (Section: p. 192.)

Question 2. (c) Demand exceeds the available biological capacity under the accounting model.

Written part. It has an ecological deficit of 4.5-2.8=1.7 global hectares per person. Credit comparison, subtraction, units and deficit interpretation. (Section: p. 194.)

Question 3. (d) Avoiding waste reduces the need for later treatment and disposal.

Written part. Prevention, reuse, recycling and disposal, respectively. One mark for each classification. (Section: p. 195.)

Question 4. (b) The concept integrates all three dimensions and future consequences.

Written part. Examples: electricity use/emissions; comfort or accessibility; cost savings/payback; upfront cost versus long-term savings. One mark for each relevant, measurable item. (Section: p. 196.)

Facts at a glance

1. Environment and ecosystem components p. 187

$$\text{Organism} \rightarrow \text{population} \rightarrow \text{community} \rightarrow \text{ecosystem}$$

2. Energy flow, food chains and decomposers p. 188

$$\text{Producer} \rightarrow \text{primary consumer} \rightarrow \text{secondary consumer}$$

3. Population growth and limiting factors p. 190

$$\Delta N = \text{births} + \text{immigration} - \text{deaths} - \text{emigration}$$

4. Relationships among organisms p. 191

$$\text{Mutualism } (+,+)$$
$$\text{commensalism } (+,0)$$
$$\text{parasitism } (+,-)$$

5. Pollution: sources, pathways and effects p. 192

$$\text{Source} \rightarrow \text{pathway} \rightarrow \text{receptor} \rightarrow \text{effect}$$

6. Ecological footprint and biocapacity p. 194

$$\text{Ecological deficit} = \text{footprint} - \text{biocapacity}$$
$$\text{if positive}$$

7. Waste management and the hierarchy p. 195

$$\text{Prevent/reduce} \rightarrow \text{reuse} \rightarrow \text{recycle} \rightarrow \text{recover} \rightarrow \text{dispose}$$

8. Sustainable development and integrated decisions p. 196

$$\text{Environment} + \text{society} + \text{economy}$$

Notation

Symbols are listed by section. The same letter may name different quantities in different sections; read the section's Fact.

Sources

ENV101 elective: ecosystems, relationships among organisms, pollution, ecological footprint, waste management and sustainable development. The Arabic source lectures are explained here in English.

The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. Check lecturer-specific derivations and conventions. The practice tests are original study material, not predictions of an examination.

Professional Development Seminar

Elective lecture themes: teaching as a profession, teacher formation, professional learning, portfolios, standards and educational technology. This follows the teacher-development focus of the supplied file; it is not a generic career-skills course.

1 Teaching as a profession

Aims. Explain two criteria that make teaching a profession; Support each criterion with a distinct classroom example.
Before this section: A professional criterion is a defining feature, not simply an employment benefit.

The source distinguishes a profession through specialized knowledge, extended preparation, social purpose, ethical responsibilities and organized practice. Teaching is more than delivering facts: it develops learners and contributes to society. A strong exam response connects each professional criterion to a teaching example.

Idea 1. A profession combines expertise with responsibility for how that expertise affects others. Teaching therefore involves both knowing a subject and making informed, ethical decisions about learners. An exam answer becomes convincing when it connects each professional feature to an observable teaching action.

Fact 1.

$$\text{Criterion} + \text{explanation} + \text{teaching example}$$

Why it holds. Specialized preparation explains why professional judgement requires more than familiarity with subject facts. Ethical responsibilities explain why competence must be used fairly and for a social purpose.

Method 1.

1. Define profession using expertise, preparation, social purpose and ethical responsibility.
2. Select two distinct criteria relevant to the question.
3. Explain each criterion in plain language.
4. Attach a concrete teaching example to each explanation.

Recognising it. For reasons teaching is a profession, give distinct criterion-example pairs instead of repeated general praise.

Pr 1. Specialized knowledge includes subject expertise and pedagogy; ethical responsibility includes fair assessment and respect for learners.

Solution. Subject expertise concerns what is taught; pedagogy concerns how learners are helped to understand it. Together they illustrate specialized professional knowledge rather than information delivery alone. Fair assessment and respect for learners illustrate

ethical responsibility. The examples support two different criteria, so they provide more evidence than repeating expertise in several forms.

Common error. Do not assume being paid alone makes an activity a profession.

In the examination. Use two complete criterion-explanation-example pairs when four marks are available, rather than a long unsupported list.

Pr 2. Does being paid alone establish that an activity is a profession?

Pr 3. Give two reasons teaching is considered a profession, each with an example.

Pr 4. Which pair best shows two different professional criteria?

(a) Subject-and-pedagogy preparation, plus fair assessment

(b) Subject knowledge, plus knowing many subject facts

(c) Being busy, plus having a long working day

2 Professionalization and the teacher role

Aims. Describe teacher roles through observable professional actions; Explain how assessment evidence guides subsequent instruction.
Before this section: A role describes a function; an action shows how that function appears in practice.

The lecture describes development from pre-professional through semi-professional toward growing professionalization. Teacher roles extend from presenting knowledge to guiding learning, assessing progress, modelling values and supporting society. Historical labels describe development, not a reason to dismiss present expertise.

Idea 2. Teacher roles describe purposes expressed through actions. An assessor interprets evidence, a guide helps learners choose strategies, and a model demonstrates professional values. A role label alone leaves the reader guessing; an action and its effect show that the role is understood.

Fact 2.

$$\text{Role} \rightarrow \text{observable professional action}$$

Why it holds. The same classroom activity can serve different roles depending on its purpose and follow-up. Describing a learner effect connects a teacher action to its professional function.

Method 2.

1. Identify the role asked about, such as assessor or learning guide.
2. Describe a specific action the teacher takes in that role.
3. Explain how the action supports learners.

4. For historical comparisons, discuss changes in preparation or organization rather than dismissing current expertise.

Recognising it. When asked for roles beyond presenting content, identify an action that changes how learning is supported.

Pr 5. A teacher acting as an assessor uses evidence from student work to adapt instruction.

Solution. As an assessor, the teacher examines student work to identify what learners understand. If several answers show the same error, that evidence indicates a learning gap. Adapting instruction in response connects the assessor role to better teaching decisions. As a guide, the teacher supports reasoning through questions or strategies instead of supplying every answer.

Common error. A list of role names without explanation earns less evidence of understanding.

In the examination. After every role name add a verb and object: diagnoses a misconception, gives targeted feedback or guides strategy choice.

Pr 6. Why is gives a test alone a weak explanation of the assessor role?

Pr 7. Describe two teacher roles beyond explaining subject content.

Pr 8. A teacher asks a learner to compare two solution strategies rather than immediately giving an answer. This most directly shows:

(a) Avoiding all responsibility for learning

(b) Guiding reasoning

(c) Merely storing information

3 Teacher formation: before and during service

Aims. Distinguish pre-service preparation, induction and continuing development; Match a support activity to its career stage and purpose.
Before this section: Teachers develop before entering practice and throughout their careers.

Teacher formation is a continuing process. Pre-service preparation establishes knowledge and skills; induction supports entry to actual teaching; continuing professional development addresses changing needs and improves practice. Training is one method of development, not the whole process.

Idea 3. Teacher formation is a continuing sequence, but support has different purposes at different stages. Initial preparation builds foundations before entry; induction helps a new teacher use them in a real workplace. Comparing both timing and purpose makes the distinction clearer than treating one as simply more training.

Fact 3.

Pre-service → induction → continuing development

Why it holds. Pre-service learning builds foundations before entry to the profession. Induction bridges those foundations and the demands of early independent practice.

Method 3.

1. Locate the person before employment, in early practice or in established service.
2. Name the matching formation stage.
3. Explain the specific knowledge or support needed at that stage.
4. Describe how the activity supports practice rather than treating attendance as proof of improvement.

Recognising it. Stage-comparison questions require a timing statement and a purpose statement for each stage.

Pr 9. A mentor supporting a newly appointed teacher belongs to induction.

Solution. The example teacher is newly appointed and is entering real classroom practice. Mentor support at this point is induction, bridging initial preparation and workplace demands. Later peer observation and inquiry are examples of continuing development. The method may look similar across stages, but the career context and purpose determine the classification.

Common error. Graduation marks an entry point, not the end of professional learning.

In the examination. In a comparison give both timing and purpose for each stage; stating before and after alone is incomplete.

Pr 10. Is every teacher workshop automatically pre-service preparation?

Pr 11. Compare pre-service preparation with induction in two clear points.

Pr 12. A newly appointed teacher receives workplace mentoring while beginning classroom duties. Why is this induction rather than pre-service?

(a) Mentoring can never be used before employment

(b) All learning stops after induction

(c) It supports entry into actual professional practice

4 Preparation, practice and professional standards

Aims. Explain how supervised field experience links theory with practice; Construct a feedback cycle using observable evidence.

Before this section: Preparation includes subject knowledge, pedagogy and supervised teaching; standards describe expected practice.

Teacher preparation combines subject knowledge, pedagogical understanding and supervised practice. Field experience connects theory with real learners and classroom decisions. Professional standards make expectations explicit and support self-evaluation. Licensing is a recognition process whose current requirements must be checked with the responsible body.

Idea 4. Reflection becomes professional improvement when it leads to a tested change. An observed lesson supplies evidence, but evidence must be interpreted, used to revise practice and checked again. The next lesson is therefore part of the cycle, rather than a separate event after feedback ends.

Fact 4.

$$\text{Knowledge} \rightarrow \text{supervised application} \rightarrow \text{feedback} \rightarrow \text{improvement}$$

Why it holds. Specific evidence makes the improvement target clearer than a vague judgement. Checking the revised lesson reveals whether the chosen change actually helped.

Method 4.

1. Identify the teaching expectation or intended learner outcome.
2. Collect relevant evidence from observation and student work.
3. Choose one specific change suggested by that evidence.
4. Apply the change and evaluate the next attempt against the same expectation.

Recognising it. An improvement-cycle question needs evidence, a focused need, a targeted action and a follow-up evaluation.

Pr 13. After teaching a supervised lesson, a trainee reviews observations and student work, then revises the next lesson.

Solution. The trainee first teaches a supervised lesson, putting planned ideas into practice. Observation notes and student work show what occurred and what learners achieved. The trainee compares that evidence with the intended learning and revises the next lesson. The cycle is completed by checking whether the revision improves the next observed outcome.

Common error. Do not invent current licence requirements from an older lecture.

In the examination. When standards are mentioned, link one expectation to evidence; do not invent current licensing rules from a lecture.

Pr 14. Why does feedback without a subsequent change and evaluation leave the cycle incomplete?

Pr 15. Give a four-step improvement cycle for a trainee after an observed lesson.

Pr 16. After an observed lesson, a trainee changes the task but collects no follow-up evidence. What is missing?

(a) Evaluation of whether the change helped

(b) Any possibility of changing practice

(c) A requirement to repeat the original lesson unchanged

5 Plan continuing professional development

Aims. Write a professional-development plan linking need, goal, activity and evidence; Distinguish participation evidence from evidence of impact.
Before this section: A development goal should address an identified teaching or learning need.

Professional development is purposeful and sustained. It begins with a learning need, sets a clear goal, chooses suitable activities and evaluates effects on practice and learners. Workshops, peer observation, professional reading and collaborative inquiry can all contribute.

Idea 5. A development plan is a chain from a real teaching need to evidence of improvement. A workshop can supply ideas, but the plan also needs classroom application and a way to judge results. A focused goal makes it possible to tell whether the activity was useful.

Fact 5.

$$\text{Need} \rightarrow \text{goal} \rightarrow \text{activity} \rightarrow \text{evidence of impact}$$

Why it holds. An activity becomes relevant when it addresses the diagnosed need. Before-and-after learner evidence can show impact more directly than participation records alone.

Method 5.

1. Use classroom evidence to name one specific development need.
2. Write a focused goal and a realistic review period.
3. Choose a learning activity and classroom application that address the goal.
4. Compare relevant evidence before and after the application to judge progress.

Recognising it. When asked for a development plan, link need, goal, activity and evidence within a defined review period.

Pr 17. A teacher notices weak student explanations, studies questioning strategies, tries them for four weeks and compares the quality of student responses.

Solution. The teacher notices weak student explanations, identifying a specific need. Studying questioning strategies is the development activity, but study alone does not show impact. Trying the strategies for four weeks puts the learning into classroom practice. Comparing response samples before and after can reveal whether explanations became clearer and better supported.

Common error. Attendance certificates show participation, not necessarily improved teaching.

In the examination. Keep all four plan parts linked to the same need; an unrelated workshop does not satisfy a coherent development plan.

Pr 18. What does a workshop attendance certificate demonstrate directly?

Pr 19. Write a short development plan for a teacher whose students struggle to explain answers.

Pr 20. A teacher attends a questioning workshop. Which next step best tests its value for weak student explanations?

(a) Collect another certificate without classroom application

(b) Apply selected strategies and compare explanation samples over time

(c) Count the slides shown in the workshop

6 Portfolios and reflective evaluation

Aims. Select portfolio evidence for a defined goal or standard; Explain how reflection turns an artifact into evidence of development.

Before this section: An artifact is a document or sample; evidence requires an explanation of what it demonstrates.

A professional portfolio is a purposeful selection of evidence linked to goals and standards, with reflection on what the evidence demonstrates. It can include plans, student work, feedback and records of improvement. A large unorganized collection is less useful than a selective, explained one.

Idea 6. A portfolio is an argument supported by selected artifacts. A lesson plan can show intended alignment, while student work can show something about the outcome. Commentary connects the evidence to a professional claim and acknowledges what one artifact alone cannot prove.

Fact 6.

$$\text{Evidence} \;+\; \text{reflection} \;+\; \text{link to a goal}$$

Why it holds. An artifact becomes useful evidence when its relevance to a goal or standard is explained. Reflection connects intention, outcome and a next learning step rather than simply decorating a collection.

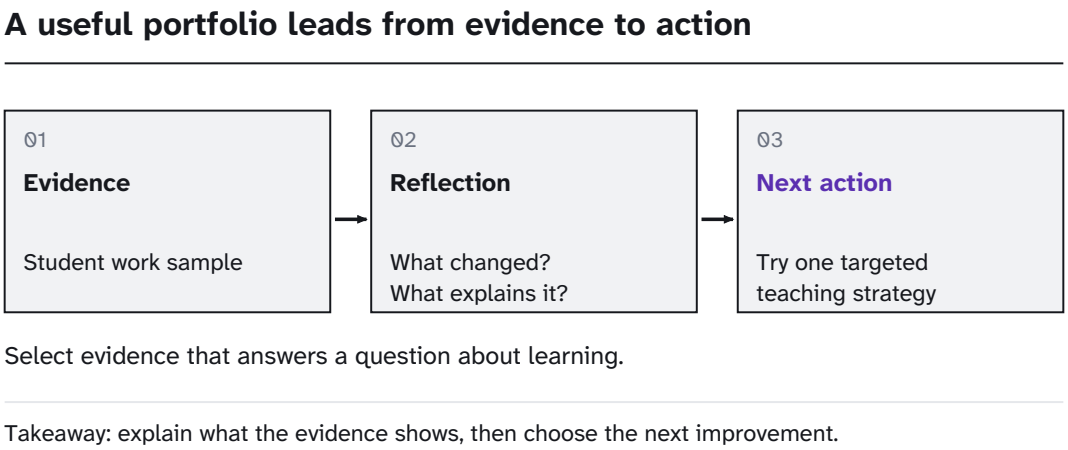


Figure 8. Figure 08. A portfolio connects selected evidence to reflection and a targeted next action. A student-work sample becomes more useful when the teacher explains what it shows and how teaching will change.

Method 6.

1. State the portfolio purpose or professional goal.
2. Select a small number of relevant artifacts and organize them clearly.
3. Explain what each artifact shows and what its limitations are.
4. Reflect on the outcome and identify a next improvement, protecting learner privacy.

Recognising it. When naming portfolio items, pair each with the specific aspect of practice it can demonstrate.

Pr 21. A lesson plan paired with student work and a reflection on changes shows more than a lesson plan alone.

Solution. The lesson plan shows what the teacher intended learners to do. Student work shows something about the resulting learning and where difficulties remained. A reflection compares the plan with that outcome and explains why a change was made. The three pieces together support a development claim more clearly than an unexplained plan alone.

Common error. Do not include identifiable student information unnecessarily; follow the institution privacy rules.

In the examination. For each item finish the sentence this demonstrates... and connect it to a goal or standard; anonymize learner material as required.

Pr 22. Does adding more unrelated documents necessarily make a portfolio stronger?

Pr 23. Name two portfolio items and explain what each can demonstrate.

Pr 24. A lesson plan alone most directly shows:

(a) That every student achieved the objective

(b) That no future improvement is needed

(c) What learning and teaching were intended

7 Ethics, collaboration and learning communities

Aims. Rewrite personal criticism as evidence-based constructive feedback; Connect ethical collaboration with learner respect and improvement.

Before this section: Observation is what can be seen or recorded; interpretation and personal judgement are different.

Professional learning can be collaborative: colleagues discuss evidence, share practice and solve problems. Ethical practice includes respect, fairness, honesty and responsibility. Feedback should focus on observable practice and improvement, rather than personal attacks.

Idea 7. Constructive feedback describes a changeable practice rather than labelling a person. The recipient needs to know what was observed, how learners were affected and what could be tried next. Respect and specificity make challenge more useful without requiring uncritical agreement.

Fact 7.

$$\text{Observation} + \text{effect} + \text{constructive next step}$$

Why it holds. Observable evidence gives the colleague something concrete to examine. An actionable suggestion turns criticism into a possible improvement step.

Method 7.

1. Identify an observable feature of teaching rather than a personal label.
2. Describe its apparent effect on learners with available evidence.
3. Suggest one feasible next step connected to that effect.
4. Use respectful language and follow confidentiality and reporting responsibilities.

Recognising it. To rewrite harsh feedback, replace personal judgement with observation, learner effect and a feasible next action.

Pr 25. A colleague gives feedback that a task confused learners and suggests clearer success criteria, rather than calling the teacher incompetent.

Solution. Calling a teacher incompetent attacks the person and gives no specific route to improvement. The example instead identifies that the task confused learners. Suggesting clearer success criteria connects the feedback to a teaching action the colleague can try. A

later check of learner understanding can show whether that change helped, completing the improvement conversation.

Common error. Collaboration is not simply agreeing with everyone or sharing private student details.

In the examination. Structure scenario answers as observation, learner effect and next step; avoid unsupported claims about personal character.

Pr 26. Is constructive feedback the same as agreeing with every colleague decision?

Pr 27. Rewrite this feedback constructively: Your lesson was bad.

Pr 28. Which statement is most useful as the observation part of feedback?

(a) Several learners asked what the task goal was

(b) You are a careless person

(c) Your lesson was bad in every way

8 Educational technology and evaluation of impact

Aims. Choose an educational tool to serve a specific learning objective; Evaluate technology through diagnosis, teacher response and learner evidence.

Before this section: A tool supports an activity; learning depends on how the activity and follow-up are designed.

Educational technology should serve a learning goal. Teachers need to select suitable tools, design learning activities, assess outcomes and consider access and ethical use. Technical operation alone does not establish teaching value. The seminar mentions digital competencies; programming instruction is excluded from this pack as requested.

Idea 8. Technology helps learning through what it enables teachers and learners to do. A quiz can reveal an error pattern quickly, but learning benefit depends on how that pattern changes the next teaching step. Evaluation therefore follows the full sequence from objective to evidence and response.

Fact 8.

$$\text{Learning need} \rightarrow \text{tool choice} \rightarrow \text{activity} \rightarrow \text{evaluation}$$

Why it holds. Plausible distractors can make a misconception visible in response patterns. A fresh question after targeted teaching tests improvement more directly than click counts.

Method 8.

1. Name the learner objective and the misconception or difficulty to investigate.
2. Choose a suitable accessible tool and design questions that reveal reasoning.
3. Inspect response patterns and adapt teaching accordingly.

4. Use a new question or response sample to judge whether understanding improved.

Recognising it. For an educational-technology proposal, state the learning purpose, diagnostic task, teacher response and evidence of value.

Pr 29. A short digital quiz can reveal misconceptions immediately, but the teacher must use the results to adapt instruction for it to support learning.

Solution. The brief digital quiz can make response patterns visible quickly. Plausible incorrect options help reveal which misconception may be present. The teacher must interpret those results and respond, for example with targeted explanation or discussion. A fresh question then checks learning; the number of clicks or time on screen alone does not establish educational value.

Common error. More screen time is not itself evidence of better learning.

In the examination. State the learning goal before the tool name and finish with evidence of learning; programming content is outside this course pack.

Pr 30. What is missing if a teacher runs a digital quiz but never uses its results?

Pr 31. Propose a simple technology use to diagnose misconceptions, and state how you would evaluate its value.

Pr 32. A digital quiz reveals a common misconception. Which action best turns the result into a learning benefit?

(a) Judge success only by how many students clicked

(b) Reteach the issue and check understanding with a new question

(c) Keep the response chart but make no teaching change

Hints

Use one hint at a time, then return to the problem.

Pr 3 (p. 209) (1) Select two distinct criteria, such as specialized preparation and ethical responsibility. (2) Explain each criterion briefly in your own words. (3) Attach one teaching example to each: show what the teacher learns or does and why that action illustrates the chosen criterion.

Pr 7 (p. 210) (1) Choose two roles beyond delivering subject explanations, ensuring they are meaningfully different. (2) For each role, name one observable action, such as diagnosing a learning gap or supporting strategy choice. (3) Explain how that action helps learning. A title without an action is only a partial answer.

Pr 11 (p. 211) (1) Set up two headings in your notes: before entry and early practice. (2) Under each, identify its main learning or support purpose. (3) Turn these into two parallel comparisons covering when support occurs and what it helps the teacher do. Do not classify solely by whether a workshop or mentor is present.

Pr 15 (p. 213) (1) Start with reviewing observation notes and student work rather than jumping directly to a new activity. (2) Identify one specific weakness and choose a change that addresses it. (3) Arrange the four-step cycle so the change is applied and then evaluated with new evidence. Keep every step linked to the same improvement need.

Pr 19 (p. 214) (1) Describe the need precisely: students give answers without sufficient explanation. (2) Set a focused goal and choose a relevant learning activity plus classroom application. (3) Specify a review period and compare response samples using clear explanation criteria. Ensure the evidence measures the original need.

Pr 23 (p. 216) (1) What aspects of teaching should your two items provide evidence for? Select two distinct aspects before choosing artifacts. (2) Consider a planning document and an anonymized learner-work sample; identify the different practice or outcome each can demonstrate. (3) Add a short explanation linking each artifact to its claim. Avoid claiming that a plan alone proves the intended learning occurred.

Pr 27 (p. 217) (1) Remove the personal or vague judgement from the original sentence. (2) Identify a specific observable difficulty and its effect on learners, using a clearly stated example rather than pretending to have evidence you did not collect. (3) Suggest one relevant next step in respectful language, such as clarifying success criteria and checking understanding before independent work.

Pr 31 (p. 218) (1) Choose a simple tool that can reveal reasoning or misconceptions, rather than starting with novelty. (2) Explain how the response pattern will guide a teaching action. (3) State how a new question or response sample will test improvement, while considering access and the defined learning objective.

Answers

Pr 2 (p. 209) No. Professional status also involves specialized preparation, knowledge, responsibility and organized standards of practice.

Pr 3 (p. 209) Specialized preparation: teachers study subject knowledge and pedagogy. Ethical duty: teachers assess fairly and protect learner dignity. Two marks for each criterion-example pair.

Pr 4 (p. 209) (a) Correct. Preparation and ethical responsibility are distinct criteria with clear teaching links.

Pr 6 (p. 210) It names an activity but does not explain how evidence is interpreted or used to support learning.

Pr 7 (p. 210) Assessor: identifies learning gaps using evidence and feedback. Learning guide: supports students in choosing strategies and solving problems. Two marks per role with a concrete action.

Pr 8 (p. 210) (b) Correct. The teacher supports the learner in selecting and evaluating a strategy.

Pr 10 (p. 211) No. A workshop during employment may be induction support or continuing development, depending on career stage and purpose.

Pr 11 (p. 211) Pre-service occurs before entry and develops foundational subject and pedagogical knowledge. Induction occurs during early practice and provides mentoring and support in the real workplace. One mark each for stage and purpose in both cases.

Pr 12 (p. 211) (c) Correct. Career stage and purpose define induction in this scenario.

Pr 14 (p. 213) It identifies an issue but does not show whether the teacher acted on it or improved practice.

Pr 15 (p. 213) Review evidence; identify one specific weakness; plan and apply a targeted change; evaluate the next lesson using evidence. One mark per coherent step.

Pr 16 (p. 213) (a) Correct. An action was taken, but its effect has not been assessed.

Pr 18 (p. 214) Participation. Improved teaching or learning needs additional evidence from practice and outcomes.

Pr 19 (p. 214) Need: weak explanations. Goal: improve evidence-based responses. Activity: learn and apply structured questioning. Evidence: compare student response samples before and after a defined period. One mark per linked component.

Pr 20 (p. 214) (b) Correct. This connects learning, classroom use and relevant evidence of impact.

Pr 22 (p. 215) No. Relevance, clear organization and reflection matter more than volume alone.

Pr 23 (p. 216) A lesson plan can demonstrate planning and alignment. An anonymized student-work sample with commentary can demonstrate learning and the effect of feedback. Two marks per item and explanation.

Pr 24 (p. 216) (c) Correct. A plan documents intentions and design; outcomes require additional evidence.

Pr 26 (p. 217) No. It can challenge practice, but uses evidence, respect and actionable suggestions.

Pr 27 (p. 217) Example: Several students could not identify the task goal; stating the success criteria and checking one example before independent work may help. Credit observable evidence, effect, actionable suggestion and respectful language.

Pr 28 (p. 217) (a) Correct. This reports an observable event that can inform an improvement discussion.

Pr 30 (p. 218) The evidence-to-action step: diagnosis has not been used to adapt instruction or check improvement.

Pr 31 (p. 218) Use a brief concept quiz with plausible distractors. Review response patterns, reteach the misconception and check with a new question. Evaluate whether explanations or transfer improve, not only whether students clicked. One mark each for tool, diagnosis, response and evaluation.

Pr 32 (p. 218) (b) Correct. This links diagnosis to action and evidence of improved understanding.

Professional Development Seminar, Test A

20 marks. Suggested time 20 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. Which is a professional criterion emphasized in the lecture?

   (a) No responsibility to society
   (b) Learning stops at graduation
   (c) No preparation
   (d) Specialized knowledge and ethical standards

2. Using student work to improve instruction best illustrates the teacher as:

   (a) A storage clerk only
   (b) An assessor and learning guide
   (c) A person who never changes plans
   (d) A passive spectator

3. Support during a teacher first years of practice is mainly:

   (a) Only pre-service admission
   (b) Induction
   (c) Retirement
   (d) Waste management

4. The central value of supervised field experience is:

   (a) Eliminating evaluation
   (b) Replacing all theory
   (c) Connecting theory to practice with feedback
   (d) Avoiding learners

Part II. Show your reasoning.

1. Give two reasons teaching is considered a profession, each with an example.

2. Describe two teacher roles beyond explaining subject content.

3. Compare pre-service preparation with induction in two clear points.

4. Give a four-step improvement cycle for a trainee after an observed lesson.

Professional Development Seminar, Test B

20 marks. Suggested time 20 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. Which is the strongest development goal?

   (a) Improve one identified practice and evaluate its effect
   (b) Collect the most certificates
   (c) Become better somehow
   (d) Attend anything

2. What makes a portfolio stronger?

   (a) No evaluation
   (b) More unrelated documents
   (c) Evidence linked to goals and reflection
   (d) Only decorative pages

3. Constructive professional feedback should be:

   (a) Only praise regardless of evidence
   (b) Unrelated to teaching
   (c) Personal and humiliating
   (d) Based on observation and useful next steps

4. The best reason to choose an educational tool is:

   (a) It supports a defined learning objective
   (b) It has the most animations
   (c) Everyone must use the same tool
   (d) It is newest

Part II. Show your reasoning.

1. Write a short development plan for a teacher whose students struggle to explain answers.

2. Name two portfolio items and explain what each can demonstrate.

3. Rewrite this feedback constructively: Your lesson was bad.

4. Propose a simple technology use to diagnose misconceptions, and state how you would evaluate its value.

Professional Development Seminar, Test A: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (d) Professional practice combines expertise, preparation and ethical obligations.

Written part. Specialized preparation: teachers study subject knowledge and pedagogy. Ethical duty: teachers assess fairly and protect learner dignity. Two marks for each criterion-example pair. (Section: p. 208.)

Question 2. (b) Assessment information guides the next teaching decision.

Written part. Assessor: identifies learning gaps using evidence and feedback. Learning guide: supports students in choosing strategies and solving problems. Two marks per role with a concrete action. (Section: p. 209.)

Question 3. (b) Induction bridges initial preparation and independent professional practice.

Written part. Pre-service occurs before entry and develops foundational subject and pedagogical knowledge. Induction occurs during early practice and provides mentoring and support in the real workplace. One mark each for stage and purpose in both cases. (Section: p. 210.)

Question 4. (c) Practice tests and develops professional knowledge in authentic settings.

Written part. Review evidence; identify one specific weakness; plan and apply a targeted change; evaluate the next lesson using evidence. One mark per coherent step. (Section: p. 212.)

Professional Development Seminar, Test B: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (a) A focused goal connects action to measurable change.

Written part. Need: weak explanations. Goal: improve evidence-based responses. Activity: learn and apply structured questioning. Evidence: compare student response samples before and after a defined period. One mark per linked component. (Section: p. 213.)

Question 2. (c) The reader can see what the evidence proves and how practice developed.

Written part. A lesson plan can demonstrate planning and alignment. An anonymized student-work sample with commentary can demonstrate learning and the effect of feedback. Two marks per item and explanation. (Section: p. 214.)

Question 3. (d) Evidence-based feedback supports improvement while respecting the colleague.

Written part. Example: Several students could not identify the task goal; stating the success criteria and checking one example before independent work may help. Credit observable evidence, effect, actionable suggestion and respectful language. (Section: p. 216.)

Question 4. (a) Pedagogical purpose should guide selection.

Written part. Use a brief concept quiz with plausible distractors. Review response patterns, reteach the misconception and check with a new question. Evaluate whether explanations or transfer improve, not only whether students clicked. One mark each for tool, diagnosis, response and evaluation. (Section: p. 217.)

Facts at a glance

1. Teaching as a profession p. 208

Criterion + explanation + teaching example

2. Professionalization and the teacher role p. 209

Role → observable professional action

3. Teacher formation: before and during service p. 210

Pre-service → induction → continuing development

4. Preparation, practice and professional standards p. 212

Knowledge → supervised application → feedback → improvement

5. Plan continuing professional development p. 213

Need → goal → activity → evidence of impact

6. Portfolios and reflective evaluation p. 214

Evidence + reflection + link to a goal

7. Ethics, collaboration and learning communities p. 216

Observation + effect + constructive next step

8. Educational technology and evaluation of impact p. 217

Learning need → tool choice → activity → evaluation

Notation

Symbols are listed by section. The same letter may name different quantities in different sections; read the section's Fact.

Sources

Elective lecture themes: teaching as a profession, teacher formation, professional learning, portfolios, standards and educational technology. This follows the teacher-development focus of the supplied file; it is not a generic career-skills course.

The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. Check lecturer-specific derivations and conventions. The practice tests are original study material, not predictions of an examination.

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